

Liquidity Traps: A Unified Theory of the Great Depression and the Great Recession[†]

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This review of liquidity traps unifies three landmark economic downturns—the US Great Depression, the Great Recession, and Japan’s Long Recession—into a single analytical framework. We examine various forces that drive natural interest rates negative: temporarily (such as banking crises and debt overhangs) or permanently (such as demographic shifts and inequality). When policy rates hit the zero lower bound, conventional monetary tools lose traction. Under a standard monetary policy regime, counterintuitive paradoxes emerge: Greater price flexibility deepens recessions, and positive supply shocks become contractionary. We show how policy effects—including the size of fiscal multipliers, forward guidance, and these paradoxes—depend critically on the monetary-fiscal regime and on central bank credibility. The paper explains how regime changes, such as Franklin D. Roosevelt’s 1933 abandonment of the gold standard and balanced-budget dogmas, successfully reversed deep slumps by credibly shifting expectations. We examine whether secular-stagnation forces are likely to assert themselves in the coming decades. (JEL E32, E42, E43, E52, E62, G01, G21)

1. Introduction

“Understanding the Great Depression is the Holy Grail of Macroeconomics,” said Ben Bernanke, then a Princeton professor of economics, in 1995.¹ This paper reviews the modern literature on the liquidity trap: the awkward situation in which the central bank has cut its policy rate to zero and cannot cut it further because of the zero lower bound (ZLB) on

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¹ See Bernanke (1995).

short-term nominal interest rates. With falling inflation, rising unemployment, and the key policy tool constrained by the ZLB, the government seeks alternative measures.

Two episodes of a liquidity trap changed the course of history: the Great Depression and the Great Recession. These historical turning points had far-reaching consequences; there is a reasonably well-established consensus among historians that the Great Depression played a decisive role in the rise of Adolph Hitler, ultimately triggering World War II.² More generally, recent economic research has found that surges in populism, both on the right and left, are highly correlated with macroeconomic crises. Looking over the past 120 years, Funke, Schularick, and Trebesch (2023) find that the two largest spikes in populist support occurred in the years following the start of two major events: the Great Depression and the Great Recession.³ These populist surges have often preceded the collapse of democratic institutions, dictatorships, and war.

In our view, crises that inflict such profound damage should be central to macroeconomics—after all, the Great Depression effectively gave birth to the field. However, after the Great Inflation of the 1970s, attention shifted away from studying large-scale downturns, first briefly, then more permanently. Several scholars argued that research on business cycle stabilization was essentially complete around the turn of the century and that macroeconomic research priorities should shift toward other questions.⁴

The Great Recession of 2008 rekindled interest in the questions that first animated the discipline, and we think that this renewed focus is important for at least two reasons. First, the potential human tragedy of crises such as the Great Depression or the Great Recession is so vast that even small gains in understanding can yield large social returns. Second, this line of research has unusually large and immediate public policy impact on real-world outcomes: The US Federal Reserve, for example, staffed by highly skilled PhD economists, meets regularly to shape policies that may help avert or exacerbate crises. These meetings draw directly on economic research, including some reviewed here, a substantial part of which originates from current or former research economists at the Federal Reserve.

We define the liquidity trap as follows: A liquidity trap is a general equilibrium in which the short-term nominal interest rate has collapsed to zero *and* the fiscal-monetary policy regime implies a further decline in the nominal rate—if not for the ZLB.

This definition does not imply that a central bank “can do nothing” or that expansionary monetary policy is like “pushing on a string.”⁵ It only suggests that, in a given monetary and fiscal policy regime, the central bank has reached the ZLB and would choose lower policy rates if not for the lower bound. Even with short-term rates at zero, the central bank can increase aggregate spending by changing expectations about future interest rates, inflation, and output, as well as directly intervene in financial markets. The power of central banks to manage expectations suggests a less pessimistic view of the effectiveness of monetary policy relative to the early literature inspired by Keynes (1936). However, changing expectations can be challenging

²See, for example, Kershaw (1998); Kershaw (2000); Evans (2003); Bullock (1952); and Bracher (1970).

³See also Guriev and Papaioannou (2022); and Funke, Schularick, and Trebesch (2016).

⁴A prominent example is Lucas (2003).

⁵The idea that monetary policy loses its power when *long-term* interest rates are sufficiently low is central in Keynes (1936). See Eggertsson and Petracchi (2021) for a discussion of the relationship between the original and modern definitions of the liquidity trap. The term is attributed to Robertson (1940). The metaphor comparing monetary policy at the ZLB to pushing on a string originated from Democratic Congressman Thomas A. Goldsborough's congressional hearing on the Banking Act in 1935. See Wood (2006) and Sandilands (2001).

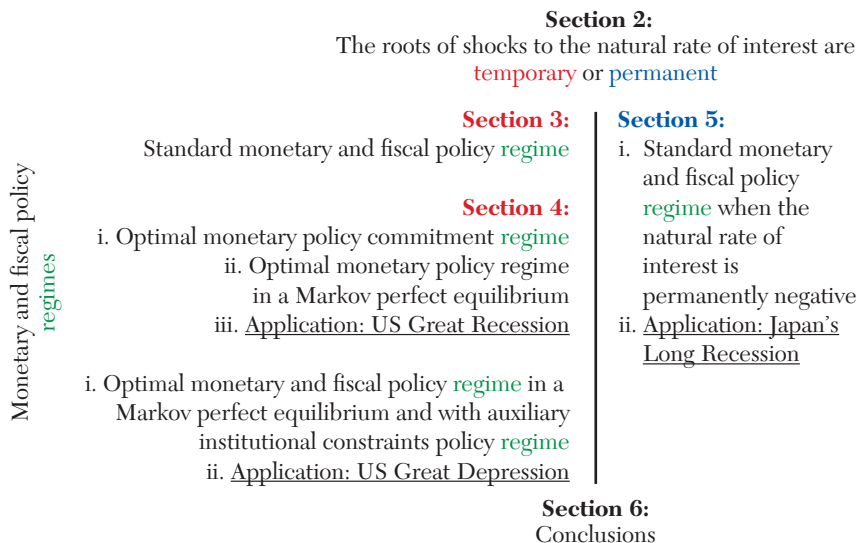


Figure 1. Two Organizational Principles of the Paper

if the central bank lacks credibility or the underlying shocks are permanent. In those cases, Keynes's policy prescriptions still resonate today.

Our review has two central organizational principles, illustrated in Figure 1. The first is the *nature of the underlying shocks* that led to a slump, be it the Great Depression, the Great Recession, or Japan's Long Recession. The left column of Figure 1 shows sections where the ZLB is binding due to temporary shocks, while the right column shows sections where the ZLB is binding due to permanent shocks. We ask: What are their driving forces? Do these forces respond to policy? Can these forces be characterized in a parsimonious way?

Our second organizational principle is the assumed *monetary and fiscal policy regime*. Figure 1 shows how each section is organized from top to bottom based on assumptions about the policy regimes. A monetary and fiscal policy regime specifies how policymakers set policy today and, no less importantly, in the future. The policy regime shapes expectations today, which has a first-order impact on today's aggregate demand and supply.

To see why these organizational principles are central, consider a classic question of macroeconomics in recession at the ZLB. What is the government spending multiplier? This statistic answers the following question: If the government increases its spending by one dollar, by how many dollars does aggregate GDP increase? We cannot address the question unless we specify the monetary and fiscal policy regime and determine whether the shock is short-lived or permanent.

Consider a shock that makes the ZLB *temporarily* binding, and suppose that the policy regime is *such that* the interest rate is fixed at the ZLB for the duration of the shock and reverts back to steady state as soon as the shock is over. In this scenario, a central result is that the government spending multiplier can be very large.

In contrast, if monetary policy is flexible and can compensate for its inability to cut rates today with promises of future interest rate reductions, then the spending multiplier is much

smaller. The decrease in the multiplier is due to the central bank tightening policy in response to a fiscal expansion.

The ability of the monetary authority to compensate for current rate cuts with commitments to future rates depends crucially on two assumptions: first, that shocks are temporary; second, that the central bank has the credibility to determine the private sector's expectations about future policy.

Elements of a unified theory of the US Great Depression, Great Recession, and Japan's Long Recession emerge from examining how different policy regimes, economic mechanisms, and shock processes generated these ZLB episodes. Although researchers have formulated numerous approaches to modeling ZLB events, our analysis of specific policy regimes and their underlying shock processes serves a distinct purpose. We aim to explain why policymakers chose their specific policies during these three landmark episodes, how they implemented these choices, and how these decisions shaped macroeconomic outcomes.

From this comparative perspective, we draw some tentative lessons at the end of the introduction.

We divide this review into four main sections, each addressing a broad theme aligned with the two central organizing principles discussed above. To maintain perspective on the overall structure, we encourage the reader to refer to the road map in Figure 1 as we outline the remainder of this review.

What are the key theoretical building blocks of a liquidity trap? Section 2 identifies the two. The first building block is the shocks, that is, forces that generate a negative natural interest rate. The natural interest rate is the equilibrium real interest rate under the assumption that wages and prices are fully flexible.⁶ We distinguish between fast-moving and slow-moving forces. Fast-moving forces include firm/household debt overhang, banking crises, and asset bubbles that can all generate temporary negative natural rates. Slow-moving forces include demographics (slowdown in population growth or increased life expectancy), a permanent increase in inequality, and lower productivity growth. These forces can generate permanently negative natural interest rates.

The second building block of the liquidity trap is the ZLB. The ZLB arises from a simple fact: You will not lend a dollar unless you receive at least one dollar in return. At a negative interest rate, it would be more profitable to sit on the money (abstracting from storage costs). This constraint stems from the existence of paper currency and we review several possible ways to relax it. We start by laying out the bare bones of the two building blocks in Section 2 with minimalistic specifications of the monetary and fiscal policy regimes, examined in detail in subsequent sections.

Even with only two elementary building blocks and a highly stylized policy regime, Section 2.3 yields a striking prediction first formalized by Krugman (1998): Under certain conditions, printing money has no effect on prices. This theoretical prediction, built on logical reasoning rather than historical patterns, faced significant skepticism once presented. As Krugman (2021) later reflected, he initially aimed to debunk the “old Keynesian fairy tale” that monetary policy could ever lack the ability to increase aggregate demand through printing money—a conviction aligned with the dominant narrative of Friedman and Schwartz (1963)

⁶We only consider real shocks, in this review, as a reason for the ZLB to be binding. We do not consider the possibility of self-fulfilling beliefs generating a liquidity trap, a separate branch of the literature. The seminal paper in this literature is Benhabib, Schmitt-Grohé, and Uribe (2001). This important branch of the literature merits a separate review.

at that time. Yet, when subjected to rigorous microfoundations, the “fairy tale” became reality under specific conditions. Remarkably, the data confirms Krugman’s (1998) prediction: Japan increased money supply by about 1000 percent (1998–2021) while prices declined 3 percent, and the United States increased money supply by 500 percent (2008–20) with inflation remaining consistently below the Federal Reserve’s 2 percent target.

The review then shows the consequences of different policy regimes under various assumptions about the underlying shocks. We start by focusing on a temporary negative natural rate of interest.

Section 3 introduces the textbook canonical new Keynesian model, which we will use to establish a number of the results that also apply in more general setting. In line with our two main organizational principles, we first limit our attention to (i) the *Standard monetary and fiscal policy regime* under (ii) the assumption that the ZLB is only temporarily binding. A simple way to describe the monetary aspect of this policy regime is that monetary policy follows the classic Taylor rule regime: The central bank raises its policy rate if inflation is above its inflation target and cuts it if below target (with some complementary weight on output relative to target). The regime, however, can find itself stuck at the ZLB for sufficiently large deflationary shocks. We then explore various alternative marginal changes in policy instruments, taking the monetary policy regime as given.

The standard monetary and fiscal policy regime has been the focus of much of the existing literature. We impose the necessary conditions for a liquidity trap, reviewed in Section 2, and explore the following questions. Suppose that the monetary policy *regime* remains fixed. How effective are the substitutes for interest rate cuts, ranging from classic aggregate demand or supply policies to unconventional monetary policies directly intervening in financial markets? Can the underlying shocks be directly addressed? Are there policies that may circumvent the ZLB itself?

Once the natural rate is temporarily negative so that the ZLB is binding, the standard monetary and fiscal policy regime forces policymakers to confront a topsy-turvy world in which many traditional economic laws are turned on their head, establishing three major results. First, the fiscal multiplier at the ZLB is significantly higher than at a positive interest rate. Second, we show the paradox of price flexibility, suggesting that higher price flexibility makes the recession even worse so that traditional market price mechanisms may not be relied on to bring about stability. Third, we show the paradox of toil, which says that while positive supply-side reforms improve the economy’s production capacity, they may *contract* demand when the economy does not fully utilize its capacities already. We briefly review an emerging empirical literature for each of the three results.

We next establish a connection between the fiscal multiplier just derived and the original Keynesian tradition. Keynes’s student Kahn (1931) was the first to derive a fiscal policy multiplier. To examine this connection more thoroughly, we extend the new Keynesian model and develop a simple heterogeneous agent new Keynesian (HANK) model.⁷

The old Keynesian multiplier works through a chain reaction: One person’s spending becomes other people’s income, some of which they spend generating more income, triggering

⁷ Early contributions focusing on analytic solutions include Galí, López-Salido, and Vallés (2007), Bilbiie (2008), Cúrdia and Woodford (2010), and Eggertsson and Krugman (2012). Quantitative analyses assuming that heterogeneity is due to uninsurable idiosyncratic risk as in Bewley (1977) include Guerrieri and Lorenzoni (2017) and Kaplan, Moll, and Violante (2018), the latter having coined the HANK acronym.

successive rounds of spending. While this mechanism is absent in the canonical new Keynesian model, it becomes transparent in the model we develop that introduces debt-constrained agents alongside unconstrained agents. The number of constrained agents maps into the marginal propensity to consume in the old Keynesian model, bridging the two traditions and amplifying both the multipliers and the paradoxes in a modern HANK setting. Section 3 concludes by evaluating tax proposals and reforms to the payment system that eliminate the ZLB constraint. We also review the practical limitations of these approaches.

Section 4 considers the same model as in Section 3 but changes one of the two central organizing principles: the policy regime. Instead of assuming the standard monetary and fiscal policy regime, we ask, What is the optimal policy regime? How is the design of policy affected by the ability of the central bank and fiscal authorities to commit to future policy and set their policies to maximize a joint objective?

We could analyze several different configurations of policy regimes. However, this section is not aimed at providing an exhaustive list of abstract theoretical characterizations of all possible policy regimes. Instead, we concentrate on a subset of policy regimes most relevant for understanding actual policymaking in the US Great Recession, the US Great Depression, and Japan's Long Recession.

We first consider two monetary policy regimes that are polar opposites. The *optimal monetary policy regime under commitment* is at one end of the spectrum: The central bank maximizes welfare and can fully commit to future short-term interest rate policy, which substantially improves outcomes relative to the standard monetary policy regime. The *optimal monetary policy in a Markov perfect equilibrium* is at the other end. In a Markov perfect equilibrium (discretion), the central bank shares the same objective as under commitment but can only formulate strategies as functions of the current state variables. Without fiscal policy backup, this precludes policies such as forward guidance about future interest rates due to credibility problems. We show that the predicted equilibrium is identical to that of the standard monetary policy regime.⁸

One lesson from Section 4, assuming a credible optimal monetary policy regime, is that the ability of the monetary policy authorities to commit to future interest rate policy has a fundamental influence on other policy interventions. Consider the fiscal multiplier. Suppose the central bank can adopt the optimal monetary policy regime under commitment, allowing it to commit to low future real interest rates at the ZLB. In this case, expansionary fiscal policy leads to tighter monetary policy, since less monetary easing is needed to stabilize the economy. The result is a smaller fiscal multiplier that resembles the one at positive interest rates.

If monetary authorities successfully adopt the optimal monetary policy regime under commitment at the ZLB, they will largely avoid a severe recession. Not only are multipliers orthodox, but generally, more traditional economic logic applies, so that the paradoxes assuming the standard monetary and fiscal policy regime no longer apply.

⁸The optimal monetary policy in a Markov perfect equilibrium is often referred to as optimal policy under discretion. We prefer the Markov perfect equilibrium language because it becomes more natural to consider alternative policy regimes with additional institutional constraints, such as the gold standard or other well-defined "commitment technologies." One such technology is the government's ability to pay back debt in nominal value: No country that can print currency has chosen outright default over indirect default via inflation. Markov perfect equilibrium language also allows for the introduction of well-defined verbal "commitment technologies" that may fall short of the central bank's ability to fully commit to policy in all states but provide valuable perspective on what can be achieved.

The empirical evidence we review suggests that neither the optimal monetary policy regime in a Markov perfect equilibrium nor the optimal monetary policy regime under commitment describe in a satisfactory way the actual policy regime in the United States during the Great Recession. We base this on statistical evidence from the literature, narrative evidence from declassified Federal Reserve Open Market Committee minutes, and speeches from policymakers. However, we argue that the two polar opposite policy regimes give useful insight into the challenges faced by policymakers and what they were trying to achieve. A more general specification is needed to account for the data, which the literature has not yet reached a consensus on. We offer comments on possible avenues for future research that may capture a more general monetary policy regime consistent with empirical observations from the Great Recession. Based on the latest evidence we review, our assessment is that an empirically realistic policy regime should demonstrate properties that align with large spending multipliers and the paradox of toil while being consistent with a nontrivial role of forward guidance.

Next, we consider an optimal monetary *and* fiscal policy regime in a Markov perfect equilibrium and take into account institutional constraints. We develop this policy regime to understand the role of policy in explaining the US Great Depression and the subsequent recovery. Three institutional constraints were of primary importance during that time: the gold standard, a commitment to a balanced budget, and a limit on the size of the government. An optimal monetary and fiscal policy regime in a Markov perfect equilibrium assumes that the government can credibly pay back its debt's nominal value, as in Calvo (1978) and Lucas and Stokey (1983).

We define a regime change as eliminating some (or all) institutional constraints that bind the hand of the policymaker. The model we review formalizes Temin and Wigmore's (1990) hypothesis that regime change implemented by Franklin Delano Roosevelt (FDR) in 1933 ended the Great Depression. We argue that it also sheds light on the second phase of the Great Depression, characterized by the "mistake of 1937" and the subsequent recovery that resumed before World War II spending began. We show that a simple new Keynesian model can replicate, in broad brush, the evolution of key variables during this period.

Section 5 returns to one of the two organizing principles by examining a crucial distinction between temporary and permanent liquidity traps. While previous sections analyzed temporary negative natural rates, we now consider an economy in which both short- and long-run natural rates are negative under the standard monetary and fiscal policy regime. This analysis builds on Summers's (2014) revival of Hansen's (1939) secular stagnation hypothesis. Formalizing this idea requires a fundamental reformulation of the aggregate demand and supply sides of the new Keynesian model, first attempted by Eggertsson and Mehrotra (2014).⁹ While some policy implications of Sections 3 and 4 carry over to this setting, others require significant modification.

The review yields the following key policy implications:

- (i) A sufficiently credible and flexible monetary policy can largely offset a temporary drop in the natural interest rate by adopting the optimal monetary policy regime under commitment. The force of credible monetary policy in the face of temporary shocks represents one of the main policy implications of this review. Implementing it is also a central challenge for policymakers.

⁹Later published in Eggertsson, Mehrotra, and Robbins (2019), which includes a quantitative model.

- (ii) Carefully designed fiscal support becomes critical if the monetary policy regime lacks credibility to support a robust recovery. In this case, fiscal policy becomes one of the central lines of defense that is robust across different models.
- (iii) Unconventional policies such as credit easing (quantitative easing, QE) play a significant role. QE proved particularly effective during severe financial market disruptions, such as after the 2008 global financial crisis. Even without financial disruption, QE can influence interest rates through long-term security purchases. However, the literature has not yet reached a consensus on its strength or the mechanism through which it operates.
- (iv) Monetary policy can lose much of its power if the natural interest rate remains negative in the short and long run. In this case, carefully designed fiscal support becomes *essential*.

As governments grapple with the possible aftermath of the 2020s inflation surge, a fundamental question emerges: Has it temporarily masked the secular decline in the natural interest rate? Long-term demographic and inequality trends driving secular stagnation remain largely unchanged, pointing to further downward pressures. A second fundamental question is the effect of the US policy regime change triggered by Donald Trump taking presidential office in January 2025. The scope of the regime change is unusually large, with an uncertain impact on inflation and the natural interest rate. Proposed policies include trade restrictions, reduced immigration, tax cuts for high earners, and cuts in government spending. While reduced immigration, a further increase in inequality, and spending cuts could lower the natural interest rate, increasing government debt could push the rate higher. Beyond the effect of the policy regime change, artificial intelligence (AI) could boost productivity and raise the natural interest rate, although its impact remains highly speculative.

The long-term evolution of the natural interest rate represents one of the central macroeconomic unknowns of our time.

2. *The Root Causes of the Liquidity Trap: Two Building Blocks*

Section 2 reviews the two building blocks of a liquidity trap. Section 2.1 defines the natural rate of interest and then reviews the forces that can cause it to become negative, either temporarily or permanently, forming the first building block. The second building block, reviewed in Section 2.2.1, is the existence of paper currency, which gives rise to the ZLB on the short-term nominal interest rate. If the natural rate is negative, the central bank must set negative policy rates to maintain price stability, which is impossible due to the ZLB. Section 2.3 reviews significant empirical predictions derived from the two building blocks.

2.1 *The First Building Block: A Negative Natural Rate of Interest*

2.1.1 *Understanding the Natural Rate of Interest: From Wicksell to Woodford*

The natural interest rate plays a fundamental role in our review as the first building block of the liquidity trap. For a central bank that targets price stability, a negative natural rate of interest is a necessary condition for the ZLB to be binding.

To understand why a negative natural rate is a necessary condition for a liquidity trap, we find it helpful to review the original and modern formulations of the concept. Knut Wicksell, a Swedish economist, first proposed the idea in 1898, defining it as the interest rate required for prices to be stable (Wicksell 1898). According to the modern definition proposed by Michael Woodford, it is the real interest rate (r) if all prices and wages are flexible. Since prices are sticky in the new Keynesian model, it is purely a fictional construct that cannot be directly observed (Woodford 1999).

Woodford demonstrates that a central bank achieves complete price stability, that is, a constant price level, if it sets the policy rate (i) equal to the natural interest rate, r^n , as will be evident in Section 3. With constant prices, a negative natural rate implies a negative nominal interest rate, violating the ZLB. A negative natural rate is a *necessary condition* for a liquidity trap, assuming the central bank has a weakly positive inflation target ($\pi \geq 0$) but not a *sufficient* condition. A central bank that targets 2 percent inflation can offset a negative natural rate as low as -2 percent, since at the ZLB, $r = i - \pi^e = -2$ percent, where π^e represents expected inflation. We next address the following question: What are plausible conditions for the natural rate of interest to turn temporarily or permanently negative? To answer this, we can abstract from price-level determination and, in large part, simplify the analysis by treating output as exogenous. We make this abstraction with the understanding that one can interpret the exogenous path for output as being generated from a model with flexible prices and is thus only a function of exogenous shocks such as technology shocks, and that it will become endogenous in coming sections once we introduce price rigidities. Since all prices are flexible in this section, the real and natural rate are one and the same thing. Yet, since this section provides the foundation of the natural rate for later sections, the real and natural rate are two different variables. We will typically refer to the real rate as the natural rate in this section, since that is the role the real rate we derive here will play.

Before continuing, we provide a broad overview of some key patterns in the data from the three major landmark episodes from economic history that motivate our analysis.

2.1.2 *Key Data from Three Landmark Episodes: Common Footprints*

The Great Depression and the Great Recession in the United States, as well as Japan's Long Recession, left the same footprints in the data.

In the United States, the Great Depression and the Great Recession began with asset price collapses and banking crises. The Dow Jones Industrial Average plummeted 89 percent from its September 1929 peak to its July 1932 trough, and 54 percent from its October 2007 peak to its March 2009 nadir. Short-term nominal interest rates collapsed to near zero in 1932 during the Great Depression and again in December 2008 during the Great Recession.

Japan's Long Recession mirrored this pattern, beginning with a spectacular implosion of an asset price bubble. Malkiel (2010) estimated that at the apex of the bubble in 1990, the value of Tokyo's Imperial Palace grounds exceeded that of all California real estate. The bubble's 1992 implosion generated a banking crisis. The Nikkei dropped by 80 percent from its 1989 peak to its 2003 trough, while the Bank of Japan cut its policy rate below 1 percent by 1995. In 1999, the Bank of Japan officially implemented a zero interest rate policy; see Section 2.3.

After both US crises, the Dow Jones Industrial Average eventually exceeded its previous nominal heights. It took 25 years after the Great Depression and five and a half years after the

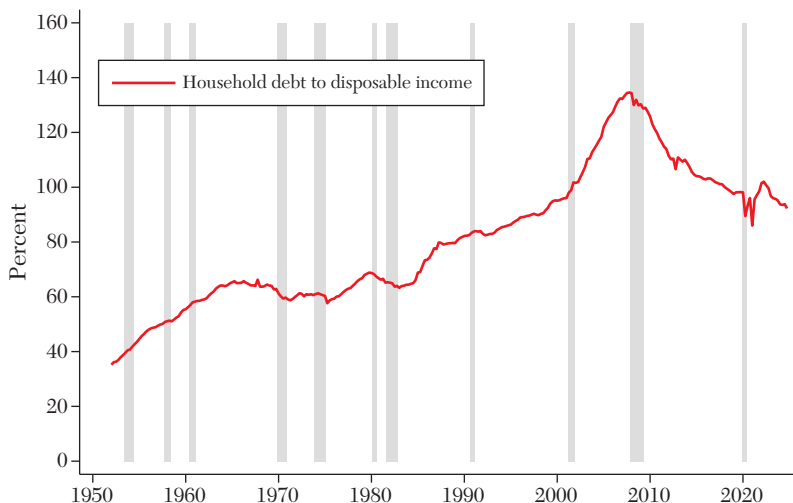


Figure 2. Ratio of Household Debt to Disposable Personal Income, 1952–2024

Source: Data from FRED series (Federal Reserve Bank of St. Louis 2024a) and (Board of Governors of the Federal Reserve System 2025h).

Great Recession. In contrast, Japan's Nikkei took 34 years to reach its 1989 peak, but as of this writing, it has dipped below it again.

We now build the first pillar of our analysis: forces that can temporarily or persistently generate a negative natural interest rate.

2.1.3 *Fast-Moving Forces and Temporary Negative Natural Rates: Debt Overhang, Banking Crises, Asset Bubbles*

Fast-moving forces that trigger a liquidity trap are the stuff of newspaper headlines: bank runs with panicked customers lining up outside banks, spectacular asset price collapses, large numbers of home foreclosures, firms filing for bankruptcy, mass layoffs, and so on. In the following, we show how these forces can be captured using an extension of the model in Eggertsson and Krugman (2012), which focuses on the overaccumulation of household debt. The more general framework incorporates insights from two influential branches of the literature that trace the fast-moving forces to shocks to firms' balance sheets and banking crises.

Modeling a Minsky Moment as Unsustainable Household Debt.—The first fast-moving force can be motivated by Figure 2, which shows a rapid increase in household debt from less than 100 percent of disposable income in 2000 to more than 130 percent on the eve of the 2008 crisis.¹⁰

¹⁰The rapid accumulation of private debt has been carefully documented by several authors, notably in important work by Mian and Sufi (2009).

We start by modeling the crisis as an abrupt realization that household debt accumulation has gone too far, often referred to as a Minsky moment. This idea is based on the work of Minsky (1986). He argued that there existed a recurrent cycle of instability in which calm periods led to complacency about debt, and hence increased leverage, which in turn set the stage for a crisis leading to rapid deleveraging. The natural interest rate turns negative if the deleveraging event is large enough. Since one person's debt is someone else's asset, we need a model with borrowers and savers to capture this idea.¹¹

Imagine a world with two types of agents: savers (s) and borrowers (b). For simplicity, they differ only in their time preferences, with $\beta^s > \beta^b$. We assume a unit mass of agents, divided equally between borrowers and savers, each receiving income Y (so total output is Y).¹² Their decision problem is the following:

$$(1) \quad \max_{C_t(i), D_t(i)} E_t \sum_{t=0}^{\infty} \beta^{(i)t} \ln C_t(i),$$

subject to

$$(2) \quad D_t(i) = (1 + r_{t-1})D_{t-1}(i) - Y + C_t(i) - T_t(i),$$

where E_t is an expectation operator, $C_t(i)$ is consumption, $i = \{b, s\}$, $D_t(i)$ is one-period real debt (so a negative D_t denotes an asset), r_t is the real interest rate at time t , and Y is the per period exogenous output endowment corresponding to output per capita.¹³ We assume the borrower's taxes, $T_t^b(i)$, remain constant at T^b , while only the saver's taxes vary. This assumption simplifies the model, ensuring Ricardian equivalence. Agents face an exogenous debt capacity $\tilde{D}^{high}$:

$$(3) \quad (1 + r_t)D_t(i) \leq \tilde{D}^{high}.$$

For a sufficiently low value of $\tilde{D}^{high}$, the impatient agent will eventually exhaust their borrowing capacity, which we assume they have exhausted by time t .¹⁴

While the borrower is at the corner solution (borrowing as much as the limit permits), the saver satisfies an interior optimality condition. The saver then pins down the short-term real interest rate:

$$(4) \quad \frac{1}{C_t^s} = (1 + r_t)\beta^s E_t \frac{1}{C_{t+1}^s}.$$

¹¹While the model below is based on Eggertsson and Krugman (2012), Guerrieri and Lorenzoni (2017) developed a similar idea with idiosyncratic income risk as in Bewley (1977).

¹²We assume a "continuum of agents" that implies that agents are sufficiently numerous to take prices and interest rates as given.

¹³The borrower and saver receive the same income.

¹⁴The value $\tilde{D}^{high}$ is strictly lower than the present discounted value of each agent's output. Constraint (3) is often stated in terms of a simple debt limit, $D_t < D^{\lim}$. However, we find this less natural since the borrowing limit presumably depends on the borrower's ability to repay in the subsequent period and, therefore, should account for the interest payment due at that time.

In solving the model, the first step is to find its steady state. It follows immediately from the consumption Euler equation of the saver that in the steady state:

$$(5) \quad \bar{r} = \frac{1 - \beta^s}{\beta^s},$$

where $\bar{r}$ is the steady-state real interest rate. We can now solve for the steady-state consumption of the borrower, who is up against his borrowing constraint

$$(6) \quad \bar{C}^b = Y - \frac{\bar{r}}{1 + \bar{r}} \tilde{D}^{high},$$

consuming his income net of interest rate payment. Conversely, the saver consumes his endowment plus the interest the debtor pays on his debt:

$$(7) \quad \bar{C} = Y + \frac{\bar{r}}{1 + \bar{r}} \tilde{D}^{high}.$$

We do not attempt to model the sources of the debt capacity, but we review the literature that explores this question below. We think of its reduction as a proxy for a more general view about the perceived “safe” level of borrowers’ debt capacity, considering, for example, default risk, moral hazard, and, for now, limiting ourselves to households. Debt-driven crisis theories typically propose that this safe level can change abruptly. We model this as an unanticipated reduction in the household’s debt capacity: $\tilde{D}^{high} \rightarrow \tilde{D}^{low}$. The reduction could, for example, reflect a sudden realization that default risk or moral hazard in the mortgage market has proved to be more severe than previously understood.

The model transitions to a new steady state in a single period, adopting $\tilde{D}^{low}$ instead of $\tilde{D}^{high}$, while retaining the same steady-state real interest rate. We refer to the adjustment period following the shock as the short run (S) and the new steady state as the long run (L). The key insight is that the debtor must deleverage (i.e., cut spending) immediately in the short run to satisfy the new constraint on debt capacity.

Since aggregate output remains constant, the saver must increase spending to offset the borrower’s spending cuts needed to meet the new borrowing limit. Manipulation of short-term budget constraints (2) yields expressions for the long-term consumption of the saver, $C_L^s = Y + (1 - \beta^s) \tilde{D}^{low}$ and the short-term consumption, $C_S^s = Y - \tilde{D}^{low} / (1 + r_s) + \tilde{D}^{high}$.¹⁵

Substituting this result into (4) gives an expression for the real rate, that is, the natural rate of interest):

$$(8) \quad 1 + r_s = \frac{1}{\beta^s} \frac{Y + \tilde{D}^{low}}{Y + \tilde{D}^{high}}.$$

The natural rate r_s can turn negative when $\tilde{D}^{low}$ is sufficiently small relative to $\tilde{D}^{high}$.

¹⁵We assume there is an equal number of savers and borrowers and, for simplicity, normalize the total to 1. Then, at any time, t , we have $Y = \frac{1}{2}C_t^b + \frac{1}{2}C_t^s$. In the long run, the saver satisfies $C_L^s = Y + (1 - \beta^s)\tilde{D}^{low}$. The short-run consumption of the saver is $C_S^s = 2Y - C_t^b$. The borrower must reduce spending to meet $D_S = \tilde{D}^{low} / (1 + r_s)$; thus the borrower’s budget implies $C_S^b = Y + \tilde{D}^{low} / (1 + r_s) - \tilde{D}^{high}$ and the resource constraint yields $C_S^s = Y - \tilde{D}^{low} / (1 + r_s) + \tilde{D}^{high}$. Substitute C_L^s and C_S^s into the consumption Euler equation for further details.

In summary, a deleveraging shock triggers a drop in the natural interest rate for the following reasons. Borrowers reduce their spending to satisfy the reduction in their borrowing limit. Savers must increase their spending by an equivalent amount to compensate for this drop because aggregate output remains fixed and consumption absorbs all production. Only a lower real interest rate can induce savers to spend more. The natural interest rate then turns negative if the shock is large enough.

A straightforward mechanism amplifies the drop in the natural interest rate as shown in Eggertsson and Krugman (2012). A fall in the price level increases the real value of nominal debt, shifting wealth from debtors to creditors, so debtors need to cut their spending even more to satisfy their debt limit. We discuss the empirical importance of this mechanism during the US Great Depression in Section 4.3.3, drawing on Hausman, Rhode, and Wieland (2021).

Authors relying on representative agent models often capture the increased desire to save, for example, Christiano, Eichenbaum, and Rebelo (2011), as a rise in β^s . Equation (8) shows that this is equivalent to widening the spread between $\tilde{D}^{low}$ and $\tilde{D}^{high}$, thus providing microfoundations for preference shocks.

There are four reasons why an exogenous reduction in $\tilde{D}$ provides a more useful microfoundation for explaining a negative natural interest rate than exogenous shifts in β^s , which are typical of first-generation ZLB models. First, a researcher can easily observe changes in debt limits with observable data. For example, Figure 2 shows that household debt peaks during the crisis and then contracts, consistent with our hypothesis, and provides a rough estimate of how much debt needs to decline for the deleveraging cycle to conclude. Second, while the duration of a β^s shock is pinned down only by assumption or by matching aggregate data, the deleveraging model predicts that interest rates recover once households reach sustainable debt levels, even if the debt limits remain permanently low. This prediction places an upper bound on the duration of the deleveraging cycle. Third, emphasizing borrowing and lending frictions encourages a deeper examination of banking relationships, generating directly measurable variables (e.g., interest rate spreads) that help discipline numerical simulations. Finally, these microfoundations naturally link to the rich literature on firm–bank ties and balance-sheet shocks in major crises, but we turn to these approaches next.

Minsky Moment as a Shock to Firms' Balance Sheets.—A second candidate for a fast-moving force is firms' overaccumulation of debt. A modest extension of the model captures this idea.

The literature has a long tradition of tracing business cycles to variations in firms' ability to borrow from the banking sector.¹⁶ Two classic papers that integrate this perspective into general equilibrium business cycle models, giving rise to a large literature, are Bernanke and Gertler (1989) and Kiyotaki and Moore (1997). Bernanke–Gertler emphasizes the role of firms' net worth in mitigating the cost of external financing due to moral hazard. In contrast, Kiyotaki–Moore emphasizes the role of firms' collateral in securing bank lending.¹⁷

To see how the model we have presented can capture these ideas, even if in reduced form, imagine that the borrower now corresponds to an entrepreneur (firm). The entrepreneur (borrower) has access to investment opportunities in the form of capital. At the same time,

¹⁶Bernanke (1983) is a classic reference that builds on Friedman and Schwartz (1963).

¹⁷For a recent review of this literature and how it evolved following the Great Depression, see Gertler and Gilchrist (2018). Two other influential papers in this literature that review its origins are Bernanke, Gertler, and Gilchrist (1999) and Gertler and Kiyotaki (2010).

the saver can only deposit his savings in banks that pay risk-free interest but lend the deposits to the entrepreneurs.¹⁸ The shock, $\tilde{D}^{high} \rightarrow \tilde{D}^{low}$, now captures a decline in firms' borrowing capacity, a reduced-form way of capturing lower collateral values of the assets firms can pledge to secure loans, for example, due to a fall in asset prices (Kiyotaki–Moore) or a reduction in firms' net worth, which limits their ability to borrow from banks due to moral hazard as the firm has less skin in the game (Bernanke–Gertler). The net result is a temporary reduction in the natural interest rate.

Minsky Moment as a Reduction in Bank Lending Capacity.—A third candidate for a fast-moving force is an abrupt reduction in banks' ability to extend credit. Again, a modest extension of the model can capture this idea.

In the wake of the Great Recession, a second generation of models emerged in the tradition of Bernanke–Gertler and Kiyotaki–Moore, focusing on the distress in the banking sector as a key source of instability¹⁹

We do not need to change the notation in the model we presented to consider the model as a narrative for reducing bank lending capacity. Instead, we reinterpret $\tilde{D}$ to represent bank lending capacity, which may unexpectedly decline. To make this interpretation more concrete, we can integrate a banking sector into the model, following the existing literature. We refer to one example in next section. The shock, $\tilde{D}^{high} \rightarrow \tilde{D}^{low}$, is now interpreted as a reduction in the bank's capacity for lending rather than a decrease in households' or firms' debt capacity.

It is helpful to introduce the notion of spread, that is, the difference between bonds that carry risk and those considered safe. To do so, we introduce a spread function that allows us to consider all three forces simultaneously: shocks to households' or firms' borrowing capacity and shocks to the banking sector that limit their ability to extend loans.

Minsky Moment and the Spread between Risky and Risk-Free Assets.—The spread between risky and risk-free bonds plays a central role in the literature on financial frictions. We can extend the model to illustrate that an increase in spreads proxies a drop in the natural interest rate. Data on spreads in the Great Recession and its aftermath raise profound questions about the origin of the liquidity trap.

Figure 3 shows the spread between Moody's Baa corporate bonds and 10-year Treasuries, which spiked sharply during the early part of the Great Recession. We can extend the model to match the increase in spreads. The spread increase is then a proxy for a drop in the natural interest rate from any of the three sources we have reviewed: over-leveraged households, firms, or distress in the banking sector.²⁰

Imagine that borrowers and savers now face different interest rates (indexed by b and s). The savers' interest rate is risk-free from bank deposits, while the borrowing rate reflects

¹⁸This extension is worked out in detail in appendix 2.2 in Eggertsson and Krugman (2012).

¹⁹Prominent examples include Cúrdia and Woodford (2010), Gertler and Kiyotaki (2010), and Brunnermeier and Sannikov (2014). See Gertler and Gilchrist (2018) for a review of this and related literature.

²⁰The evolution of these spreads raises a central question about the nature of the Great Recession liquidity trap: temporary or permanent. We discuss this question at the end of this section.

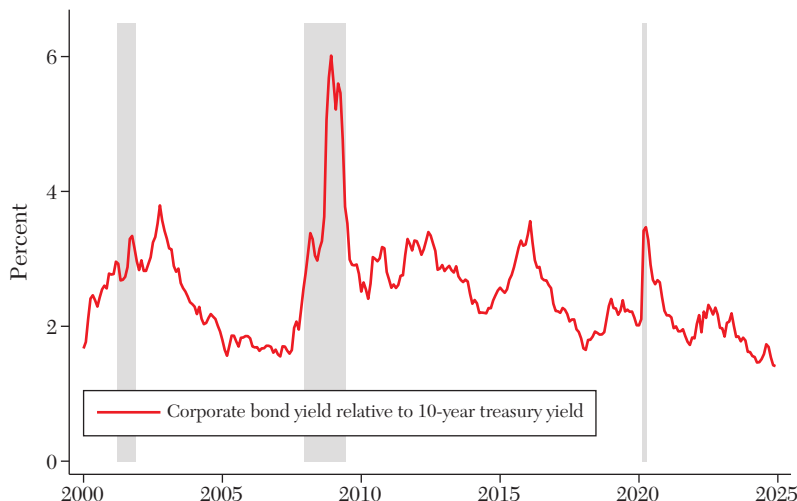


Figure 3. Moody's Baa Corporate Bond Yield Relative to the 10-Year Treasury Constant Maturity, 2000–2024

Source: Data from FRED series BAA10Y (Federal Reserve Bank of St. Louis 2024b).

idiosyncratic and aggregate risk. The function $\phi(\cdot)$ relates these two rates. Benigno, Eggertsson, and Romei (2019) derive this function from a microfounded banking model:²¹

$$(9) \quad 1 + i_t^b(j) = (1 + i_t^s)\phi\left(\frac{D_t^b(j)}{\tilde{D}_t^b(j)}, \frac{D_t}{\tilde{D}_t}, \varsigma_t\right),$$

where the borrower no longer faces a strict limit on debt capacity. Instead, if borrowing, $D_t^b(j)$, exceeds the risk-free borrowing capacity, $\tilde{D}_t^b(j)$, of the household, j , the borrowing rate increases smoothly. One part of the overall spread comes from the part of $\phi(\cdot)$ that captures the risk of idiosyncratic default of each household. D_t refers to aggregate lending of the banking system, which captures the risk associated with bank lending that is independent of the risk of individual borrowers and cannot be diversified. The term ς_t captures other characteristics of the underlying banking model, banks' leverage ratios, and the cost of raising equity, which also influences the spread.²² Benigno, Eggertsson, and Romei (2020) show that the natural rate of interest can be expressed analytically as a function of the arguments in the spread function, that is, the higher the spread, the lower the natural rate of interest, even if the dynamics of their model become more complicated than when the borrowing limit is strict.

²¹ They build on earlier literature, such as Cúrdia and Woodford (2010), that also develops a banking model giving rise to a spread function but focuses on shocks to banks' intermediation technology.

²² This is shown in Benigno, Eggertsson, and Romei (2020), based on Justiniano, Primiceri, and Tambalotti (2015) and Jermann and Quadrini (2012).

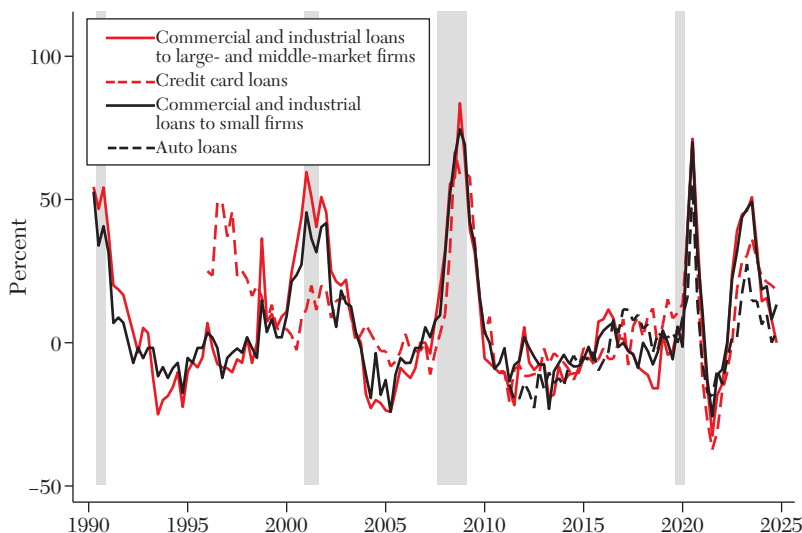


Figure 4. Loan Officer Survey of Lending Standards: Net Percentage of Domestic Banks Tightening Standards, by Loan Category, 2000–2024

Source: Data from FRED series (Board of Governors of the Federal Reserve System 2025e, g, f, d).

A Minsky moment, which triggers higher spreads and a lower natural interest rate, can now be traced to any of the three sources we have reviewed, that is, household or firm debt or reduction in bank lending capacity. We share Gertler and Gilchrist's (2018) assessment that all three forces played essential roles not only in the US Great Recession but also in the Great Depression and Japan's Long Recession.

By 2013—several years after the onset of the 2008 financial crisis—most spreads had normalized and households reversed the earlier increase in household debt (Figures 2 and 3). Lending standards, measured by the Senior Loan Officer Survey (Figure 4), indicated that credit conditions had normalized. Federal Reserve stress tests showed that banks' balance sheets recovered from 2011 onward.²³ The debt-deleveraging hypothesis predicts that such normalization should cause the natural interest rate to return to a positive level, potentially generating inflation or an economic boom if the Federal Reserve keeps rates at zero.

Policy rates remained at zero in late 2013, with the economy showing no signs of inflationary pressure or an economic boom. The lackluster recovery made researchers concerned about missing elements in the general framework, prompting exploration of alternative factors driving low interest rates beyond the fast-acting ones generally considered responsible for the financial crisis and the subsequent Great Recession.

In the Introduction, we emphasized that one of the organizational principles of this review is to consider (i) fast-moving forces giving rise to temporary reductions in the natural rate and

²³See, e.g., Board of Governors of the Federal Reserve System (2011) capital analysis and review and similar reports issued in the following years.

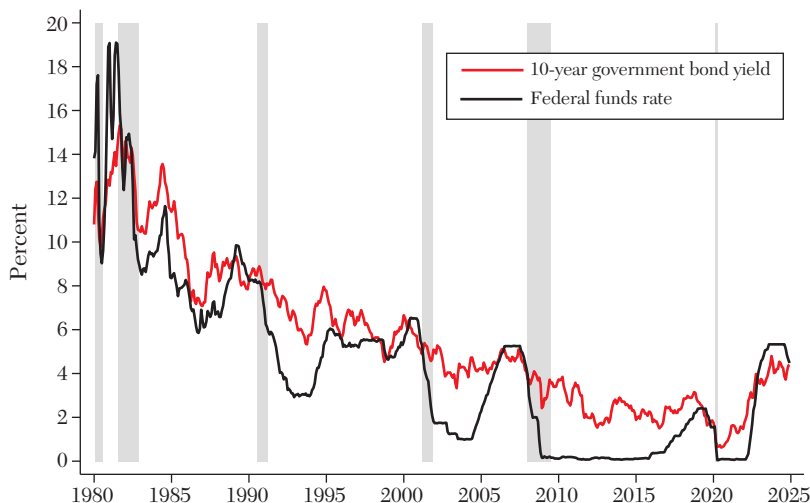


Figure 5. Short- and Long-Term Interest Rates, 1980–2024

Source: Data from FRED (Board of Governors of the Federal Reserve System 2025a) and the Organization for Economic Co-operation and Development (2025).

(ii) persistent forces causing the natural rate of interest to remain negative in both the short and long run, which we now turn to.

2.1.4 *Slow-Moving Forces and Permanently Negative Natural Rates: Demographics, Inequality, and Secular Stagnation*

Several plausible slow-moving forces can trigger a permanent decline in the natural rate of interest. Here, we consider a simple alternative model to generate a permanently negative natural rate of interest. Permanently negative natural rates pose a more fundamental policy challenge to policymakers. This is a theme we return to in Section 5, here focusing on the determination of the natural rate of interest.

Figure 5 shows that short-term interest rates fell to zero after the Great Recession, continuing the general multi-decade decline evident across long-term interest rate measures. The long-term trend suggests that slower-moving forces were also at work, though temporary swings in short-term rates may have masked their effects. Much of the period 1980 to 1990 was characterized by declining inflation. From around 1990 onward, interest rates continued to fall despite stable inflation.

The secular stagnation hypothesis posits the possibility of a permanently negative natural rate. The modern interpretation of this hypothesis originates from the novel reading of Hansen (1939) by Summers (2014).²⁴

²⁴See further discussion of historical origins of the secular stagnation hypothesis in Section 5.

Conventional new Keynesian models face a fundamental limitation in capturing this hypothesis because they impose, by construction, a positive steady-state interest rate. Both the representative agent and the representative saver–borrower models require a positive steady-state interest rate, $\bar{r} = \frac{1 - \beta^s}{\beta^s}$. However, this constraint is not derived based upon some fundamental economic principle and is an artificial consequence of assumption that households are infinitely lived.

We now review Eggertsson and Mehrotra (2014), the first paper to model secular stagnation in general equilibrium.²⁵ Unlike the literature discussed previously, in which the natural interest rate falls temporarily, secular stagnation implies a permanent reduction in the natural interest rate. As in other parts of Section 2, we treat output as exogenous to isolate how other factors determine the natural interest rate.

Consider an overlapping generations model, inspired by Samuelson (1958), with three generations: young (y), middle-aged (m), and old (o). We augment Samuelson’s model with a constraint on the borrowing capacity of the young. Moreover, we assume that only middle-aged people receive income endowment Y . Consider the representative household of a cohort born at time t . This household has the utility function

$$(10) \quad \max_{C_t^y, C_{t+1}^m, C_{t+2}^o} E_t \{ \ln(C_t^y) + \beta \ln(C_{t+1}^m) + \beta^2 \ln(C_{t+2}^o) \},$$

subject to flow budget constraints when young, middle-aged, and old, as well as a borrowing constraint:

$$(11) \quad C_t^y = D_t^y,$$

$$(12) \quad C_{t+1}^m = Y - (1 + r_t)D_t^y + D_{t+1}^m,$$

$$(13) \quad C_{t+2}^o = -(1 + r_{t+1})D_{t+1}^m,$$

$$(14) \quad (1 + r_t)D_t^y \leq \tilde{D}_t.$$

Assuming the young borrow to capacity, the optimality conditions are

$$(15) \quad C_t^y = \frac{\tilde{D}_t}{1 + r_t}, \quad \frac{1}{C_t^m} = \beta E_t \left(\frac{1 + r_t}{C_{t+1}^o} \right), \quad C_t^o = -(1 + r_{t-1})D_{t-1}^m.$$

In the above formulation, D_t^y represents the debt that young people accumulate during the period t . In contrast, $\tilde{D}_t = D_t^y(1 + r_t) = C_t^y(1 + r_t)$ represents the borrowing capacity of young people, assuming that the borrowing limit is binding. The repayment value of their debt determines the borrowing capacity in the next period and may vary exogenously with time.

²⁵The authors later combine the original framework with a quantitative evaluation in Eggertsson, Mehrotra, and Robbins (2019).

Suppose that the size of each generation is given by N_t and we define population growth as $1 + g_t = \frac{N_t}{N_{t-1}}$. Equilibrium in the bond market requires that borrowing by the young equals the savings of the middle-aged, so that $N_t D_t^y = -N_{t-1} D_t^m$ or

$$(16) \quad \underbrace{(1 + g_t) D_t^y}_{L_t^d} = \underbrace{-D_t^m}_{L_t^s}.$$

The left side represents the demand for savings, L_t^d , while the right side represents the supply, L_t^s . The first equation in (15) expresses the demand for the loans. We derive the loan supply by combining the middle equation in (15) with equations (11)–(14), assuming perfect foresight and that borrowers face a binding borrowing constraint:

$$(17) \quad L_t^d = \frac{1 + g_t}{1 + r_t} \tilde{D}_t, L_t^s = \frac{\beta}{1 + \beta} (Y_t - \tilde{D}_{t-1}).$$

Equating the demand and supply for loans yields

$$(18) \quad 1 + r_t = \frac{1 + \beta}{\beta} \frac{(1 + g_t)(\tilde{D}_t)}{Y_t - \tilde{D}_{t-1}}.$$

This formula fundamentally differs from representative-agent or borrower-saver models, as the steady-state natural interest rate can be either positive or negative. Equation (18) demonstrates that a permanent debt-deleveraging shock to $\tilde{D}_t$ permanently affects the interest rate, as does a slowdown in population growth, g_t .

This formulation highlights a more general principle: Any force affecting the supply of savings relative to the demand for loans can affect the natural interest rate, even in a steady state. Several examples of these types of force are shown in Eggertsson and Mehrotra (2014) and Eggertsson, Mehrotra, and Robbins (2019). These include (i) an increase in life expectancy, (ii) an increase in inequality, (iii) a fall in the relative price of investment, and (iv) a slowdown in the growth rate of overall productivity, all of which can lower the natural rate. Meanwhile, increasing government debt or pay-as-you-go social security puts upward pressure on it.

Another hypothesis for persistently low interest rates is the *global savings glut*, a term coined by Bernanke (2005). Bernanke argued that following the Asian financial crisis of the late 1990s, many developing countries—particularly in Asia—began accumulating large quantities of savings and foreign exchange reserves as a form of self-insurance against future financial crises. Although capital is traditionally expected to flow from developed to developing economies, the savings glut hypothesis stresses the flows move in the opposite direction. Capital flowed from developing countries to the United States, putting downward pressure on US interest rates. The analysis of Eggertsson et al. (2016) aligns closely with our discussion, demonstrating how savings–investment imbalances in one country can become a “contagious malady,” generating declines in the natural interest rate elsewhere. Building on Bernanke’s argument, we note that if investment–savings imbalances intensify in countries outside the United States, for instance, due to aging in Japan or China, open financial accounts will transmit these effects internationally. The global savings glut thus provides an additional candidate for our list of potential drivers of secular stagnation.

Caballero and Farhi (2018) offer another interesting complementary perspective that opens up the possibility of a permanently negative real interest rate via a different mechanism. They

attribute this to a shortage of safe assets rather than demographic or technological factors. We comment further on this interesting alternative hypothesis on various occasions later in the paper. While the policy implications are similar in many cases, they also differ to some extent.

Following Eggertsson (2004) a relatively robust literature has emerged that tries to quantify the various forces that could be responsible for lower interest rates.²⁶

At this stage, we assess that the literature has not yet reached a consensus on reliably forecasting the natural interest rate or on how previous estimates have been influenced by the multitude of events following the COVID-19 pandemic and the disruptive policy regime in the United States in 2025.

2.2 Building Block Two: The ZLB and the Curse of Cash

Paper currency gives rise to the ZLB and is, therefore, along with a negative natural rate of interest, a root cause of a liquidity trap.²⁷ We now introduce paper money and the nominal price level—the price of goods in units of money. The main conclusions are: i) If the natural rate of interest is negative, the central bank is unable to target a stable price level due to the ZLB, and ii) under certain conditions, increasing the money supply has no effect.

2.2.1 Paper Currency and the ZLB

To show how paper currency gives rise to the ZLB, we build on the borrower–saver model already explained in Section 2.1.3.

The saver household trades in two new additional assets compared to the problem we considered in Section 2.1.3: nominal bonds, B_t , and paper currency, M_t . For simplicity, we keep the borrower’s problem unchanged. Another simplification is borrowed from Werning (2015). We consider a zero liquidity-limit economy ($D \rightarrow 0$). Under this assumption, then $C_t^b = C_t^s = Y$. Alternatively, the analysis is equivalent to a different limiting case: The number of borrowers approaches zero, so the model reduces to a representative-agent model.

The price level, P_t , is the cost of consumption units in dollars. Nominal bonds are defined as follows: One dollar of B_t at time t yields household $1 + i_t$ dollars with certainty in period $t + 1$. Consistent with our previous notation, a negative value for D_t is a risk-free asset purchased at time t that pays with certainty $1 + r_{t+1}$ of the consumption unit in the next period.

The saver’s budget constraint is²⁸

$$(19) \quad B_t + M_t - D_t = (1 + i_{t-1})B_{t-1} - P_t(1 + r_{t-1})D_{t-1} \\ + M_{t-1} + P_t Y - P_t C_t^s - T_t^s.$$

This budget constraint shows that the absence of arbitrage prevents the risk-free nominal interest rate from becoming negative. Suppose the nominal interest rate is negative in period

²⁶Several authors have attempted to quantify contributors to lower natural interest rates, such as demographics: Carvalho, Ferrero, and Nechio (2016); Gagnon, Johannsen, and López-Salido (2021), inequality Auclert and Rognlie (2018); Straub (2019); Mian, Straub, and Sufi (2021), or multiple factors at once Farhi and Gourio (2018); Del Negro et al. (2017); Summers and Rachel (2019); Mian, Straub, and Sufi (2021); Platzer and Peruffo (2022)

²⁷The title of this section, “The Curse of Cash,” is borrowed from the title of a book by Kenneth Rogoff (Rogoff 2017).

²⁸Since the borrower holds neither money nor nominal bonds, and in equilibrium $C_t^s = C_t^b = Y$, we omit the superscript s .

$t - 1$. Then, the household knows with certainty that holding paper money dominates lending the money at a negative nominal rate, since at time t , $(1 + i_{t-1})B_{t-1} < M_{t-1}$. Sitting on the money is better than lending it. Therefore, the absence of arbitrage requires:

$$i_t \geq 0.$$

We extend the model to include the benefit of holding real money balances in the saver's household's utility function. We denote real money balances by $m_t \equiv M_t/P_t$, where M_t is cash and P_t is the price at time t of the consumption good in terms of dollars. This part of the utility is additive separable from other elements of the utility function given by the function $\varphi(m)$, with $\varphi_m(m) > 0$ up to a certain satiation point m^* . Above this point having more real money balances does not increase utility, so $\varphi_m(m) = 0$ for all $m \geq m^*$. The satiation point captures the idea that additional money does nothing to facilitate transactions once people have enough cash. The optimal substitution between real money balances and spending leads to a static first-order condition:

$$(20) \quad \frac{\varphi_m\left(\frac{M_t}{P_t}\right)}{u_c^s(Y)} = \frac{i_t}{1 + i_t}.$$

For $m_t \leq m^*$ we can then back out the money demand at $i_t \geq 0$ by inverting $\varphi_m\left(\frac{M_t}{P_t}\right)$. However, once $i_t = 0$ and $m > m^*$, the demand for money is indeterminate. If the interest rate is zero, the saver is indifferent between holding a risk-free one-period nominal bond or money: Both are assets denominated in dollars with the same monetary value in the next period, and neither serves any purpose other than to store wealth when $m_t > m^*$. We assume that money demand takes the form

$$(21) \quad \begin{aligned} \frac{M_t}{P_t} &= \varphi_m^{-1}\left(\frac{i_t}{1 + i_t}u_c(Y)\right) \\ &= \begin{cases} \nu_y Y - \nu_i i_t & \text{when } m \leq m^* \text{ so } i_t \geq 0 \\ \text{indetermined} & \text{when } m > m^* \text{ so } i_t = 0 \end{cases}. \end{aligned}$$

We assume the coefficients $\nu_y \geq 0$ and $\nu_i \geq 0$ in the last equation. Money demand, often labeled the liquidity preference–money supply (LM) curve, is a standard representation in textbooks, which since 2008 also reflect that short-term interest rates cannot go below zero, so the LM curve is flat at $i_t = 0$. The logic of the textbook LM curve is that higher income leads to higher demand for money, while higher nominal interest rate increases the opportunity cost of holding cash, thus reducing demand. Once interest rates are zero, demand becomes infinitely elastic.²⁹

Even if people trade paper currency, short-term nominal interest rates can deviate from the ZLB, but only to a limited extent, for example, due to storage cost and the role of bank reserves

²⁹Obtaining this particular form for money demand by inverting $\varphi_m(m)$ requires special assumptions. However, since the actual slope of the LM curve above zero is irrelevant to the analysis, we will not dwell on this.

held at the central bank for interbank transactions, which provide them with a valuable service, see Section 3.2.7.

2.2.2 Price Level Determination When the Natural Rate Is Temporarily Negative

This subsection shows how nominal prices behave when the natural interest rate turns temporarily negative. We employ an elemental policy regime—a fixed future money supply and several other simplifying assumptions. The analysis builds on Krugman's (1998) elegant example, with minor adjustments to map into our previous analysis.

We advise readers to consider the logic of this model carefully due to its rare simplicity, relative to the insight it offers. It features a single reduced-form one-time unexpected shock and almost no dynamics. Yet all of the key insights, which include an out-of-sample prediction that went strongly against the conventional wisdom at the time, were later confirmed in the data. Moreover, the key insights were confirmed and extended to much richer models, featuring a full stochastic environment with a liquidity trap extending over arbitrarily long time and more realistic dynamics. This will become apparent later in the review.

The ZLB fundamentally alters the determination of the price level in this framework. Under certain conditions, printing money does not affect prices precisely when a policy instrument is needed to stabilize prices and, more generally, output. In what follows, we simplify the model into two equations and specify the policy regime. We assume an unexpected shock in the initial period, which we label *S* (for short run). In the subsequent period, labeled *L* (for the long run), the economy returns with certainty to the steady state. After households observe the shock at the beginning of period *S*—before making any consumption decisions—they have perfect foresight.

The saver's household satisfies again the pricing equation for the risk-free one-period asset given by equation (4) but now, in addition, a pricing equation for the nominal bonds:

$$(22) \quad 1 + i_t = \frac{1}{\beta^s} \frac{P_{t+1}}{P_t} \frac{u_c(C_t^s)}{u_c(C_{t+1}^s)} = \frac{1}{\beta^s} \frac{P_{t+1}}{P_t}.$$

Under perfect foresight we can combine equations (4) and (22):

$$(23) \quad 1 + i_t = (1 + r_t) \frac{P_{t+1}}{P_t},$$

often coined the Fisher equation. We can now formally define an equilibrium.³⁰

Krugman's model achieves remarkable insights into the core issues of price level determination at the ZLB with minimal algebra through three key simplifying assumptions. The first is a simplified money demand function where the LM curve stands vertical at positive interest rates and flat at zero:

$$(24) \quad \frac{M_t}{P_t} = \begin{cases} \nu^{-1} Y & \text{when } m \leq m^* \text{ so } i_t \geq 0 \\ \text{indetermined} & \text{when } m > m^* \text{ so } i_t = 0 \end{cases}.$$

³⁰A perfect foresight equilibrium is a set of paths for short-term interest rates, i_t , the price level, P_t , real money balances, m_t , which solve equations (21)–(23), given the exogenous process for the natural rate of interest, r_t , determined by equation (8) and a sequence M_t coupled with the identity $m_t \equiv \frac{M_t}{P_t} = \frac{M_{t-1}}{P_{t-1}} \frac{M_t}{M_{t-1}} \frac{P_t}{P_{t-1}}$.

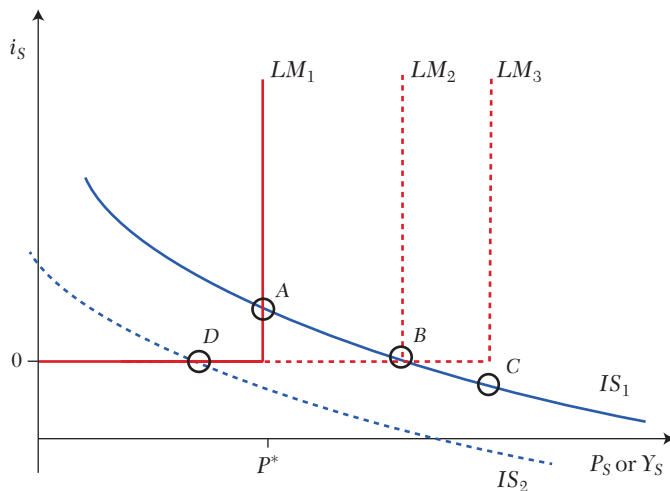


Figure 6. The Liquidity Trap

This simplifying assumption is well justified, as it does not affect the results but saves on algebra. We adopt the notation $\nu^{-1} \equiv \nu_y$ relative to equation (21). We can interpret equation (24) as a cash constraint, that is, a certain fraction of output, ν^{-1} , needs to be bought by paper currency instead of alternative payment methods. We can also interpret this equation as a simplistic variation of the classic quantity theory of money where ν is the velocity of cash. We get $M_t \nu = P_t Y$, which is the form of the quantity theory its main proponent, Milton Friedman, laminated on his car license plate. Now ν has the interpretation of being the velocity, how quickly money circulates in the economy, given nominal spending $P_t Y$. The monetarist claim was that ν was constant.³¹ The key difference, relative to Friedman's quantity theory, is that M_t only refers to paper currency in this review, not to broader monetary aggregates.

The LM curve is plotted in Figure 6 and takes the same form every period. The most fundamental aspect of this curve is that it is *flat* at zero. A flat LM curve means that the supply of money is indeterminate once the short-term interest rate is zero; in this case, the variation of the “quantity theory of money” just sketched out does not hold since the velocity of money is strictly decreasing at the ZLB. The reason is that people now can choose to hold money as an asset, that is, $M_t \nu > P_t Y$, see Figure 6.³² If the cash constraint is binding, then the curve is vertical.³³

³¹ See the interview with John Taylor, retrieved on January 31, 2025 at https://www.youtube.com/watch?v=V_ah1bEo_ww.

³² A monetarist would reasonably object that this is not a proper quantity theory of money, as M should refer to some broader monetary aggregate. Eggertsson and Petracchi (2021) reconcile the view of the monetarist and the Keynesians in a microfounded model where the reconciliation is obtained by using a microfounded money demand function, showing that what separates the two schools is different assumptions about the underlying monetary policy regime.

³³ The assumption a vertical LM curve instead of the more plausible assumption that people economize on holding cash the higher is the nominal interest rate, $\eta_i > 0$ is a clever modeling device from Krugman (1998). Nothing of substance changes if instead we assume $\eta_i > 0$ in 24 except adding a few unnecessary steps of algebra to derive the long-run price level.

The second simplifying assumption is that there is a one-time unexpected exogenous shock in period 0, denoted short-run S , summarized by r_s , the natural interest rate. The shock reverts to the steady state, $\bar{r}_L$, where bar stands for steady state and L for long run. Once households observe the shock, they have perfect foresight. The natural interest rate drops due to an unexpected one-time increase in β^s to $\beta_s^s > \beta^s$ at time S , meaning people are less willing to spend in the short term for a given real interest rate. It is a reduced-form representation summarizing any of the proposed forces triggering a temporary negative natural rate of interest in Section 2.1 exemplified by equation (8).

The third simplifying assumption is that the monetary policy regime fixes the money stock at some value $\bar{M}_L$.

Under these three assumptions, the model can be solved without almost any algebra, revealing several insights. In period 1, the long run, L , the exogenous real interest rate is positive at its steady state, $\bar{r}_L = \frac{1 - \beta^s}{\beta^s}$. We can then immediately guess and verify the solution for the price level from equation (24) assuming the cash constraint is binding:³⁴

$$(25) \quad \bar{P}_L = \nu \frac{\bar{M}_L}{Y}.$$

All the gripping action occurs in the short run (S), where the intersection of the LM and investment–saving (IS) equation determines equilibrium. The LM equation takes the same form as in equation (24), and we write it when collecting the LM and IS equations in (26) slightly differently than before.

A convenient way of writing the LM equation is as two inequalities, where at least one is strictly binding each time, giving the piecewise linear function in Figure 6. To obtain the IS equation, we substitute the long-run price level from equation (25) into the Fisher/IS equation (23), so we obtain the pair of equations

$$(26) \quad \begin{array}{ll} \text{LM} & \frac{M_S}{P_S} \geq \nu^{-1} Y \quad \text{and} \quad i_S \geq 0 \\ \text{IS} & 1 + i_S = (1 + r_s) \frac{\bar{P}_L}{P_S} \end{array}$$

The equilibrium is determined where the LM and IS cross, as shown in Figure 6. This intersection determines the nominal interest rate, i_S , and the price level, P_S . The policy that can affect the outcome is the short-term money supply, M_S . Several other variables are exogenous or determined by the policy regime. The policy regime fixes the long-term price level, $\bar{P}_L$, while Y is an exogenous endowment. Meanwhile, the natural rate of interest, r_s , is determined by an exogenous shock to β^s or, more generally, by equation (8) if there is borrowing and lending.

First, consider an equilibrium without any movement in r_s . Suppose the previous price level was $P_- = \bar{P}$. We define full price stability at point A, where the two curves intersect if the central bank sets the money supply at its steady-state level, $M_S = \bar{M}_L$. From there the central

³⁴Recall that we have ruled out equilibria due to self-fulfilling expectations. The assumption of vertical LM curve allows us to skip a few unnecessary algebra steps without losing any additional insights

bank can lower the interest rates through open market operations by printing money (M_s) and exchanging it for short-term government bonds (B_s).³⁵

The simplest way to understand the determination of the price level is to recognize that the equilibrium real interest rate, r_s , is purely exogenous and that the endogenous variables (i_s , P_s) must move for the IS equation in (26) to be satisfied. If an equilibrium exists, the market will always deliver this real interest rate regardless of the behavior of nominal interest rates, prices, and money supply.

Consider an increase in money supply, M_s , through open market operation, shifting the LM curve from LM_1 to LM_2 . The IS and LM curve now intersect at point B. The IS equation in (26) shows that P_s must increase proportionately to the fall in i_s . Thus, while actual inflation increases, *expected* inflation drops. These price dynamics are a property of a policy regime that fixes the future money supply, a form of price-level targeting. The price level must increase in the short run since the policy regime fixes the long-term price level. When the central bank lowers the nominal interest rate by increasing the money supply, the only way to keep the real interest rate at its exogenously given value is via lower inflation expectations. And this happens with inflation $P_s/P_- > 1$ in the short run once the central bank lowers the interest rate.

Consider an additional increase in the money supply to LM_3 . As Figure 6 shows, this implies a negative nominal interest rate at point C. Point C cannot be an equilibrium as it violates the ZLB, making the cash constraint slack. Accordingly, the equilibrium remains at point B. No further interest rate cuts are possible; they have reached zero, so the LM curve kinks. Any increases in the money supply beyond LM_2 , such as LM_3 , do not affect the price level. The only thing that happens is that households hold money as an asset instead of a government bond (with zero interest rate).

The most interesting case occurs when there is an unexpected shock so that r_s is negative. The dashed line in Figure 6 shows the IS curve shifted by a negative natural rate to IS_2 . The equilibrium is now at point D, below what the central bank defines as price stability, point A. At point D there is deflation, $P_s/P_- < 1$.

Deflation occurs at point D because with the future price level fixed and the nominal interest rate already at zero, the only way to achieve a negative *real interest rate* is by creating expected inflation. Short-term prices must fall relative to long-term prices since nominal interest rates cannot fall further. Since the curves intersect at the ZLB in this deflationary equilibrium, increasing the money supply has no effect. Shifting the LM curve to LM_2 or LM_3 does nothing since spending is no longer constrained by the money supply. Thus, the central bank loses control over the price level. The LM curve loses relevance, and the economy stays at point D regardless of how much money the central bank prints in the short run.

The key—the very core—of this result is that households expect any increase in M_s to be *temporary*. They expect that any increase will always be reverted to $\bar{M}_L$ as soon as the shock is over.

The need for the current price level to fall in response to the shock hints at potential problems in more general settings, which we confirm when prices do not adjust instantaneously. The simple model shows signs of the problem once we consider a limiting case with fixed

³⁵We adopt this simplified scenario for pedagogical reasons, following traditional textbook treatments of interest-rate control, although we consider modern institutional arrangements later. Section 2.3 discusses a scenario in which the central bank pays interest on bank reserves to manage the policy rate.

prices. This limiting case leads to an identical IS–LM graph as in Figure 6 but with the output Y_5 on the x-axis instead of the price level. In this case, output is falling instead of inflation, and the central bank cannot increase aggregate demand. Monetary policy cannot cut the nominal interest rate to turn the tide, while printing money does not affect output, which is conditional on the policy regime. Although these two examples may seem unique, they reveal an insight that extends to a broad range of models and is not unique to this policy regime. More generally, a sufficiently negative natural rate of interest results in a combination of a fall in inflation and output in more general settings.

We conclude by observing that Krugman (1998) identifies a straightforward solution to the decline in prices or output. The government can stabilize the price level if it firmly “commits to being irresponsible,” in Krugman’s (1998) memorable phrase, and convinces the public that it will permanently increase the money supply so that $\bar{P}_L = \nu \frac{\bar{M}_L^{high}}{Y}$. For sufficiently high $\bar{M}_L^{high}$, the IS curve returns right back to where it started.

In summary, Krugman concludes that a credible increase in inflation expectation always allows the central bank to reach point A. However, following up on this promise requires a one-time increase in the price level once the ZLB is no longer binding. Similarly, a corresponding monetary expansion can prevent output from declining with sticky prices. In later sections, we will explore this idea in much greater depth when considering alternative policy regimes and where we analyze policy in terms of interest rate policy, which is more consistent with current practice in central banks.

2.2.3 Price Level Determination When the Natural Rate Is Permanently Negative

The ZLB imposes even tighter restrictions on inflation dynamics when we consider a permanent reduction in the natural interest rate.

Let us revisit the overlapping generations model from Section 2.1.4 but introduce the price level as in the last subsection. The steady-state Fisher equation is³⁶

$$(27) \quad 1 + \bar{r} = (1 + \bar{r})\bar{\Pi}.$$

Unlike in our previous example, $\bar{r}$ is no longer guaranteed to be greater than zero. The ZLB then implies a lower bound on steady-state inflation:

$$(28) \quad \bar{\Pi} \geq \frac{1}{1 + \bar{r}}.$$

This inequality shows that the ZLB sets a floor for steady-state inflation based on the natural interest rate. Suppose, for example, that the natural interest rate permanently drops to –5 percent annually. Then inflation must rise approximately above 5 percent when prices are flexible; otherwise, equilibrium does not exist, that is, $\bar{\pi} = \bar{\Pi} - 1 \geq \frac{1}{1 + \bar{r}} \approx 5$ percent. The problem of existence suggests major problems when the natural interest rate becomes

³⁶As in the last subsection, we adjust the household’s budget constraint to allow trade in one-period risk-free nominal bonds, assuming they are traded in zero supply so that actual borrowing is denominated in the real bond. The middle-aged household now also satisfies a Euler equation for risk-free nominal bonds, which determines the nominal interest rate given by $\frac{1}{C_t^m} = \beta E_t \left(\frac{1+i_t}{C_{t+1}^o} \frac{P_t}{P_{t+1}} \right)$. Combining that equation with the middle equation in (15) and evaluating in steady-state yields the Fisher equation stated in the text.

negative if prices or wages are rigid. We show in Section 5 that in this case, a secular stagnation equilibrium emerges, which features a permanent recession where inflation stays below the central bank's target and output below potential—for an arbitrary duration of time and with no end in sight.

2.3 *When Printing Money Is Irrelevant: Empirical Evidence from Japan*

The modern liquidity trap literature made a striking prediction at its very beginning, over 25 years ago, which was met with a great deal of skepticism at the time: Increasing the money supply is irrelevant at the ZLB if future policy expectations are entrenched. From 1998 to 2021, the Bank of Japan increased the money supply by over 1,000 percent while prices declined by 3 percent, confirming this prediction.

The framework we developed in the last section is a remarkable example of successful model-based reasoning that carried greater predictive power than looking at historical correlations. Krugman (1998) highlighted a fundamental empirical prediction: An increase in the money supply has *no effect* on prices and output *as long as expectations regarding the future money supply remain fixed*.

The academic community initially met this prediction with great skepticism, perhaps because it did not have a historical counterpart. Of course, this prediction is particularly striking because it proved correct despite being out of consensus.

Krugman's irrelevance result might seem special, as it relies on a policy regime in which the future money supply remains completely fixed. Committing to a fixed future money supply does not align well with how most central banks typically communicate future policy. Moreover, Krugman assumes that the central bank increases the money supply solely by purchasing short-term, risk-free government bonds. Despite these seemingly restrictive assumptions, Krugman's irrelevance result is, in fact, much more general.

Eggertsson and Woodford (2003) extend Krugman's irrelevance result in two ways. First, they propose a general interest rate rule rather than assuming a fixed future money supply. Their specification includes interest rate rules commonly employed in practice, such as the Taylor rule. Second, while Krugman (1998) assumes that the government increases the money supply exclusively by purchasing short-term government bonds, Eggertsson and Woodford (2003) go further and demonstrate that *how* the government increases the money supply is irrelevant. For example, increasing the money supply through purchases of long-term government bonds with positive interest rates has no effect, provided that expectations about future monetary policy remain unchanged.

More generally, Eggertsson and Woodford (2003) allow the central bank to purchase any financial asset with arbitrary state-contingent payoffs. The most relevant assumptions of their result for this review are as follows. First, for asset markets: Markets are complete and the amount the central bank purchases of an asset does not alter that asset's payoffs across different states of the world. Second, for monetary policy: The central bank's money supply expansion at the ZLB does not affect expectations about the future monetary policy regime once the ZLB is no longer binding. That is, the expected interest rate reaction function that the central bank follows, which is governed by some policy rule $i_t = \phi\left(\frac{P_t}{P_{t-1}}, Y_t, \tilde{\xi}_t\right)$, where $\tilde{\xi}_t$ collects all exogenous shocks to the model, remains unchanged regardless of asset purchases or increases in the supply of money at the ZLB.

Krugman's irrelevance result remains unchanged: Any increase in the money supply has no effect, regardless of how the central bank implements it, irrespective of the specific assets the government purchases.³⁷

As we opened the section, professional economists greeted Krugman's prediction, as well as that of Eggertsson and Woodford, with great skepticism. Given the radical nature of these predictions and how directly they contradicted monetarism, the skepticism is not very surprising, in retrospect. Indeed, as we mentioned in the introduction, Krugman initially set out to disprove what later became his famous prediction.

Kenneth Rogoff, commenting on Krugman's paper motivated by Japan's liquidity trap, nicely summarized the commonly shared view in the profession at the time:³⁸

No one should seriously believe that the Bank of Japan would face any significant technical problems in inflating if it puts its mind to the matter, liquidity trap or not. For example, one can feel quite confident that if the Bank of Japan were to issue a 25 percent increase in the current money supply and use it to buy back 4 percent of government nominal debt, inflationary expectations would rise.

This reflected a professional consensus at the time that Japan's liquidity trap resulted mainly from the Bank of Japan's unwillingness to take necessary steps to generate inflation. It was a consensus largely informed by Friedman and Schwartz's (1963) influential account of the Great Depression, which became the mainstream narrative of that period. The basic message was that money supply contraction created recessions, most notably the Great Depression, and monetary expansion would cure them. Friedman's prophecy of the 1970s Great Inflation in his 1968 American Economic Association Presidential Address (Friedman 1968) further strengthened this basic proposition. It was grounded in Friedman's famous dictum that "inflation is always and everywhere a monetary problem"—and by extension, deflation as well. Milton Friedman addressed Japan's recession in similar terms in a *Wall Street Journal* op-ed on December 17, 1997:³⁹

There is no limit to the extent to which the Bank of Japan can increase the money supply if it wishes to do so. Higher monetary growth will have the same effect as always. After a year or so, the economy will expand more rapidly; output will grow and, after another delay, inflation will increase moderately.

In retrospect, many observers were overly optimistic. Figure 7 shows the evolution of Japan's monetary base since September 1998. Despite the QE policy, which started in March 2001 and more than doubled the money supply by March 2006, the price level continued to fall and inflation expectations remained unchanged.⁴⁰

Later, Milton Friedman added another detail to his proposal relative to Rogoff's. He emphasized that the Federal Reserve could print money and buy long-term bonds even if the short-term nominal interest rate was zero. Eggertsson and Woodford's (2003) irrelevance

³⁷ Eggertsson and Woodford (2003) present their irrelevance result as a generalization of Wallace's (1981) irrelevance result to an environment with sticky prices, monetary frictions, and the ZLB. Economists sometimes considered Wallace's result to be of limited practical interest, as it implied that all open market operations were irrelevant. In contrast, in Eggertsson and Woodford (2003) this irrelevance applies exclusively at the ZLB. They discuss in detail the logic behind the irrelevance result and possible caveats.

³⁸ See the commentary in Krugman (1998).

³⁹ See Friedman (1998).

⁴⁰ For data on inflation expectations, see Eggertsson and Ostry (2005), who measure them with data from Surveys of Professional Forecasters.

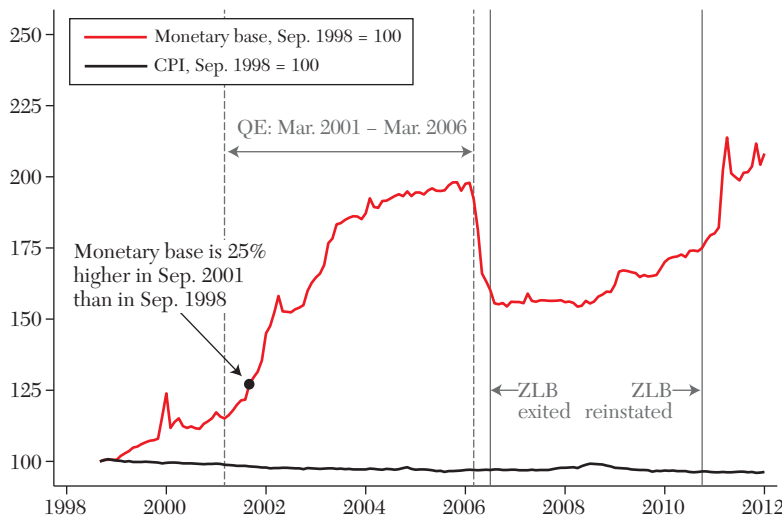


Figure 7. Monetary Base and Price Level in Japan, 1998–2012

Source: Data on monetary base (Bank of Japan 2025) and the CPI from the OECD (Organization for Economic Co-operation and Development 2022).

results suggest this policy is ineffective. We evaluate this proposal in the context of empirical evidence in Section 3.4.2 from the Great Recession.

The irrelevance results of Krugman (1998) and Eggertsson and Woodford (2003) explain the experience of the Bank of Japan, contrary to the conventional wisdom. The Bank of Japan ended QE in 2006 because it gained confidence that the economy was recovering and inflation was rising. As anticipated by the irrelevance results, without the Bank of Japan “credibly committing to being irresponsible,” even doubling the money supply did not affect realized inflation at the time or even inflation expectations. Market participants correctly predicted that the Bank of Japan would contract the money supply at the first sign of inflation—which it did. Figure 7 displays the contraction at the first hint of inflation in 2006.

Looking over a more extended period, from September 1998 to April 2021, the Bank of Japan increased the money supply by 1,103 percent. It did so not only by buying short-term government bonds as in its initial phase. In the aftermath of the Great Recession in 2008, the Bank of Japan also started purchasing long-term government bonds and a broad range of assets, such as exchange-traded funds to support equity markets, real estate investment trusts to stimulate the property market and corporate bonds, and corporate paper to lower credit spreads. The result of all these interventions can hardly count as a victory for the quantity theory of money. The price level fell by 3 percent during the same period.⁴¹

⁴¹ It is beyond the scope of this survey to enter the discussion of what is the most appropriate measure of money supply. We rely on the monetary base here, for this is the policy instrument that the central bank can control directly, as opposed to, for example, deposit holding of private citizens in banks that the central bank cannot directly control. What remains clear is that central banks throughout the world used their central policy instrument, the monetary base, to purchase a broad spectrum of assets, and moreover, several countries extended deposit insurance to accounts that had not been covered prior to the crisis.

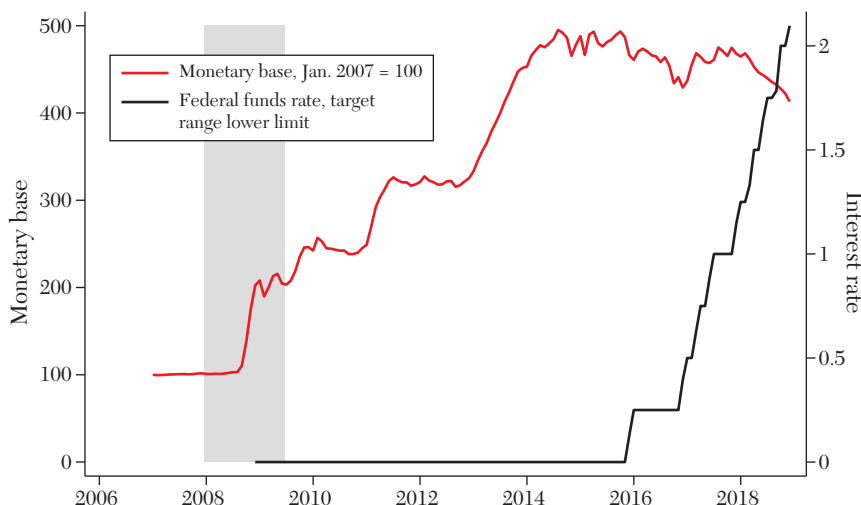


Figure 8. Monetary Base and Federal Funds Target Range Lower Limit in the United States, 2007–2018

Sources: Data from FRED (Board of Governors of the Federal Reserve System 2025b,c).

Remarkably, even though Japan's experience had demonstrated the limited effect of QE on prices after the Bank of Japan raised interest rates in 2006, 23 prominent economists, financial writers, and investors signed an open letter to Federal Reserve Chairman Ben Bernanke in November 2010. In this letter, they voiced concerns about the Federal Reserve's plan for a second round of quantitative easing (QE2), predicting that it would lead to inflation and currency debasement (see Asness et al. 2010). This letter highlights how deeply Friedman and Schwartz's narrative and the quantity theory of money had influenced—and perhaps continues to influence—the broader academic and financial community.

A caveat to Krugman's prediction makes the irrelevance result starker. Economists define the monetary base as the currency in circulation plus the reserves that commercial banks hold at the central bank to facilitate interbank transactions. Since 2008, several central banks have gained the authority to pay *interest on reserves*, including the Federal Reserve in the fall of that year. This new capability implies that expanding the monetary base through reserve increases remains irrelevant at the ZLB and when it no longer binds. Figure 8 illustrates that after the Federal Reserve cut its interest rate to zero in December 2008, the monetary base increased by a factor of five by the time it raised the federal funds rate seven years later, in 2015. Unlike the Bank of Japan, the Federal Reserve did not need to contract the monetary base to raise the policy rate. Instead, it raised interest rates on reserves, which increased other measures of short-term nominal interest rates. Under this new operational framework, the conclusion becomes even clearer. What matters in the short run is not some measure of the money supply, but rather the central bank's current and expected short-term policy rates.

It would be incorrect to draw lessons from these irrelevance results that monetary policy cannot stimulate spending, since they rely on explicit assumption that can be violated in practice. It is more constructive to view them as isolating which margins the central bank can

operate on once at the ZLB, and which seem less promising, instead of interpreting them as saying monetary policy has no power.

Section 3.4 considers how various policies may be effective because they violate the assumptions imposed on asset markets in Eggertsson and Woodford (2003). In contrast, Section 4.3 examines policy regime changes that significantly impact expectations of future interest rates, thus violating the assumptions about the monetary policy regime. Nevertheless, the Japanese experience should serve as a humble reminder that implementing expansionary policies is far less straightforward than many economists claimed when Japan's Long Recession first took hold.

2.3.1 *Resolving Contradictions in the Literature*

The irrelevance results of Krugman (1998) and Eggertsson and Woodford (2003) may seem inconsistent with several other results in the literature. For example, Auerbach and Obstfeld (2005), in their paper "The Case for Open-Market Purchases in a Liquidity Trap," argue that open market operations in government bonds will increase aggregate demand along the line Rogoff argued, which may appear to contradict both irrelevance results. However, they do not contradict them, since Auerbach and Obstfeld (2005) assume that open market operations result in a permanent increase in the money supply, in contrast to what the Bank of Japan implemented. As Krugman (1998) already showed and we reviewed in the last section, a permanent increase in the money supply will indeed increase aggregate demand. Krugman coined this solution with the memorable phrase "committing to being irresponsible." Hence, the two results align. However, the expansionary effect of committing to a higher money supply in the future has nothing to do with the open market operations carried out at the time of the ZLB. All that matters are expectations about the money supply once economic conditions normalize. Open market operations before that date do not need to have any bearing on the central bank's actions in the future.

Buiter (2003) suggests "helicopter drops" of money, and Svensson (2000) proposes foreign exchange interventions to stimulate demand. Galí (2020) argues for fiscal expansion financed by increases in the money supply, contrasting this approach with bond issuance. The irrelevance results imply that the composition of aggregate government liabilities—whether money or bonds—is inconsequential at the ZLB, provided it does not affect expectations about future policy. To reconcile these studies (and numerous other policy proposals in the same vein) with the irrelevance results, we must recognize that many implicitly assume the proposed policies involve either a *permanent* increase in the money supply or an alteration of the interest rate *reaction function*, the latter being how Eggertsson and Woodford frame their result. In Krugman's terms, such proposals represent a "commitment to being irresponsible."

3. *Temporary Negative Natural Rates under the Standard Policy Regime: When the World Flips Upside-Down*

Section 3 analyzes inflation, output, and interest rates in a liquidity trap, assuming (i) a temporarily negative natural rate and (ii) the Standard monetary and fiscal policy regime. These assumptions land us in a topsy-turvy world where several accepted economic wisdoms are turned on their heads, leading to substantial policy implications. They include a significant role of fiscal policy, the counterintuitive effects of supply shocks, and the destabilizing role of price

flexibility. While primarily focused on theoretical results, we also review empirical work and discuss implications for the policy debate during the Great Recession.

We use the canonical new Keynesian model common in graduate textbooks for our analysis.⁴² This model represents the core of modern monetary policy analysis. The framework integrates new Keynesian thinking that emerged following the Great Inflation of the 1970s with real business cycle models that were developed around the same time.⁴³ We focus on the textbook representative agent version for tractability, but show that our central results are amplified in the increasingly common HANK model.

3.1 *The New Keynesian Framework for Analyzing the Liquidity Trap*

In the canonical new Keynesian model, a temporary reduction in the natural interest rate generates a liquidity trap with a sharp drop in output and a decline in inflation, replicating the depth of the Great Recession and Great Depression for reasonably calibrated parameters.

3.1.1 *Data and Baseline Simulations from Two Landmark Episodes*

During the Great Depression, US output dropped by approximately 30 percent, with a clear turning point in April 1933, when the economy started to recover (see red line in panel A of Figure 9). In comparison, output fell by less than a third of that in the Great Recession (7.5 percent) (panel B).

Effective policy response may partly explain the difference between the depth of the Great Depression and the Great Recession. Consistent with this interpretation, there is less evidence of a drastic turning point during the Great Recession. US policymakers were better prepared. At the helm of the Federal Reserve, for example, was Ben Bernanke, the former Princeton professor we cite in the introduction, a world-renowned specialist on the Great Depression who earned a Nobel Prize for his work in that field, part of which is discussed in Section 2, with other world-known scholars that had done highly recognized work on the Great Depression in other key economic advisory roles at the White House.⁴⁴

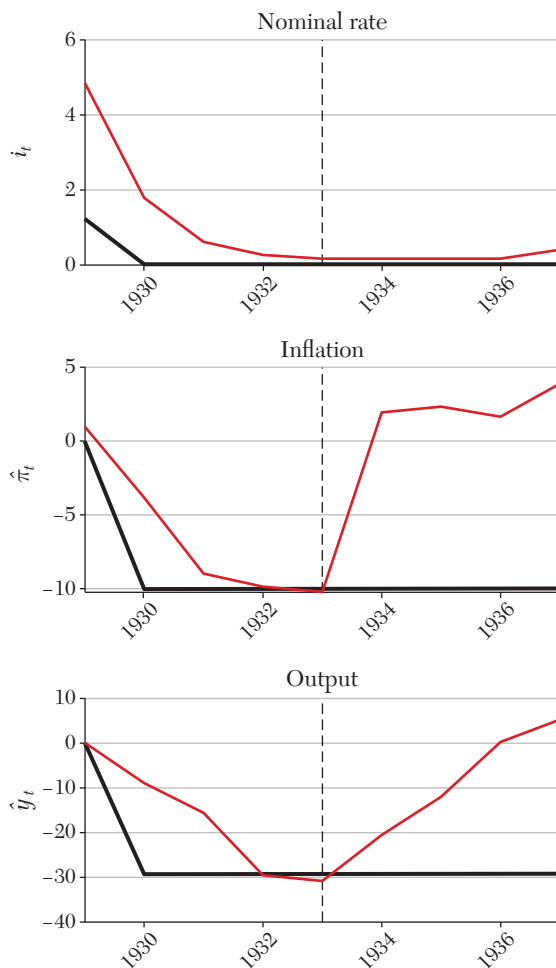
Inflation's reaction differs between the two episodes. Deflation reached 10 percent per year in 1932 during the Great Depression, but inflation dropped only modestly below the Fed's inflation target during the Great Recession. The solid black line shows the simulated path based on a calibration of the new Keynesian model.

⁴²For textbook treatments, see Woodford (2003) and Galí (2015), central contributors to this research paradigm. In its modern form, the canonical new Keynesian model—a microfounded second-order expansion of welfare and the investment-saving–aggregate supply (IS–AS) equations in deviation from the natural rate—first appeared in Woodford's late 1990s papers (e.g., Woodford 1996, 1999). See also Clarida, Gali, and Gertler (1999), which reviews this literature's origins.

⁴³Seminal new Keynesian papers are collected in Mankiw and Romer (1991). The foundational real business cycle paper is Kydland and Prescott (1982), with precedents in Brock and Mirman (1972) and earlier deterministic models. For a review, see King and Rebelo (1999).

⁴⁴Remarkably, in two most influential economic advisory roles at the White house at the time were two scholars whose work on the Great Depression is widely cited. It is discussed and features in this review. Christina Romer was the Chair of the Council of Economic Advisors while Lawrence Summers was the Director of the National Economic Council.

Panel A. Great Depression



Panel B. Great Recession

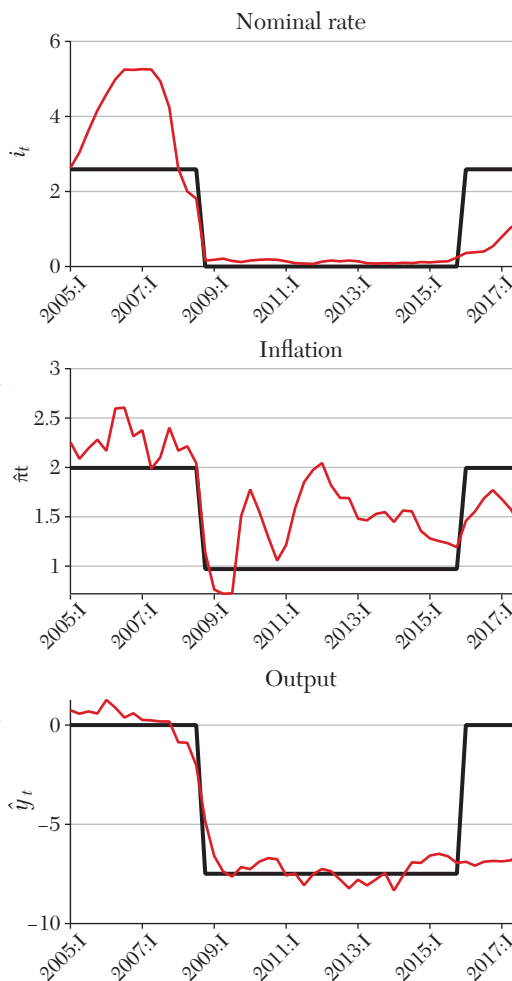


Figure 9. Simulating the Great Depression and the Great Recession in the New Keynesian Model

Notes: The Great Depression and the Great Recession in the model (solid black line) and the data (thin red line). *Source:* Great Depression data adapted from Eggertsson (2012b). Great Recession data from FRED: Real Gross Domestic Product US Bureau of Economic Analysis (2025c), Personal Consumption Expenditures: Chain-type Price Index US Bureau of Economic Analysis (2025a), and Federal Funds Effective Rate Board of Governors of the Federal Reserve System (2025a). Output shows deviation from the linear trend.

3.1.2 The Canonical New Keynesian Model and the Standard Policy Regime

Section 2.1 presented models in which the temporary drops of the natural rate of interest are endogenous while output is exogenous. Now, we flip the script by analyzing the liquidity trap in the new Keynesian model. Output is endogenous while the natural interest rate is exogenous.

This review presents the standard new Keynesian model in terms of output and interest rate deviations from their efficient values instead of natural values. The new Keynesian model was initially developed to analyze monetary policy. The textbook model represents output and real interest rates as deviations from their natural equilibrium—the values reached if all prices are flexible. These natural values conveniently summarize all exogenous shocks. These shocks are independent of policy, since monetary policy only plays a role if prices are rigid.

When analyzing monetary and fiscal policy together, the natural equilibrium is less useful. Fiscal policy instruments, such as government spending, change the natural output and interest rate, so the natural output and interest no longer summarize all exogenous shocks that are independent of policy. We define *the efficient equilibrium* to address this limitation. It is the equilibrium if the policymaker can set the fiscal policy instruments, assuming flexible prices, to achieve the first-best outcome, a property our model will have. The efficient output and interest rate summarize the exogenous shocks independently of both monetary and fiscal policy. The natural interest rate has the same interpretation as the efficient rate if fiscal policy is unchanged.⁴⁵

The new Keynesian model has two fundamental equations: The IS equation and the new Keynesian Phillips curve (AS); we report them here in their log-linear approximated form, as their derivation is available from several sources. Once we substitute into the IS equation a policy rule for interest rates, we refer to the resulting equation as aggregate demand (AD).⁴⁶

A central property of all Keynesian models is that output is determined by how many goods people want to buy, given their expectations about future output, inflation, the nominal interest rate, and the exogenous shocks. Firms are then obligated to produce however many goods

⁴⁵ Households derive utility from a consumption aggregate defined over a continuum of goods of measure one,

$$C_t \equiv \left[\int_0^1 c_t(i)^{\frac{\theta-1}{\theta}} di \right]^{\frac{\theta}{\theta-1}},$$

where $\theta > 1$ is the elasticity of substitution between goods. In addition, households derive disutility from supplying labor to a continuum of industries (also of measure one), with preferences given by:

$$E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\frac{1}{\sigma_c}} - 1}{1 - \frac{1}{\sigma_c}} + \frac{G_t^{1-\frac{1}{\sigma_g}} - 1}{1 - \frac{1}{\sigma_g}} - \int_0^1 \frac{N_t(i)^{1+\omega}}{1+\omega} di \right] \xi_t,$$

where $N_t(i)$ denotes labor supplied to each industry, i , ω is the inverse of the Frisch elasticity of labor supply, G_t , government spending contributing to utility that enters utility identically to C_t and has the same Dixit–Stiglitz preference structure. The parameter σ_c represents the intertemporal elasticity of substitution of consumption. In this section, the utility of government spending plays no role except in defining efficient allocation. Still, it becomes relevant when we explore optimal policy regimes in Section 4. We define the interest rate elasticity of aggregate demand as $\sigma \equiv \sigma_c \frac{\bar{C}}{\bar{Y}}$. Given the specification of aggregate consumption, demand for each good is

$$y_t(i) = \left(\frac{c_t(i)}{C_t} \right)^{-\theta},$$

where θ is the demand elasticity of goods of type i , for simplicity, production is linear in labor, implying $y_t(i) = N_t(i)$. Each firm retains its price with probability α and re-optimizes with probability $1 - \alpha$ to maximize profits. We adopt the same pricing framework as Benigno, Eggertsson, and Romei (2020), so $\hat{\pi}_t$ measures inflation in deviation from the central bank's inflation target. These assumptions yield a slope of the Phillips curve given by

$$\kappa = \frac{(1 - \alpha)(1 - \alpha\beta)}{\alpha} \frac{\sigma^{-1} + \omega}{1 + \omega\theta}.$$

For further details, see Woodford (2003), chapter 3.1, whose notation is closest to ours.

⁴⁶ The exact characterization we use to illustrate fiscal instruments is from Denes, Eggertsson, and Gilbulkh (2013).

people want to buy. In the new Keynesian framework, we express this fundamental property by the IS equation, which comes from combining households' optimal consumption decisions with the economy's resource constraint. Throughout our analysis, all variables are expressed as natural logarithmic deviations from their steady-state values:

$$(29) \quad \hat{Y}_t = E_t \hat{Y}_{t+1} - \sigma(\hat{i}_t - E_t \hat{\pi}_{t+1} - \hat{r}_t^e) + \mathcal{F}_t^{IS},$$

where $\hat{Y}_t \equiv \ln Y_t - \ln \bar{Y}$ is the output in log deviation from steady state, $\hat{i}_t \equiv \ln(1 + i_t) - \ln(1 + \bar{i})$ is the short-term nominal interest rate in log deviation from steady state, while $\hat{\pi}_t \equiv \ln \Pi_t - \ln \bar{\Pi}$ is inflation in log deviation from the central bank's inflation target. The parameter $\sigma > 0$ measures the sensitivity of aggregate demand to real interest rates, and $\mathcal{F}_t^{IS}$ is a composite of fiscal variables defined when we analyze fiscal policy. Still, we keep track of these terms to economize on calculations later. The variable $\hat{r}_t^e$ is the efficient rate of interest, summarizing all exogenous shocks as we assume that $\hat{Y}_t^e = 0$. The efficient interest rate may vary with time due to the time-varying impatience of the household, see, for example, the variable ξ_t in footnote 45.

Expressed as in equation (29), the IS equation says how much output people demand. As shown in the equation, this depends on how high an income people expect to have in future period $E_t Y_{t+1}$ and on the real interest rate, $i_t - E_t \pi_{t+1}$, which is the price of buying goods today relative to tomorrow. A reduction in the real rate makes spending today cheaper, thus increasing aggregate demand. The demand also depends on the fiscal composite term, $\mathcal{F}_t^{IS}$, which we get to later, and the efficient rate of interest, $\hat{r}_t^e$, but our central focus will be on substantial drops in the efficient rate.

We interpret a decline in the efficient interest rate as a reduced-form representation of the economic forces driving the natural interest rate formalized in Section 2. Occasionally, when confusion is unlikely, we interchangeably use the terms “natural rate of interest” and “efficient rate of interest.” However, to avoid ambiguity, the term “natural interest rate” will always refer to its value absent variation in fiscal policy, in which case the two are the same.

The most important property of the new Keynesian model, which differentiates it from models of older vintages, is the central role of expectations. In this review, it plays the most crucial role through the demand side, that is, the IS equation, where expectations play a central role and are determined *endogenously as a function of the policy regime*.⁴⁷ Forward iteration of equation (29) yields

$$(30) \quad \hat{Y}_t = -\sigma E_t \sum_{j=0}^{\infty} [(\hat{i}_{t+j} - \hat{\pi}_{t+1+j} - \hat{r}_{t+j}^e - \sigma^{-1} \mathcal{F}_{t+j}^{IS})],$$

which says that output people want to buy depends on the expected path of interest rate, expected inflation, the efficient rate of interest, and the fiscal composite term over the infinite horizon. Since long-term interest rates depend on current and expected future short-term interest rates, an alternative interpretation is that the IS curve depends on long-term interest rates.⁴⁸

⁴⁷ In developing the IS–LM model Hicks (1937) highlights that treating expectation as exogenous is the main weakness of the model.

⁴⁸ A large part of the modern literature presumes that only the real interest rate matters and that economic agents form expectations rationally. Accordingly, it matters little from the perspective of demand if the central bank cuts the nominal rate or raises expected inflation; other things are constant. Recent literature has started to empirically and theoretically question this assumption; see, e.g., survey papers by Weber et al. (2022) and Eusepi and Preston (2018).

The nominal interest rate is subject to the ZLB, as discussed in detail in Section 2.⁴⁹

$$(31) \quad \hat{i}_t \geq i^{zlb}.$$

Because we express the variables as deviations from steady state, the ZLB implies that the short-term nominal interest rate, in log-deviation from steady state, must exceed $i^{zlb} \equiv -\ln(1 + \bar{r}) = -\ln[(1 + \bar{r})\bar{\Pi}]$. This notation has a few valuable properties, which we will discuss shortly.

The new Keynesian Phillips curve, derived from firms' optimal price setting, is

$$(32) \quad \hat{\pi}_t = \kappa \hat{Y}_t + \beta E_t \hat{\pi}_{t+1} + \mathcal{F}_t^{AS},$$

where $\kappa > 0$ measures the sensitivity of inflation to output, but firms set their prices infrequently and expect them to remain fixed for extended periods, as in Calvo (1983). The variable β is the household time discount factor typically calibrated close to 1, satisfying the restriction $0 < \beta < 1$. The composite variable $\mathcal{F}_t^{AS}$ represents the effect of fiscal policy instruments (and, possibly, other exogenous shocks) and will be defined once relevant.

Again, it is helpful to forward this equation to yield

$$\pi_t = E_t \sum_{s=t}^{\infty} \beta^{s-t} [\kappa \hat{Y}_{t+s} + \mathcal{F}_{t+s}^{AS}].$$

A second property of the new Keynesian Phillips curve is that firms set their prices intermittently, with probability $1 - \alpha$ in each period. No less importantly, given the price they have set, they are obligated to sell their customer whatever they demand at this price. Since firms adjust prices only occasionally, they must account for future market conditions. The fact that firms only occasionally revisit their prices is well documented empirically.⁵⁰ Consequently, when setting prices, firms forecast future economic conditions. If they anticipate a prolonged economic slump (and deflation), they will set lower prices, amplifying deflationary pressures.

The most critical assumptions in this section are the specification of the monetary and fiscal policy regime and that the shock is temporary. We define the standard fiscal and monetary policy regime as follows:

ASSUMPTION 1A (A1a): *A standard monetary and fiscal policy regime is:*

(i) *Monetary policy regime:*

$$\hat{i}_t = \max(i^{zlb}, \hat{r}_t^e + \phi_{\pi} \hat{\pi}_t + \phi_y \hat{Y}_t),$$

where ϕ_{π} and ϕ_y determine the central bank's reaction to the deviation of inflation and output from the target.

⁴⁹A higher inflation target reduces i^{zlb} , giving the central bank more room to cut rates.

⁵⁰See the classic paper by Nakamura and Steinsson (2008), whose headline finding, in the context of subsequent literature, indicates that the typical Consumer Price Index (CPI) item's price remains fixed for approximately nine months.

- (ii) *Fiscal policy regime: There can be exogenous variation in $\mathcal{F}_t^{AS}$ and $\mathcal{F}_t^{IS}$. The budgetary impacts of these variations are financed by lump-sum taxation unless otherwise stated.*

The monetary policy rule is a family of policy rules that differ only in the implicit steady-state inflation target, $\bar{\pi}$.⁵¹

In this review we consider various fiscal policies. In this section we mostly assume that current or future lump-sum taxes will offset all fiscal interventions. Initially, we abstract from fiscal policy but keep track of $\{\mathcal{F}_t^{IS}, \mathcal{F}_t^{AS}\}$ for results shown later.⁵²

We can state the standard monetary and fiscal policy regime in an alternative form, which proves helpful for some applications but results in a less familiar AS–AD representation. Unless stated otherwise, the equilibrium results presented in Section 3 remain identical whether we assume the policy regime defined by A1a or A1b. Eggertsson and Woodford (2003) propose this alternative policy regime:

ASSUMPTION 1B (A1b): A *Standard monetary and fiscal policy regime* is:

- (i) *Monetary policy regime: the central bank commits to adjusting the nominal interest rate so that*

$$\begin{aligned}\hat{\pi}_t &= 0, & \text{for } \forall t \text{ if possible} \\ \hat{i}_t &= i^{zlb}, & \text{otherwise.}\end{aligned}$$

- (ii) *Fiscal policy regime: There can be exogenous variation in $\mathcal{F}_t^{AS}$ and $\mathcal{F}_t^{IS}$. These variations are financed by lump-sum taxation unless otherwise stated.*

If the central bank satisfies its inflation target, then in equilibrium, using IS and AS equations, $\hat{i}_t = \hat{r}_t^e$.⁵³ This is not feasible if $\hat{r}_t^e < i^{zlb}$, hence the qualification in A1b: “if possible.” It says that if the central bank cannot achieve its inflation target due to the ZLB, it will keep the interest rate at the ZLB.⁵⁴ An equilibrium can now be defined, but it is productive to put more structure on the shock, $\hat{r}_t^e$, along with some restriction on parameters to ensure the existence

⁵¹ It is increasingly common to include the efficient rate of interest in the Taylor rule for a better empirical fit, see, e.g., Cúrdia et al. (2015). In our case, we include this term for modestly cleaner algebra. Given the assumed stochastic process for $\hat{r}_t^e$ in A2, the result would be unchanged if we replaced it by $\bar{r}$ in the monetary policy regime.

⁵² To be more specific, we assume that fiscal policy is sustainable in the sense that lump-sum taxes will adjust for any path of prices, output, and interest rate to ensure that the household’s transversality condition is satisfied. Our analysis relies on local approximation, so our equilibrium selection criterion only considers bounded equilibria. An alternative equilibrium selection device is the Markov perfect equilibrium considered in Section 4, which is locally unique to the steady state and yields the same result in equilibrium as the standard monetary and fiscal policy regime.

⁵³ Imposing for simplicity that $\mathcal{F}_t^{IS} = \mathcal{F}_t^{AS} = 0$, the condition is adjusted accordingly if these variables take nonzero value.

⁵⁴ As we will see in Section 4, the equilibrium implied by the standard monetary and fiscal policy regime can be micro-founded naturally by assuming that the central bank maximizes social welfare but imposing a Markov perfect equilibrium, which limits a central bank’s policy strategy to only react to current economic conditions as expressed by economic constraints it faces.

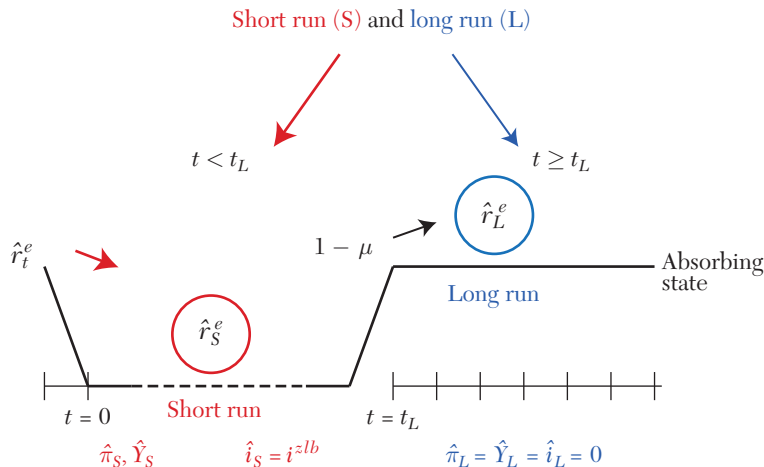


Figure 10. Assumption 2 (A2)

and uniqueness of equilibriums, before proceeding. We do this with an analytic example that we can solve in closed form.⁵⁵

3.1.3 Analytic Solution, AD–AS Diagram, Macroeconomic Crisis

Perhaps surprisingly, given the complications of the ZLB, we can reduce the model to only two equations in two unknowns and use the result to sketch an AD–AS diagram we will rely on for the following sections to show a number of conclusions. The key to obtaining this simple characterization is a convenient assumption about the temporary fall in the efficient interest rate.

ASSUMPTION 2 (A2): The Efficient Rate of Interest and Output. *The efficient interest rate, $\hat{r}_t^e$, and output, $\hat{Y}_t^e$, are the real interest rate and output when the government sets the fiscal policy instruments at values that maximize social welfare when prices are flexible. An unexpected shock at time zero results in $\hat{r}_0^e = \hat{r}_S^e < 0$. This shock persists with probability μ . Conditional on $\hat{r}_t^e = \hat{r}_S^e < 0$ at time t , the efficient interest rate reverts to the steady state, $\hat{r}_L^e = 0$, with probability $1 - \mu$ at period $t + 1$, which is an absorbing state. The shock reverts to steady state at a stochastic date denoted t_L . The efficient output level is constant at its steady state so that $\hat{Y}_t^e = 0$ at all times, and the fiscal instruments are constant at all times in the efficient equilibrium.*

Figure 10 shows that the efficient interest rate follows a special stochastic process. Initially, at $t = 0$, it declines unexpectedly to a constant value below its steady state. It then reverts to a steady state with a fixed probability of $1 - \mu$. Once the process returns to the steady state, it

⁵⁵Equilibrium is a collection of stochastic processes $\{\hat{Y}_t, \hat{i}_t, \hat{\pi}_t\}$ solving (29), (31), (32), and (A1a) or (A1b), given $\{\hat{r}_t^e\}$ and $\{\mathcal{F}_t^{IS}, \mathcal{F}_t^{AS}\}$.

stays there forever. We call the period before it reverts to steady-state short run (S) and after the long run (L). When the shock occurs, people form expectations about how long the short run will last, which is uncertain. The figure also highlights, in red, a key result for the endogenous variables that we will establish. This is key to be able to show the result in a simple AS–AD diagram. All the endogenous variables are some constant numbers in the short and the long run.

A central feature of our setup is that $\hat{r}_t^e$ follows a stochastic process with infinitely many possible contingencies. We characterize each contingency by the period t_L , in which the shock reverts to the steady state (an absorbing state). The expected duration of the shock is $\frac{1}{1-\mu}$. Unlike common perfect foresight models, our stochastic specification recognizes that even if market participants initially expect the shock to die down in just a few quarters (as market expectation suggested at the beginning of the Great Recession), the recession may persist for years. When shocks persist longer than market participants initially expect, this reflects the realization of a low probability contingency.⁵⁶

We impose A2 for theoretical clarity as it enables closed-form solutions. Section 3.2.6 shows the key results are robust to assuming an autoregressive stochastic process of order 1 (AR(1)) stochastic process. Next, we impose an assumption on the fiscal stance. This assumption is not innocuous and is later relaxed:

ASSUMPTION 3 (A3): *The fiscal stance, summarized by $\mathcal{F}_t^{IS}$ and $\mathcal{F}_t^{AS}$, follows the same process as $\hat{r}_t^e$ and is perfectly correlated with it, that is, it takes some constant value, $\mathcal{F}_S^{IS}$ and $\mathcal{F}_S^{AS}$ in $t < t_L$ and $\mathcal{F}_t^{AS} = \mathcal{F}_t^{IS} = 0$ for $t \geq t_L$.*

Assumptions A1–3 allow us to solve the model for the long run once the shock is in a steady state (that is, $t \geq t_L$) and then use this as input for the solution for $t < t_L$. For $t \geq t_L$. Using the method of Blanchard and Kahn (1980) allows us to solve equations (29), (32), and the policy rule in A1a to yield the unique bounded solution at positive interest rate:

$$(33) \quad \hat{\pi}_t = \hat{\pi}_L = 0, \hat{Y}_t = \hat{Y}_L = 0, \hat{i}_t = \hat{i}_L = \hat{r}_L^e = 0,$$

if $\phi_\pi + \frac{1-\beta}{\kappa}\phi_y > 1$.⁵⁷ The Taylor principle, $\phi_\pi > 1$ and $\phi_y \geq 0$, is a sufficient condition for this to always be satisfied. We state it as

ASSUMPTION 4 (A4):

$$(34) \quad \phi_\pi > 1, \text{ and } \phi_y \geq 0.$$

If we use the alternative definition of the standard monetary and fiscal policy regime, A1b, then A4 is irrelevant.

Using A1a and A2–A4, we can verify that as long as $\hat{r}_S^e \geq i^{zlb}$, the ZLB does not bind. In this case, there is always a unique bounded solution, and the central bank can implement its inflation target so that $\hat{i}_S = \hat{r}_S^e$. We can derive this result following the steps we used to establish the unique bounded solution in the long run.

⁵⁶An alternative interpretation is that agents hold beliefs inconsistent with the underlying stochastic process, thus deviating from rational expectations.

⁵⁷See proposition 4.3 in Woodford (2003), p. 254.

Things become interesting once $\hat{r}_S^e < i^{zlb}$. Then the ZLB binds, and we set $\hat{i}_S = i^{zlb}$. We obtain a solution for $t \leq t_L$ by solving the model backward and computing the expectations of each variable using the long-run solution as input. The resulting system can be solved using a moderately extended method of Blanchard and Kahn (1980).⁵⁸

ASSUMPTION 5 (A5):

$$(35) \quad \vartheta(\mu) \equiv (1 - \mu)(1 - \beta\mu) - \kappa\sigma\mu > 0,$$

where we make the function ϑ depend on μ to emphasize the importance of the shock persistence on the value of the function. Since $0 < \beta < 1$ then $0 < \vartheta \leq 1$. Under A1a and A2–A5, we obtain a unique bounded constant solution in the short-run $\{\hat{\pi}_S, \hat{Y}_S\}$ as a function of $\{\hat{r}_S^e, \mathcal{F}_S^{AS}, \mathcal{F}_S^{IS}\}$ and $\{\hat{i}_S = i^{ZLB}\}$ for all $0 \leq t < t_L$.

For convenience, we summarize A1–A5 in footnote 60.⁵⁹

Using the results derived so far, then under A1b we obtain the IS curve:

$$(36) \quad \hat{Y}_S = -\frac{\sigma}{1 - \mu}(\hat{i}_S - \mu\hat{\pi}_S - \hat{r}_S^e + \sigma^{-1}\mathcal{F}_S^{IS}).$$

We derive this equation in two steps. First, from equation (33), we know the unique bounded solution for inflation is $\hat{Y}_L = \hat{\pi}_L = 0$. Second, given A1a and A2–A5, a unique bounded solution exists in the short run, denoted $\hat{\pi}_S$ (see footnote 59). These results allow us to substitute out for expected output and inflation in the short run, that is, $E_S \hat{Y}_{S+1} = \mu \hat{Y}_S$, $E_S \hat{\pi}_{S+1} = \mu \hat{\pi}_S$, and solve for $\hat{Y}_S$.

⁵⁸To show that the model has a unique bounded solution we build on Eggertsson (2008) footnotes 41–44. Use the expression for the long run to obtain expectations in the short run:

$$E_{t,S} \hat{\pi}_{t+1} = \mu E_{t,S} \hat{\pi}_{t+1,S} + (1 - \mu) E_{t,S} \hat{\pi}_{t+1,L} = \mu E_{t,S} \hat{\pi}_{t+1,S},$$

where $E_{t,S}$ is the expectation formed at time t conditional on $t < t_L$, so the shock is in the low state and $\hat{\pi}_{t+1,S}$ denotes inflation at time $t + 1$ conditional on the shock remaining in its low state. We can similarly express expectations about output as $E_{t,S} \hat{Y}_{t+1,S}$. Substituting $\hat{i}_t = i^{zlb}$, equations (29) and (32) at time $t < t_L$ are expressed as:

$$\begin{aligned} \hat{Y}_{t,S} &= \mu E_{t,S} \hat{Y}_{t+1,S} - \sigma(\hat{i}_t - \hat{r}_S^e) + \sigma\mu E_{t,S} \hat{\pi}_{t+1,S} + \mathcal{F}_S^{IS}, \\ \hat{\pi}_{t,S} &= \kappa \hat{Y}_{t,S} + \beta\mu E_{t,S} \hat{\pi}_{t+1,S} + \mathcal{F}_S^{AS}. \end{aligned}$$

Using the method of Blanchard and Kahn (1980), we obtain a unique bounded solution for $\hat{Y}_S$ and $\hat{\pi}_S$ as a function of $\mathcal{F}_S^{IS}$, $\mathcal{F}_S^{AS}$, $\hat{r}_S^e$ if A5 is satisfied as proved in Proposition 2 in Eggertsson (2010).

⁵⁹Summary of assumptions:

Horizons: short run (S): $0 < t < t_L$, long run (L): $t \geq t_L$. *Duration:* The variable t_L is a random stopping point.

- | | | |
|----|--|---|
| A1 | Standard monetary and fiscal policy regime | $\begin{cases} \text{A1a Taylor Rule} \\ \text{A1b Inflation Target} \end{cases}$ |
| A2 | Efficient interest rate $\downarrow$ in S. | |
| A3 | Fiscal stance at steady state in L; may follow efficient interest rate in S. | |
| A4 | Taylor-rule coefficients: $\phi_\pi > 1, \phi_y \geq 0$. | |
| A5 | Bound on expected shock duration: $\mu < \bar{\mu}$. | |

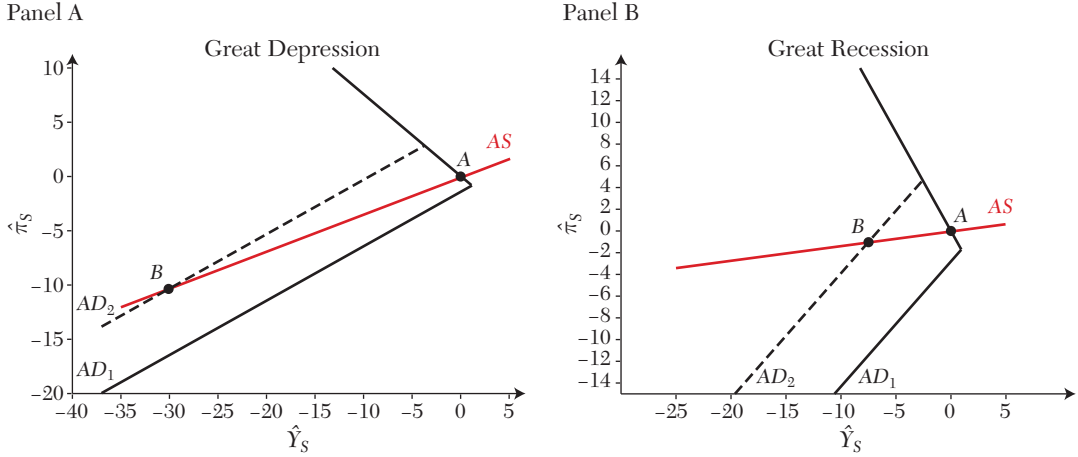


Figure 11. AS-AD Diagram for Great Depression and Great Recession

We now can solve for AD by combining the monetary policy rule in A1a and equation (36):

$$(37) \quad \hat{Y}_S = \begin{cases} -\frac{\sigma(\phi_\pi - \mu)}{1 - \mu + \sigma\phi_y} \hat{\pi}_S + \frac{1}{1 - \mu + \sigma\phi_y} \mathcal{F}_S^{IS}, & \text{if } \hat{\pi}_S \geq i^{zlb}, \\ \frac{\sigma\mu}{1 - \mu} \hat{\pi}_S + \frac{\sigma}{1 - \mu} (\hat{r}_S^e - i^{zlb}) + \frac{1}{1 - \mu} \mathcal{F}_S^{IS}, & \text{if } \hat{\pi}_S = i^{zlb}. \end{cases}$$

Similarly, we express the new Keynesian Phillips curve (32), or AS equation, as:

$$(38) \quad \hat{\pi}_S = \kappa \hat{Y}_S + \beta\mu \hat{\pi}_S + \mathcal{F}_S^{AS} = \frac{1}{1 - \beta\mu} (\kappa \hat{Y}_S + \mathcal{F}_S^{AS}).$$

In the first equation, we substitute for expected inflation; in the second, we solve for $\hat{\pi}_S$.

The AD and AS curves are plotted together in Figure 11, using the numerical calibration of the model for the Great Depression and Great Recession explained in the following section. The general shape does not depend on the parameters. The AS curve is always upward sloping, while the AD curve is downward sloping if the ZLB is not binding but kinks at the ZLB and changes slope. When the ZLB is not binding, the central bank cuts the nominal interest rate in response to a drop in inflation more than one-to-one, reducing the real rate and stimulating demand. However, the central bank cannot cut the nominal rate below zero, which generates a kink at $\hat{\pi}_S = i^{zlb}$. In this case, lower inflation *increases* the real interest rate instead of decreasing it, since the central bank cannot cut policy rates, so the real interest rate depends only on *expected* inflation, that is, $\hat{r}_S = i^{zlb} - E_S \hat{\pi}_{S+1} = \mu \hat{\pi}_S$. As inflation (expectations) falls the real interest rate increases, reducing demand.

The AD and AS curves have two possible intersections, depending on the exogenous shock $\hat{r}_S^e$. If $\hat{r}_S^e \geq i^{zlb}$, there is a regular equilibrium at point A in Figure 11, with inflation and output at steady state in the Great Depression example (left) and the Great Recession example (right). If $\hat{r}_S^e < i^{zlb}$, the shock is large enough for the ZLB to bind, as illustrated at point B. We

can solve the model analytically using the AD and AS equations, (37), (38) to yield (assuming $\mathcal{F}_S^{IS} = \mathcal{F}_S^{AS} = 0$ if the ZLB is not binding):

$$(39) \quad \hat{Y}_S = \begin{cases} 0 & \text{if } \hat{r}_S^e \geq i^{zlb} \\ \frac{\sigma(1-\beta\mu)}{\vartheta}(\hat{r}_S^e - i^{zlb}) + \frac{\sigma(1-\mu\beta)}{\vartheta}\mathcal{F}_S^{IS} + \frac{\sigma\mu}{\vartheta}\mathcal{F}_S^{AS} & \text{if } \hat{r}_S^e < i^{zlb} \text{ so } \hat{i}_S = i^{zlb} \end{cases}$$

$$(40) \quad \hat{\pi}_S = \begin{cases} 0 & \text{if } \hat{r}_S^e \geq i^{zlb} \\ \frac{\sigma\kappa}{\vartheta}(\hat{r}_S^e - i^{zlb}) + \frac{\kappa}{\vartheta}\mathcal{F}_S^{IS} + \frac{1-\mu}{\vartheta}\mathcal{F}_S^{AS} & \text{if } \hat{r}_S^e < i^{zlb} \text{ so } \hat{i}_S = i^{zlb} \end{cases}$$

The model restricts the persistence of the shock because an equilibrium does not exist if people expect the shock to persist for too long. Due to the model's powerful propagation mechanism, the shock's expected duration, $\frac{1}{1-\mu}$, cannot exceed a critical value. A5 formalizes this restriction, requiring that $\vartheta(\mu) = (1-\mu)(1-\beta\mu) - \kappa\sigma\mu > 0$, where ϑ depends critically on the persistence parameter μ . We define $\bar{\mu}$ as the critical value when this condition no longer holds, that is, $\vartheta(\bar{\mu}) = 0$. As μ approaches $\bar{\mu}$, ϑ approaches zero and the model generates increasingly severe contractions. Eventually, the economy implodes into a deflationary spiral, often called a deflationary black hole.

Figure 12 clarifies the reason for the implosion, illustrating via solid lines the case where $\mu = 0$ so the ZLB is binding with certainty for only one period. The equilibrium occurs at intersection point A. Once the ZLB binds, the vertical AD curve entirely determines output. It is, thus, demand determined. In this case, the shock, $\hat{r}_S^e$, directly determines output. *Given* output, the intersection of the vertical portion of the AD curve at the ZLB (equation (37)) and the AS curve (equation (38)) determines inflation.⁶⁰

Increasing μ (illustrated by the dashed lines in Figure 12) sheds light on how the model can generate severe contractions comparable to the Great Recession or Great Depression. It clarifies the critical role of market expectations of the persistence of the shock to the efficient rate of interest, as higher μ implies markets anticipate a longer-lasting contraction, shifting both the AD and AS curves.

The AD curve becomes flatter at the ZLB because lower *inflation expectations*, $\mu\hat{\pi}_S$, reduce output. With the central bank's policy rate constrained by the ZLB, any decline in expected inflation raises the real interest rate. A fall in inflation expectation depresses demand and magnifies the initial contraction via the IS equation, since the policy rate is fixed. Moreover, current market expectations of future output declines (represented by $\mu\hat{Y}_S$) further amplify these effects.

As μ increases, the AS curve steepens. When the AS curve is steeper, any given output gap leads to sharper deflation because firms, anticipating a prolonged deflationary environment, set lower prices once they optimally choose their price, which will be in effect for some time. In turn, these lower prices and expectations of weaker demand reduce output further,

⁶⁰Note that when the interest rate is positive, the AD curve is not vertical, though it is close to vertical. Since π_S is multiplied by the factor $\frac{\sigma(\phi_\pi - \mu)}{1 - \mu + \sigma\phi_y}$, this factor becomes smaller as μ approaches ϕ_π .

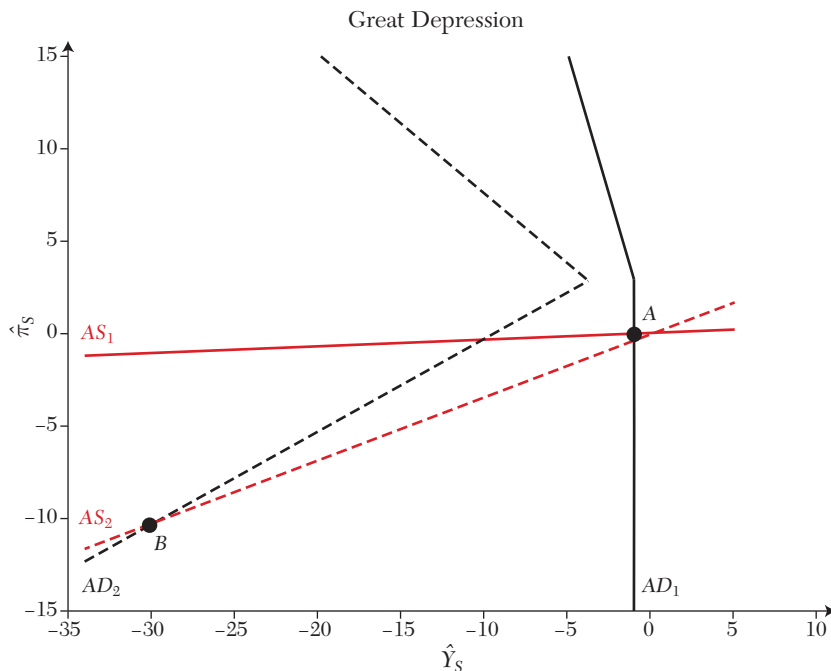


Figure 12. The Consequence of Increasing the Expected Duration of the Shock at the ZLB on AD and AS Illustrated Using the Great Depression Calibration Example

according to the IS curve. This effect is amplified by higher persistence (μ): Suppressed current output and expectations of future declines both reduce current inflation and depress inflation expectations, which, in turn, feed back into the IS equation and drive output even lower, setting in motion a self-reinforcing deflationary spiral.

The combined effect—illustrated by the intersection of the dashed curves at point B in Figure 12, representative of scenarios such as the Great Depression—is a deeper contraction accompanied by greater deflation. As μ approaches the critical threshold $\bar{\mu}$, contractionary forces feed off each other and intensify until the AD and AS curves become parallel, at which point equilibrium ceases to exist.

This discussion illustrates how a log-linear approximation of the new Keynesian model can generate arbitrarily large contractions when shocks are sufficiently persistent, potentially explaining severe historical episodes such as the Great Recession and Great Depression.⁶¹

⁶¹One might suspect from the linearized equations alone that crossing the threshold $\bar{\mu}$ indicates indeterminacy—infinitely many possible solutions. However, Eggertsson and Singh (2019) show, by solving the fully nonlinear model, that this interpretation is incorrect. They demonstrate that once μ reaches the critical value $\bar{\mu}$ a finite contraction in output occurs, and any further increase results in the nonexistence of equilibrium. In addition, they calibrate the nonlinear model using the same Bayesian methods we apply here, finding similar quantitative conclusions to those we report.

TABLE 1
PARAMETER VALUES FOR NUMERICAL EXAMPLES AT QUARTERLY FREQUENCY

Parameter	Great Recession	Great Depression	Description
<i>Parameterization based on Denes and Eggertsson (2009)</i>			
α	0.8932	0.7746	Calvo parameter (price stickiness)
β	0.9985	0.9970	Discount factor
μ	0.9184	0.9016	Markov transition probability
ω	6.8569	1.5302	Inverse of the Frisch elasticity of labor supply
σ^{-1}	3.2338	1.1529	Risk aversion
θ	6.5571	12.6956	Elasticity of substitution between different goods
$\hat{r}_S^e$	-0.0239	-0.0138	Efficient rate in the low state of Markov process
Parameter	Great Recession	Great Depression	Comment
<i>Composite and calibrated parameters</i>			
κ	0.0028	0.0087	$\kappa = \frac{(1-\alpha)(1-\alpha\beta)}{\alpha} \frac{\sigma^{-1} + \omega}{1 + \omega\theta}$
ψ	0.0991	0.3727	$\psi = \frac{1}{\sigma^{-1} + \omega}$
$\bar{\pi}$	0.0050	0.0000	Target inflation rate
$\bar{r}$	0.0065	0.0030	$\bar{r} = \frac{1 + \bar{\pi}}{\beta} - 1$
$\bar{r}^{zlb}$	-0.0065	-0.0030	$\bar{r}^{zlb} = -\ln(1 + \bar{r})$
$\bar{\tau}_S$	0.1000	0.1000	Steady-state consumption tax
$\bar{\tau}_I$	0.3000	0.3000	Steady-state income tax
ϕ_π	1.5000	1.5000	Response of policy rate to inflation
ϕ_Y	0.5000	0.5000	Response of policy rate to output gap

3.1.4 Calibrating the Model and the Role of the Inflation Target to Create Policy Space

We calibrate the model to replicate the depth of the output decline and inflation observed during the Great Depression and Great Recession in the United States, as illustrated in Figure 9.

We apply Bayesian methods for this calibration,⁶² assuming each period corresponds to one quarter. Table 1 summarizes the parameter values obtained from this procedure.⁶³ For the Great Depression, we choose parameters and shocks to match a 30 percent decline in output and a 10 percent deflation.

We plot the model-generated results alongside data from 1929 to 1937, assuming the efficient interest rate remains negative. For the Great Recession, we calibrate the parameters and shocks to match a 7.5 percent deviation of output from trend and a decline in inflation from 2 percent to 1 percent, following Del Negro, Eggertsson et al. (2017). We compare the model results with data from 2005–17, assuming the shock returns to steady state in 2015:IV, when the Federal Reserve raised rates. The precise timing of the shock's return does not directly affect equilibrium output or inflation at any point in time; what matters instead is the *expected* duration of the shock at each moment.

The AD–AS diagram was drawn up using the calibrated value to give readers a rough quantitative sense of the results. There are two differences worth highlighting. First, prices are

⁶²For details, see Denes and Eggertsson (2009) and Denes, Eggertsson, and Gilbukh (2013).

⁶³Here, $\psi \equiv \frac{1}{\sigma^{-1} + \omega}$, where ω is the inverse of the Frisch elasticity of labor supply.

more flexible in the Great Depression than the Great Recession, as shown by the fact that the AS curve is steeper; we discuss the implications of this difference in Section 3.1.5. Second, the inflation target is 2 percent during the Great Recession but 0 percent during the Great Depression. The inflation target critically affects the central bank's ability to respond to a negative efficient interest rate, as it determines the available *policy space* before hitting the ZLB, that is, where the IS curve kinks.

After the Great Inflation of the 1970s, the first signs of interest in the liquidity trap literature and the ZLB emerged after the United States conquered inflation in the early 1990s. It was rekindled by an insightful commentary by Summers (1991), who pointed out that negative real interest rates had emerged for about one-third of the time since World War II to 1991, suggesting that central banks would lose their ability to accommodate negative real rates if they adopted a zero inflation target. This led to a small but informative literature, which then grew rapidly as Japan faced low interest rates in the late 1990's.⁶⁴

The standard monetary and fiscal policy regime encompasses a family of policy regimes that differ in their inflation targets. We measure the variable $\hat{i}_t$ as the natural logarithm of the central bank's nominal interest rate relative to the logarithm of its steady-state value at quarterly frequency:

$$(41) \quad \ln(1 + \bar{i}) = \ln(1 + \bar{r}) + \ln\bar{\Pi},$$

where $\bar{\Pi} \equiv 1 + \bar{\pi}$ is the gross inflation target of the central bank and $\bar{r}$ is the steady-state real interest rate.

We define the variable $i^{zlb} < 0$ as the maximum amount by which the central bank can lower interest rates from steady state before hitting the ZLB. It determines the central bank's room for maneuver. Using this notation, the ZLB constraint requires that $\hat{i}_t \geq i^{zlb}$, where

$$(42) \quad i^{zlb} \equiv -\ln\bar{\Pi} - \ln(1 + \bar{r}) \approx -\bar{\pi} - \bar{r},$$

and where $\bar{\pi}$ is the steady-state inflation target. A policy regime with a higher inflation target gives the central bank more space to cut rates without encountering the ZLB constraint. Similarly, a higher short-term steady-state real interest rate increases the available policy space. During the Great Recession, countries that entered the financial crisis with historically higher inflation rates typically avoided hitting the ZLB constraint, as documented by Eggertsson, Lancastre, and Summers (2019) in a cross-country analysis.

Figure 13 illustrates how a higher inflation target expands the central bank's ability to respond to a negative efficient interest rate. The ZLB generates a kink in aggregate demand, marking the point at which the central bank exhausts its ability to lower rates further. We

⁶⁴Summers proposed an optimal inflation rate of 2–3 percent. Fuhrer and Madigan (1997), using a simplified forward-looking FRB–US model without explicit microfoundations, showed a zero inflation target frequently constrains policy and raises output volatility relative to a 4 percent target; Orphanides and Wieland (1998) provide similar insights. Rotemberg and Woodford (1997) indirectly addressed the ZLB by penalizing interest-rate volatility, showing policy inertia stabilizes output and inflation without breaching the ZLB, qualifying Fuhrer and Madigan's (1997) results. Woodford (1999) explicitly analyzed policy inertia under ZLB constraints, foundational for subsequent studies. Coibion, Gorodnichenko, and Wieland (2012) corroborate Woodford (1999), explicitly imposing the ZLB and incorporating steady-state inflation. Finally, Wolman (1998) was among the first to numerically impose the ZLB explicitly in a microfounded model.

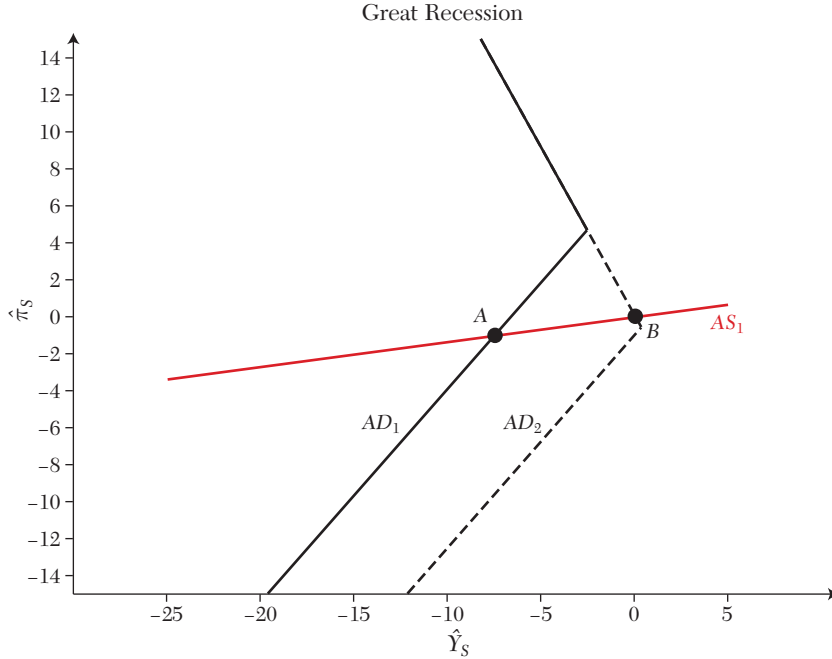


Figure 13. The Effect of Increasing the Inflation Target

derive this kink point by equating the two branches of the AD curve in equation (37), assuming for simplicity that $\phi_y = 0$, yielding:

$$(43) \quad \hat{\pi}^{kink} = -\frac{1}{\phi_\pi} [i^{zlb} - \hat{r}_S^e] \approx -\frac{1}{\phi_\pi} [\bar{\pi} + r_S^e],$$

where r_S^e is the level of the efficient interest rate in the short run. As Figure 13 shows, increasing the inflation target from 2 percent to 10 percent pushes the kink point below point B (where inflation is on target, that is, $\hat{\pi}_S = 0$). As the dashed line illustrates, increasing the inflation target shifts the kink point downward. With this change, $\hat{\pi}^{kink} < 0$ allows the central bank to reduce the policy rate sufficiently to accommodate the decline in the efficient interest rate while keeping inflation on target at point B—with additional room to spare. We draw Figure 13 using the Great Recession calibration, where the efficient interest rate falls by approximately 9 percent. The decline is of similar magnitude in the Great Depression parameterization. From equation (43), we derive a necessary condition for the central bank to achieve its inflation target:

$$(44) \quad \bar{\pi} \geq -r_S^e.$$

A fixed credible inflation target is part of how we *define* the standard monetary and fiscal policy regime. We define inflation in terms of deviations from the inflation target, that is, $\hat{\pi} = \pi_t - \bar{\pi}$, where the steady-state inflation, $\bar{\pi}$, equals the target. Our discussion suggests

that increasing the inflation target can expand the central bank's room for maneuvering in response to a negative efficient rate shock, making a higher inflation target straightforward policy advice. Krugman (1998), for example, suggested that the Bank of Japan should commit to a 4 percent inflation target for 15 years.

Policymakers face several challenges when determining if and how to raise the inflation target, $\bar{\pi}$, once an economy reaches the ZLB. First, the central bank needs a reliable estimate of how much it must raise the inflation target. If the efficient interest rate falls to -9 percent, the central bank must set an inflation target of at least 9 percent to achieve the inflation target. Estimating the efficient interest rate is challenging because policymakers cannot directly observe it. In Section 4, we discuss how central banks can solve this problem using a trigger strategy proposed by Eggertsson and Woodford (2003). Under this strategy, the central bank raises rates once observable data reach a predetermined price-level target. This approach eliminates the need for the central bank to know the exact value of the efficient rate. We relate this idea to how the Federal Reserve attempted to address the problem in practice in 2012.

Second, raising the inflation target at the ZLB creates an inherent credibility problem. In Section 4, we formally demonstrate this credibility problem and present narrative evidence that policymakers considered a significant constraint during the Great Recession. We then examine how policymakers addressed the credibility problem during the Great Depression, focusing on FDR's inflationary regime change in 1933.

Third, Eggertsson and Woodford (2003) show that committing to high inflation for an extended period is suboptimal. The calibrated model in that paper requires only a modest price level response in the optimal commitment solution. A commitment to lower future nominal rates beyond $t = t_L$ results in an output boom. This boom, combined with lower nominal rates, generates most of the expansionary effect, requiring little movement in the price level. While this result is specific to their calibration, it demonstrates that policymakers cannot take for granted that the response to temporarily negative efficient interest rates requires a major increase in inflation for an extended period.

We have already shown how increasing the steady-state inflation target can reduce output contraction. However, as we have seen, this approach may require extreme inflation commitments. In our calibration, it required close to 10 percent annual inflation to bring output back to potential.

For the remainder of this section, we take the inflation target as given and maintain the standard monetary and fiscal policy regime. Recall from A2 that this policy regime allows for variations in fiscal instruments, which we have yet to specify. Moreover, we focus on how changes in fiscal variables affect output and inflation *holding the monetary part of the policy regime fixed*. Before turning to this analysis, however, we document a fundamental and troubling property of the economic situation we have modeled: Market forces alone cannot restore the economy to a self-sustaining recovery.

3.1.5 *Why Market Forces Fail to Generate a Self-Sustaining Recovery at the ZLB: The Paradox of Price Flexibility*

So far Section 3 shows that a liquidity trap can cause significant output contraction when prices are rigid. Intuitively, one might expect that more frequent price adjustments would eliminate this output contraction through natural market self-correction. After all, it is the only new element we added to our model relative how we define the liquidity trap in Section 2,

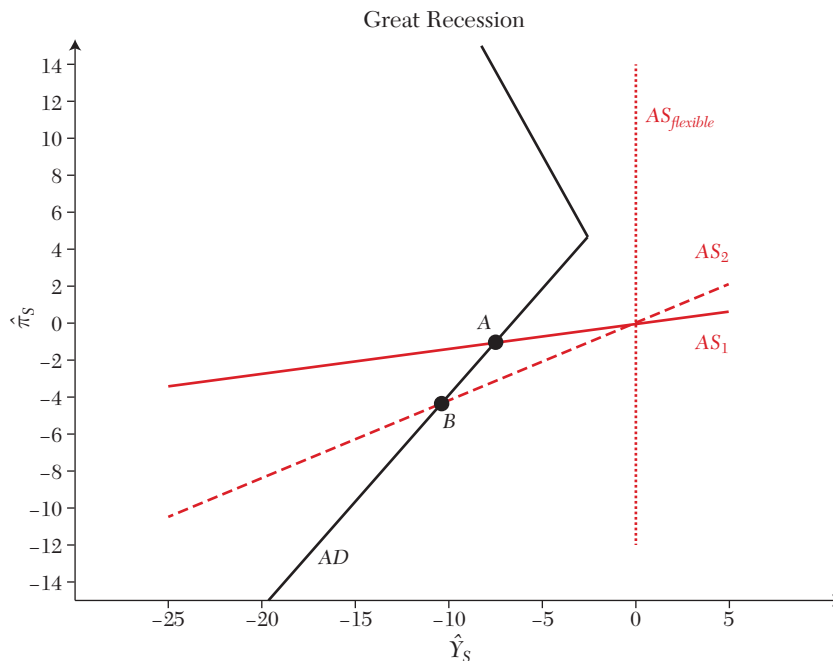


Figure 14. The Price Flexibility Paradox

Notes: The price flexibility paradox, increasing price flexibility in the Great Recession calibration (point A) by replacing the slope of the Phillips curve, κ , with the one from the Great Depression calibration (point B).

where we established the two building blocks of the liquidity trap but assume that prices were flexible. Yet this seemingly logical conclusion is incorrect. This is the paradox of price flexibility.

THE PARADOX OF FLEXIBILITY: Assume the following conditions hold: (i) the natural interest rate is temporarily negative as in A1; (ii) the standard monetary and fiscal policy regime (A2) applies; and (iii) the ZLB binds and output falls below its natural level. Under these conditions, as prices become increasingly flexible, output falls further below its natural level. That higher price flexibility reduces output is paradoxical: The natural output is defined as the level of output produced when all prices are flexible. The natural level of output is always higher than equilibrium output at the ZLB assuming the standard monetary and fiscal policy regime.

Increased price flexibility at the ZLB reduces output by lowering inflation expectations. Lower inflation expectations raise the real interest rate and thereby reduce demand. Figure 14 shows a graphical illustration of the paradox.

Consider equilibrium A, which corresponds to the Great Recession example. Now consider increasing price flexibility. Then, the AS curve rotates around the steady-state values of inflation and output and steepens. A steeper AS curve yields lower expected inflation ($\mu \hat{\pi}_s$) at the ZLB, a higher real interest rate, and thus lower demand, as point B shows.

But how can greater price flexibility intensify recessions, given that output should equal its natural level under fully flexible prices? The resolution of this paradox is that the equilibrium under fully flexible prices, in which output equals its natural level when $r_t^e < i^{zlb}$, is a mirage. It simply does not exist. Figure 14 illustrates this logic. As price flexibility increases, the AS curve steepens, intensifying the output contraction at point B. The AS curve becomes vertical in the limit of fully flexible prices (as shown by the dotted line). In this case, the AS curve never intersects the AD curve. No equilibrium exists.

The nonexistence result is not an artifact of using a log-linear approximation of the underlying nonlinear model. Instead, it is an inherent property of the standard monetary and fiscal policy regime. We can demonstrate this fundamental property under minimal assumptions: Consider a model with fully flexible prices in which output is an exogenous endowment. We assume the standard monetary and fiscal policy regime. Then, no equilibrium exists. We provide a formal mathematical proof by contradiction in the footnote.⁶⁵

At first glance, the absence of an equilibrium under fully flexible prices might seem to invalidate the model. We view it differently. Full flexibility is an idealized limit; in practice, prices change only intermittently. Incorporating empirically plausible Calvo stickiness restores equilibrium, but it can entail a severe output contraction. The “no-equilibrium” result therefore serves as a warning light: it marks the boundary where the standard policy regime breaks down and further action is required—either a monetary policy regime shift (Section 4) or the other policy measures examined in the remainder of this section.

The paradox may help explain why the Great Recession was less severe than the Great Depression. Figure 14 visually illustrates the paradox and highlights an interesting quantitative result. We compute point A using the Great Recession calibration and point B using the same calibration, except the slope of the Phillips curve, κ , is replaced with the higher value from the Great Depression calibration. This modification generates an additional 4 percentage point drop in output and a 4.5 percentage point decline in inflation. Thus, if prices during the Great Recession had been as flexible as during the Great Depression, the recession would have been more severe.

Eggertsson and Krugman (2012) coined the term *paradox of price flexibility*. They demonstrated this paradox using a spender–saver model similar to the one we used in Section 2, augmented with price rigidities and nominal debt. In their model, an unexpected decline in inflation redistributes wealth from borrowers, who have a high marginal propensity to consume (MPC), to savers, who have a low MPC, thus reducing aggregate spending. Greater price flexibility leads to a larger decline in the price level, amplifying this redistribution effect.

⁶⁵We establish this result by contradiction. Consider a nonlinear, flexible-price economy under perfect foresight with real and nominal bonds and exogenous output. We introduce the cash constraint from Section 2 to simplify the discussion of price determination. The equilibrium conditions are summarized as follows: (i) Fisher equation: $1 + r_t = (1 + i_t) \frac{P_{t+1}}{P_t}$; (ii) ZLB constraint: $i_t \geq 0$; (iii) Cash constraint: $M_t \geq \chi P_t Y$. As in Section 2, the real interest rate is exogenous. Suppose an unexpected shock makes the real interest rate negative in period 0 ($r_0 < 0$), after which it reverts to $r_t = \beta^{-1} - 1 > 0$ for all $t > 0$. The central bank controls the price level at any time $t > 0$ by choosing M_t as the ZLB is not binding. $\frac{P_t}{P_{t-1}}$ is gross inflation. Solve the model backward from any period $t > 0$. The central bank inherits P_{t-1} . It is straightforward to confirm the solution $\frac{P_t}{P_{t-1}} = 1$ satisfies (i)–(iii) for all $t \geq 1$. But fundamentally, this *includes* period 1. Thus, the central bank sets the money supply so that $P_1 = P_0$ in period 1. However, this immediately leads to a contradiction in period 0. The solution implies $\frac{P_1}{P_0} = 1$, which by condition (i) means $i_0 = r_0 < 0$, which violates the ZLB constraint (ii). Hence, there is no equilibrium consistent with this policy regime.

The paradox of price flexibility can manifest through two distinct channels: first, through the wealth redistribution effect described above, which requires heterogeneity between borrowers and savers; second, in representative agent models through the expected inflation channel, as we have seen earlier in this section. In the latter case, greater price flexibility reduces expected inflation when markets anticipate a prolonged slump, thereby increasing the real interest rate and suppressing aggregate demand.⁶⁶

3.2 *Aggregate Demand within a Standard Policy Regime: What Can Fiscal Policy Do?*

Fiscal policy is a powerful alternative to the interest rate cuts at the ZLB under the standard monetary and fiscal policy regime. Section 3.2 reviews theoretical and empirical work suggesting that government spending multipliers are substantially larger at the ZLB than at positive rates. Large fiscal multipliers imply that fiscal austerity through spending cuts reduces the tax base, potentially *increasing* deficits. At the ZLB, “confidence” shocks, interpreted as shocks to expectations about future fiscal policy, may play a major role. At the same time, at a positive interest rate the central bank can offset them. We develop a simple HANK model and show that it amplifies policy effects via old Keynesian mechanisms: Each person’s spending becomes someone else’s income, creating another round of spending and so on. We close by showing the case in which multiple tax instruments eliminate the ZLB problem and discuss the central bank’s experiments with negative interest rates.

3.2.1 *Fiscal Multipliers: Why Are They So Much Larger at the ZLB?*

A central prediction of the framework we present in Section 3.2 is that increasing government spending has a significantly greater impact on output at ZLB than under normal circumstances. While policymakers were well aware of this result in the United States, European policymakers and private and official forecasters seemed to appreciate these results less, or ignore them, during the Great Recession. The underlying policy regime, which we take to be the standard one in A1, is critical for this finding.

In 2009 UK Conservative Party leader David Cameron declared that the “age of austerity” arrived, meaning it was time to curb government spending and/or raise taxes to balance the budget. After becoming prime minister in 2010, he implemented austerity measures. Several other governments, including Greece, Spain, and Italy, followed a similar strategy. The United States never fully embraced austerity, but after an initial spending burst in 2009 the US administration faced political pressures to contain deficits, thereby limiting more decisive fiscal actions.

Blanchard and Leigh (2013) document a striking cross-country correlation from European countries. Figure 15 plots the real GDP growth forecast error for 2010 and 2011 relative to spring 2010 forecasts, when several European countries implemented fiscal consolidation packages. The chart suggests that austerity triggered a larger recession than expected. Countries that planned (and then implemented) significant austerity experienced a larger drop in growth

⁶⁶Eggertsson (2012a) first formally demonstrated the paradox in the new Keynesian model at the ZLB. Closely related observations appear in other work. For a comprehensive recent overview, see Bhattarai, Eggertsson, and Schoenle (2018), who characterize conditions under which the paradox applies in a general new Keynesian setting. They estimate a medium-scale new Keynesian model following Smets and Wouters (2007) and find that greater price flexibility within their estimation window (1966:I–2004:IV) would have resulted in higher inflation and output volatility. Recent related work includes Galí (2013), who connects this observation to Keynes. Earlier contributions include Fisher (1923), Tobin (1975), and De Long and Summers (1986).

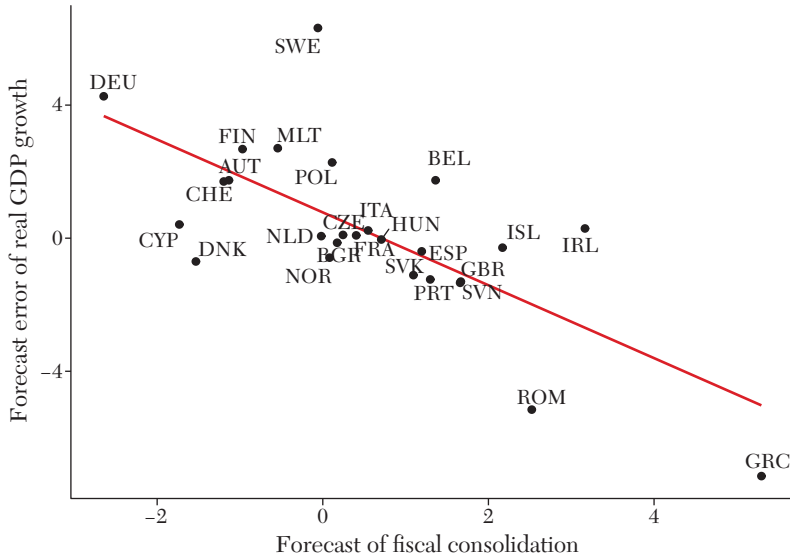


Figure 15. Forecast Errors for Real GDP Growth, 2010–2011

Notes: Forecast errors are relative to spring 2010 projections, which incorporated austerity measures for 2010–2011 designed in spring 2010. Data from Blanchard and Leigh (2012). The red line depicts the OLS regression. Country codes: AUT, Austria; BEL, Belgium; BGR, Bulgaria; CHE, Switzerland; CYP, Cyprus; CZE, Czech Republic; DEU, Germany; DNK, Denmark; ESP, Spain; FIN, Finland; FRA, France; GBR, United Kingdom; GRC, Greece; HUN, Hungary; IRL, Ireland; ISL, Iceland; ITA, Italy; MLT, Malta; NOR, Norway; NLD, Netherlands; POL, Poland; PRT, Portugal; ROM, Romania; SVK, Slovakia; SVN, Slovenia; SWE, Sweden.

than had been anticipated when these plans had been designed, relative to countries that did not implement austerity measures.

While we show that fiscal multipliers are larger at the ZLB than under normal circumstances, this result may have been overlooked by many analysts and government officials in 2010, especially in the eurozone, who appear to have based their outlook on the estimated effect of austerity measures using empirical evidence accumulated at positive short-term interest rates. US policymakers were better aware of the consequence of the ZLB; see, for example, Romer and Bernstein (2009), a note written by two key policymakers. The fiscal multiplier the Council of Economic Advisors of the US President reported in that note, 1.6, is close to those reported in this review, both empirically and theoretically, especially once heterogeneity is taken into account in Section 3.2.5.

To understand the effect of fiscal spending, we need to define the composite terms that shift the IS equation (29) and the AS equation (32). They are defined as follows:

$$(45) \quad \mathcal{F}_t^{IS} \equiv (\hat{F}_t - E_t \hat{F}_{t+1}) + \sigma \chi^s E_t (\hat{\tau}_{t+1}^s - \hat{\tau}_t^s) = (1 - \mu)(\hat{F}_S - \sigma \chi^s \hat{\tau}_S^s),$$

for the IS equation. The expression following the second equality sign uses A1–A5. Similarly, the composite term for the AS equation is

$$(46) \quad \mathcal{F}_t^{AS} \equiv \kappa \psi (\chi^I \hat{\tau}_t^I + \chi^s \hat{\tau}_t^s - \sigma^{-1} \hat{F}_t) = \kappa \psi (\chi^I \hat{\tau}_S^I + \chi^s \hat{\tau}_S^s - \sigma^{-1} \hat{F}_S).$$

Here, $\chi^s > 0$, $\chi^I > 0$, and $\psi > 0$ are coefficients.⁶⁷ $\hat{F}_t$ is the deviation of real government spending on goods and services from a steady state, expressed as a fraction of steady-state output. $\hat{\tau}_t^s \equiv \tau_t^s - \bar{\tau}$ is a sales tax levied on top of the firm's sticky price relative to the rate in steady state, and $\hat{\tau}_t^I \equiv \tau_t^I - \bar{\tau}$ represents taxes on all household income (labor income and income from firms' profits). A temporary increase in government spending increases aggregate demand in the short run, according to the IS equation, by directly increasing government consumption. A temporary cut in sales tax also increases demand by giving consumers more incentive to spend.

In the standard new Keynesian model, a cut in income taxes⁶⁸ does not affect the IS equation but affects the AS equation by increasing the incentive to work. The irrelevance of income taxes on the IS equation is not a general result. Yet, it is a useful one.⁶⁹ Income taxes play a constructive role when analyzing the impact of a reduction in government spending on the deficit because they are kept constant at their steady-state value, capturing the fact that tax revenues drop in proportion to the drop in the tax base, that is, output/income. Moreover, when analyzing the effect of supply shocks, they are a convenient reduced-form representation of general supply shocks, or what Chari, Kehoe, and McGrattan (2007) call a *labor wedge*, a shorthand label for several distortions and shocks.

In contrast to income taxes, sales taxes and government spending affect the IS and AS equations. The AS equation is affected by incentives to work: For example, high government spending in the short run absorbs more aggregate output at the expense of private consumption, thus increasing the marginal utility of income and growing people's eagerness to work. A cut in sales taxes works the same way. Neither of these supply effects is quantitatively important in the Great Recession and Great Depression calibrated examples.

We assume that all fiscal instruments except lump-sum taxes (which adjust in the background) are at steady state in the long run:

$$(47) \quad \hat{\tau}_t^s = \hat{\tau}_t^I = \hat{F}_t = 0 \text{ for } t \geq t_L.$$

However, they may now be adjusted in reaction to the shock *in the short run*:

$$(48) \quad \hat{\tau}_t^s \geq 0, \hat{\tau}_t^I \geq 0, \hat{F}_t \geq 0 \text{ for } t < t_L.$$

We define the fiscal multiplier under any policy regime in the first part of equation (49). In contrast, the expressions shown after the first and second equality are special to A1–A5:

$$(49) \quad M_F = \frac{E_0 \sum_{t=0} \beta^t \Delta \hat{Y}_t}{E_0 \sum_{t=0} \beta^t \Delta \hat{F}_t} = \frac{\frac{1}{1 - \mu\beta} \Delta \hat{Y}_s}{\frac{1}{1 - \mu\beta} \Delta \hat{F}_s} = \frac{\Delta \hat{Y}_s}{\Delta \hat{F}_s}.$$

⁶⁷ Defined as $\chi^s \equiv \frac{1}{1 - \bar{\tau}^s}$, $\chi^I \equiv \frac{1}{1 + \bar{\tau}^I}$ and $\psi \equiv \frac{1}{\sigma^{-1} + \omega}$. See footnote 45 for more details.

⁶⁸ Here, our definition of income includes labor income and firms' profits.

⁶⁹ It hinges on two assumptions. First, the new Keynesian agents are not borrowing constrained. In a model with constrained agents, income tax cuts will directly increase the spending of those constrained by a debt limit. Second, wages are assumed to be perfectly flexible, so variations in income taxes have a more substantial effect on inflation than in the case of rigid wages.

TABLE 2
GOVERNMENT SPENDING MULTIPLIERS

Multiplier	Great Recession		Great Depression	
	$\hat{i}_t = i^{zlb}$	$\hat{i}_t > i^{zlb}$	$\hat{i}_t = i^{zlb}$	$\hat{i}_t > i^{zlb}$
$\frac{\Delta \hat{Y}}{\Delta \hat{F}_S}$	1.1	0.3	2.2	0.2

A useful property of our assumption is that what the literature defines as the *dynamic multiplier* (the discounted sum in (49)) is identical to the *impact multiplier* (which is the last part of (49)) describing how much output increases *today* in response to government spending *today*.

Setting all the fiscal instruments in equation (48) to zero except for $\hat{F}_S$, we substitute (45)–(47) into equations (37) and (38) to obtain

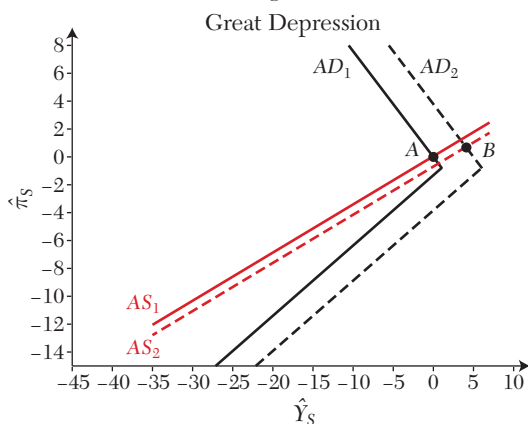
$$(50) \quad \frac{\Delta \hat{Y}_S}{\Delta \hat{F}_S} = \begin{cases} \text{if } \hat{i}_S \geq i^{zlb} : \\ \quad \mathbf{A1a} & \frac{(1 - \mu\beta)(1 - \mu) + \kappa\psi(\phi\pi - \mu)}{(1 - \mu + \sigma\phi_y)(1 - \mu\beta) + \sigma\kappa(\phi\pi - \mu)} < 1 \\ \quad \mathbf{A1b} & \sigma^{-1}\psi < 1 \\ \text{if } \hat{i}_S = i^{zlb} : \\ \quad \mathbf{A1a} \text{ or } \mathbf{A1b} & \frac{\vartheta + \sigma\mu(\sigma - \psi)}{\vartheta} > 0. \end{cases}$$

The quantitative effect of fiscal policy at positive interest rates depends on whether we specify the standard monetary and fiscal policy regime via a ZLB-constrained Taylor rule or a strict inflation target. The two are approximately equivalent, as reported in Table 2. As the value of the multiplier at positive interest rates is not the focus of this review, we refer readers to Woodford (2011) for an overview.

Using the Great Depression calibration, Figure 16 shows why the government spending multiplier is larger at the ZLB than at positive interest rates. Consider the two initial equilibriums depicted by point A, on the left for positive interest rates and the right for the ZLB. The figure shows the effect of an increase in government spending by 5 percent relative to output (which is similar in magnitude to the peak of the New Deal government spending). Both **AD** and **AS** shift rightward, resulting in a new pair of equilibriums, B. The figure shows that the expansionary effect of fiscal spending at the ZLB is larger than the same fiscal expansion at positive interest rates. In the Great Depression example, output increases by more than 10 percent in response at the ZLB and by about 0.1 percent at positive interest rates. At positive rates, the central bank increases the interest rate in response to the increase in government spending, thus raising the real interest rate, which reduces demand. In contrast, at the ZLB, the central bank does not raise the interest rate. Instead, government spending increases inflation expectations, thus reducing the real interest rate.

The two numerical examples illustrate a more general principle. As the recession becomes more severe (compare the Great Depression to the Great Recession in Table 2), the multiplier increases. This “divine coincidence” is emphasized by Christiano, Eichenbaum, and Rebelo (2011). In the formulas for both the multiplier (equation (30)) and the drop in output

Panel A. ZLB is not binding



Panel B. ZLB binding

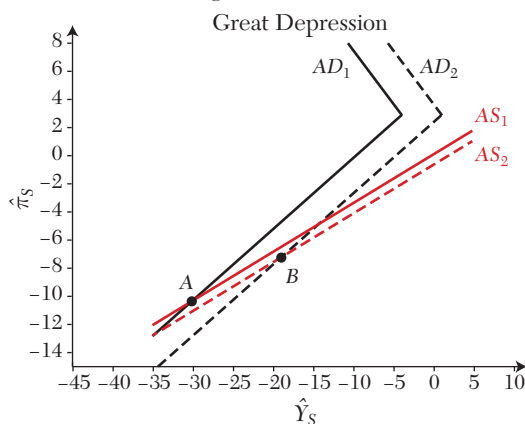


Figure 16. The Effect of an Increase in Fiscal Spending Using the Great Depression Calibration

(equation (39)), we see that lower ν will both reduce output and increase the multiplier. The multiplier is also increasing with higher interest rate elasticity or higher κ (higher price flexibility).

Understanding why government spending in the standard monetary and fiscal policy regime increases aggregate demand is straightforward. The first-order effect follows directly from the IS equation (29). Government spending increases aggregate spending as long as a reduction in private consumption doesn't fully offset it. This reduction in private consumption only occurs if the real interest rate increases, which happens under positive interest rates due to the central bank raising rates.

The dynamics are reversed at the ZLB. Government spending increases inflation expectations when the central bank is constrained and operating below both target output and inflation. Since nominal interest rates cannot fall further, higher inflation reduces real interest rates, stimulating demand through intertemporal substitution, which is a welcome development by the central bank. Additionally, expectations of higher future output stimulate current spending through the permanent-income hypothesis, as reflected in the expected future income term in equation (29). This creates a virtuous cycle that contrasts sharply with the deflationary spiral discussed earlier in response to shocks at the ZLB.

An increase in $\hat{F}_S$ not only increases aggregate demand but also shifts out aggregate supply (Figure 16). This happens because increased government spending diverts resources away from private consumption. The marginal utility of private consumption increases and increases labor supply, reduces wages, and shifts the **AS** curve out, since firms are willing to produce more at a given price level. The effect of the increase in aggregate supply is subtle at the ZLB: It is contractionary but is overwhelmed by the aggregate demand channel. We return to this paradoxical property of the model in Section 3.3.1.

As can be seen in the definitions of $\mathcal{F}_t^{IS}$ and $\mathcal{F}_t^{AS}$, sales tax cuts have the same effect as the increase in government spending once scaled by $\sigma\chi_s$. Thus, the discussion above remains largely unchanged. However, the economic mechanism by which these two policies increase demand differs, which may matter in practice. Sales tax cuts do not increase spending directly

through government consumption of goods and services. Instead, they increase private consumption by incentivizing consumers to spend more.

3.2.2 *The Fiscal Multiplier in Practice: Empirical Evidence at and Away from the ZLB*

This section briefly summarizes recent empirical studies on government spending multipliers at the ZLB and contrasts them to estimates at positive interest rates. Overall, the evidence suggests a multiplier at the ZLB around 1.5. Reviewing the large literature on government spending multipliers is beyond the scope of this paper and we focus on work most closely connected to the theme of this review.⁷⁰

Some care is needed in interpreting this evidence. Not all empirical studies make the nature of the policy regime as explicit as in we do in our Section 4.2 theoretical analysis. If monetary policy were to tighten at the ZLB in response to fiscal shocks (see especially section 4.2), the multiplier would be lower than we have derived in this section—typically lower than 1. Based on the theoretical findings we’ve reviewed, we interpret estimates of large fiscal multipliers at the ZLB (along with the negative impact of positive supply shocks as in Section 3.3) as suggesting that the actual empirical policy regime shares some key features with the standard monetary and fiscal policy regime. We revisit this topic detail in Section 4.2.2.

Considering the ZLB, empirical research generally finds multipliers around 1.5, significantly higher than the 0.3 to 0.8 range observed at positive interest rates. Studies such as those by Klein and Winkler (2021), who analyze data from 17 countries between 1917 and 2016; Miyamoto, Nguyen, and Sergeyev (2018) with Japanese data from 1980 to 2014; and Ramey and Zubairy (2018), focusing on the United States from 1889 to 2015, all report a ZLB multiplier in the vicinity of 1.5.⁷¹ Nakamura and Steinsson (2014) discuss an open economy multiplier also around 1.5, aligning with these findings under certain conditions. Chodorow-Reich (2019) finds that cross-sectional evidence points to a national multiplier of 1.7 or above, assuming that monetary policy does not offset the fiscal expansion, consistent with the standard monetary policy regime (what he terms “no monetary response”).

These studies report slightly higher multipliers than those suggested by models calibrated to the Great Recession shown in Table 2. Gordon and Krenn (2014) estimate the fiscal multipliers during 1940–41, leading up to World War II, around 2.5, modestly surpassing the multiplier implied by the Great Depression calibration reported in Table 2. We provide a possible explanation for why the model parameterization implies somewhat lower multipliers than the data for the Great Recession in Section 3.2.5. The standard new Keynesian model assumes that no agents are borrowing constrained, satisfying Ricardian Equivalence. As we will see, once this assumption is relaxed, we will arrive at a higher multiplier effect via income effects.

3.2.3 *Fiscal Consolidation at the ZLB: When Austerity Increases Rather Than Reduces Deficits*

An implication of large spending multipliers at the ZLB is that austerity policies that aim at balancing the budget via cuts in government spending or increases in sales taxes may backfire.

⁷⁰Recent overview articles include Ramey (2016), Nakamura and Steinsson (2018), and Ramey (2019).

⁷¹Ramey and Zubairy (2018) stress that this estimate excludes World War II rationing periods and large standard errors.

Large deficits following the Great Recession created political forces that drove David Cameron to declare the “age of austerity” in 2009. These political forces were present not only in the United Kingdom but also in most other industrial countries in the wake of the Great Recession. Government stimulus packages and the costs of the banking crisis drove the deficits. Consequently, numerous economists and politicians prioritized controlling public finances to “restore confidence” and facilitate a rapid recovery.

We define austerity as spending cuts or increased sales taxes, but both reduce output and the tax base (both used by the United Kingdom). To understand the effect of austerity on the deficit, we must be explicit about the government budget constraint. Consider a government that issues all its debt B_t in short-term nominal bonds. The government budget constraint is:

$$(51) \quad B_t = (1 + i_{t-1})B_{t-1} + (1 + \tau_t^s)P_t G_t - (\tau_t^I + \tau_t^s)P_t Y_t - P_t T_t^L.$$

B_t is the dollar value of the debt at time t , which is repaid with interest in the next period: $(1 + i_t)B_t$. It is convenient to write the government budget constraint in terms of its real value in period t inclusive of the interest rate as $w_t \equiv \frac{B_t}{P_t}(1 + i_t)$. Here T_t^L is a lump-sum tax, other fiscal instruments are defined following equations (45) and (46). A first-order approximation of the government budget constraint written in terms of w_t around some value $\bar{w}$ for the debt yields the following equation:

$$(52) \quad \beta \hat{w}_t = \hat{w}_{t-1} + \beta w_y \hat{y}_t - w_y \hat{\pi}_t + (1 + \bar{\tau}^s) \hat{F}_t - (\bar{\tau}^I + \bar{\tau}^s) \hat{Y}_t - \hat{T}_t^L - c_y \hat{\tau}_t^s - \hat{\tau}_t^I,$$

where both $\hat{w}_t$ and $\hat{T}_t^L$ are defined relative to steady state as a fraction of output and $c_y \equiv \frac{\bar{C}}{\bar{Y}}$. Recall that in the steady state, we approximate around $\beta = \frac{1}{1 + \bar{r}}$.⁷²

Up to this point, we have assumed that lump-sum taxes are adjusting in response to a particular fiscal policy. Due to Ricardian equivalence, the timing of lump-sum taxes is irrelevant, rendering the deficit indeterminate. We now assume lump-sum taxes in the short run are unchanged at the steady state:⁷³

$$(53) \quad \hat{T}_t^L = 0 \text{ for } t < t_L,$$

but in the long run, they adjust to stabilize public debt. That is, $\hat{T}_t^L$ adjusts so that

$$(54) \quad \hat{w}_t = \hat{w}_{t_L-1} = \hat{w}_L \text{ for } t \geq t_L.$$

As before, provided that other fiscal instruments remain constant and only lump-sum taxes adjust, the paths for inflation and output are determined independently of fiscal policy. It turns

⁷²That is, $\hat{T}_t^L = \frac{T_t^L - \bar{T}^L}{\bar{Y}}$ and $\hat{w}_t = \frac{w_t - \bar{w}}{\bar{Y}}$.

⁷³For technical reasons, we assume that lump-sum taxes are set so that public debt cannot exceed a debt limit, $\bar{w} > 0$, that can be arbitrarily high and is never reached in the simulation considered. By imposing this limit, we guarantee that lump-sum taxes are set so that the transversality condition of the representative household is always satisfied. The simplest way of ensuring this is to assume that the stochastic process governing the shock has some terminal date $T \geq t_L$ at which the shock reverts back to a steady state with probability 1.

TABLE 3 SHORT-RUN DEFICIT MULTIPLIERS OF GOVERNMENT SPENDING				
Multiplier	Great Recession		Great Depression	
	$\hat{t}_t = i^{zlb}$	$\hat{t}_t > i^{zlb}$	$\hat{t}_t = i^{zlb}$	$\hat{t}_t > i^{zlb}$
$\frac{\Delta \hat{D}_S}{\Delta \hat{F}_S}$	0.6	1.5	-0.3	1.1

out that the specification for the standard monetary policy regime Alb is more convenient for the characterization of equilibrium for what follows. The short-run deficit is then⁷⁴

(55)
$$\widehat{BD}_S = \beta w_y \hat{t}_S - w_y \hat{\pi}_S - (\bar{\tau}^I + \bar{\tau}^s) \hat{Y}_S + (1 + \bar{\tau}^S) \hat{F}_S - c_y \hat{\tau}_S^I - \hat{\tau}_S^I,$$

yielding the following expression for the deficit (given output and inflation, as determined by (39) and (40)) under the assumption that other fiscal variables are kept at a steady state:

(56)
$$\widehat{BD}_S = \begin{cases} 0 & \text{if } \hat{r}_S^e \geq i^{zlb} \\ \beta w_y i^{zlb} - w_y \hat{\pi}_S - (\bar{\tau}^I + \bar{\tau}^s) \hat{Y}_S & \text{if } \hat{r}_S^e < i^{zlb}. \end{cases}$$

Table 3 summarizes the effect of austerity measures implemented by cutting government spending. Implementing this form of austerity worsens the deficit in the Great Depression calibration but improves it using the Great Recession calibration. The effect of sales tax cuts has the same qualitative impact but with an opposite sign: Raising them reduces government revenues for the Great Depression calibration but raises them for the Great Recession calibration. Despite implementing austerity measures, several countries, such as Greece, Spain, and Portugal, initially experienced increased deficits, primarily explained by the collapse of their tax bases. A contraction in taxation revenues is consistent with the Great Depression numerical example shown in Table 3. In contrast, the United Kingdom’s fiscal austerity slowed growth but reduced the deficit, aligning more closely with the Great Recession numerical example in Table 3.

3.2.4 *Confidence Effects and Government Debt: The Nuanced Impact of Deficit Spending at the ZLB*

While we have seen that fiscal austerity can backfire and increase instead of reducing the deficit, our framework does support the idea that “confidence” may play a role at the ZLB, much beyond regular circumstances. This confidence effect can, in general, be offset at a positive interest rate, so it is special to the lower bound.

Debt accumulation resulting from the short-run fiscal policy can change expectations about long-run fiscal policy. Assessing the impact of a short-run fiscal stimulus requires considering

⁷⁴Given (33) and (54), the government budget constraint implies that $\hat{T}_L = (1 - \beta)\hat{w}_L = (1 - \beta)\hat{w}_{t_L-1}$.

its influence on expectations, or “confidence.” To model it, we abandon A3, according to which all fiscal instruments—except lump-sum taxes—remained fixed at their steady state in the long run. While isolating the specific effects of various fiscal instruments is convenient, it is unrealistic. The government cannot levy lump-sum taxation without incurring substantial costs.

For simplicity, we will assume policy regime A1b, where the central bank targets to set its policy rate to implement its inflation target, which it can achieve except if the ZLB is binding. In the absence of the ZLB, expectations about long-run values for the fiscal instruments are irrelevant for short-run output and inflation, as the central bank will offset any effect this long-run expectation may have on demand or supply in the short run. Visually, the downward-sloping arm of the AD curve above the ZLB kink becomes horizontal in Figure 16 at the inflation target where the AS and AD intersect. Since long-run taxes have no effect on the AS equation and monetary policy will offset any changes in demand to maintain the inflation target, long-run expectations play no role. The AD curve is horizontal and the ZLB not binding.⁷⁵ Hence, the confidence effect we will explore is special to the ZLB, as the monetary policy may not be able to offset changes in expectations about future fiscal policy.

If the ZLB is binding, then expectations about long-run variables start to have a significant effect, since the AS and AD equations intersect on the downward-sloping part of the AD curve. Monetary policy cannot offset changes in expectations about future fiscal variables and their possible effect on output.

Consider deviations of the fiscal instruments from the steady state in the long run. If the government runs a large deficit, it is reasonable to assume that people will change their expectations about future output, since it cannot finance itself with lump-sum taxation.⁷⁶ We also allow for long-run inflation to deviate from the original target. Following the same steps in deriving the short-run AD as in equation (37), while keeping track of long-run expectations, we obtain

$$\begin{aligned}
 (57) \quad \hat{Y}_S = & \underbrace{\frac{\sigma}{1-\mu} (i^{zlb} - \mu \hat{\pi}_S - \hat{r}_S^e) + \hat{F}_S - \sigma \chi^S \hat{\tau}_S^s}_{\text{Short-Run Demand Effects}} \\
 & + \underbrace{\left(\sigma + \frac{1-\mu\beta}{\kappa} \right) \hat{\pi}_L - \psi \chi^I \hat{\tau}_L^I - \omega \psi \chi^S \sigma \hat{\tau}_L^s + \omega \psi \hat{F}_L}_{\text{Long-Run Demand Effects}}.
 \end{aligned}$$

Assuming long-run debt is above steady state, Table 4 summarizes the effect of deficits on short-run demand under different assumptions about which variables are adjusted to pay interest on the debt in the long run. Denes, Eggertsson, and Gilbukh (2013) show some various numerical examples and simulation; we here summarize only qualitative features that public debt can have on future expectations, or “confidence,” depending on how it affects market expectations of future tax adjustment.

⁷⁵See the detailed discussion in Denes, Eggertsson, and Gilbukh (2013).

⁷⁶Our prior assumption was made for convenience so we could isolate the effect of each policy tax instrument abstracting from their impact on the budget constraint.

TABLE 4
EFFECT OF DEFICIT IN THE SHORT RUN ON AGGREGATE DEMAND
UNDER DIFFERENT ASSUMPTIONS OF WHAT VARIABLES ARE USED TO
PAY INTEREST ON THE DEBT IN STEADY STATE

Adjusting variable	$\hat{\tau}_L^l \uparrow$	$\hat{\tau}_L^s \uparrow$	$\hat{F}_L \downarrow$	$\hat{\pi}_L \uparrow$
$\frac{\Delta \hat{Y}_s}{\Delta \hat{w}_L}$	—	+	+	+

Reducing public debt can improve confidence primarily by creating expectations of lower future income taxes, which will induce people to spend more via the permanent income hypothesis. Meanwhile, if higher public debt generates expectations of higher long-run sales taxes, demand increases today to take advantage of lower sales taxes in the short run. Consider now higher debt triggering expectations of lower long-run government spending to finance interest rate payments. These expectations increase demand today, for it leaves more resources for private consumption in the long run, which will stimulate consumption spending today via the permanent income hypothesis. Finally, if higher debt triggers an increase in inflation expectations, this also increases demand in the short run.

The overall message is that policymakers should complement the effect of a short-run fiscal policy intervention on debt with a clear vision of its long-run impact on various tax instruments and public spending. In principle, the basic theory we have outlined can only answer what the effect will be, given the adjustment of different policy instruments. The model is silent, however, on which instrument will be adjusted. For this we need a theory of future governments’ policies. A more comprehensive model of government behavior is required to develop a positive theory of a policy regime that includes both short- and long-term policy instruments. We will adopt this approach in Section 4.

3.2.5 *HANK Models: How They Can Amplify Spending Multipliers and ZLB Paradoxes*

This section shows how borrowing-constrained consumers amplify fiscal policy multipliers using a simple HANK model. To do so we will rely upon a modest generalization of Eggertsson and Krugman (2012), presented in Section 2.1.3, which differs from the canonical new Keynesian model only in how the IS curve is written.

The resulting model captures key features of modern HANK models, which have received increasing attention in recent years. This allows us to connect our previous results with the original Keynesian multiplier. Under fixed prices, as in the original IS–LM model, the multipliers for fiscal spending and lump-sum taxes become identical to those in the “Keynesian cross” taught to undergraduates. This equivalence reveals that the fraction of borrowing-constrained consumers mirrors the role played by the Keynesian marginal propensity to consume—a crucial parameter for fiscal policy. The extension generates larger fiscal multipliers than our earlier derivations. It shows how lump-sum transfers (like COVID-19 stimulus checks) can become powerful policy tools, whereas they have played a trivial role so far in our theory.

Tax cuts generate direct demand effects when some households face borrowing constraints: Such households spend the entire tax cut immediately. This spending triggers a Keynesian

multiplier process, in which one household's spending becomes another's income, generating further rounds of spending (since constrained households consume all additional income). This cascade effect, first formalized by Keynes's student Kahn (1931), forms the foundation of fiscal multipliers, old and new, in the HANK tradition.

In Kahn's framework, which is still featured in modern undergraduate textbooks, households spend a fraction of c of their income (the marginal propensity to consume) and save the remainder $1 - c$.⁷⁷ Under rigid prices and a fixed interest rate, the government spending multiplier becomes⁷⁸

$$(58) \quad \frac{\Delta \hat{Y}_t}{\Delta \hat{F}_t} = \frac{1}{1 - c}.$$

This multiplier emerges from the Keynesian consumption function, in which consumption responds directly to disposable income. The same logic implies that tax cuts also affect output:

$$(59) \quad \frac{\Delta \hat{Y}_t}{\Delta \hat{T}_t} = \frac{c}{1 - c}.$$

Standard new Keynesian models, with their representative Ricardian household, imply that the time pattern of lump-sum taxes is irrelevant. Following Friedman's permanent income hypothesis (Friedman 1957), consumption reflects lifetime resources rather than current income. Yet these original Keynesian multipliers reemerge when we extend the new Keynesian model to include heterogeneous agents with differing degrees of patience and borrowing constraints.

We now return to the model of Eggertsson and Krugman (2012) from Section 2.1.3 and generalize it.⁷⁹ Instead of the number of borrowers and savers being the same, we assume that a fraction, χ , of households (which remain of mass 1) is more impatient than the remaining $1 - \chi$. We follow a useful simplification due to Werning (2015): zero liquidity limit in which the debt limit $\bar{D}$ approaches zero. Assuming identical income shares and lump-sum tax burden between agents ($\hat{Y}_t = \hat{Y}_t^b = \hat{Y}_t^s$ and $\hat{T}_t = \hat{T}_t^b = \hat{T}_t^s$),⁸⁰ aggregate spending is approximated by

$$(60) \quad \hat{Y}_t = (1 - \chi)\hat{C}_t^s + \chi\hat{C}_t^b + \hat{F}_t.$$

⁷⁷While Kahn (1931) distinguishes between propensities to consume from profits and wages, we follow the later literature in using a single MPC.

⁷⁸With consumption $\hat{C}_t = c(\hat{Y}_t - \hat{T}_t)$ and output $\hat{Y}_t = \hat{C}_t + \hat{F}_t$, substitution yields $\hat{Y}_t = \frac{1}{1-c}\hat{F}_t - \frac{c}{1-c}\hat{T}_t$. Each variable is expressed in deviation from steady state relative to aggregate output. While the IS-LM framework typically makes demand interest-rate sensitive, we abstract from this: It does not affect the multiplier computation under a fixed rate, but is often done in derivation of the Keynesian cross.

⁷⁹This approach builds on Campbell and Mankiw (1989) who interpreted the hand-to-mouth as representing people that do not participate in financial markets. Major contributors to integrate this perspective into the new Keynesian framework are Galí, López-Salido, and Vallés (2007) and Bilbiie (2008).

⁸⁰See Benigno, Eggertsson, and Romei (2020) for microfoundations to generate the result that each agent receives equal income.

Variables are expressed as deviations from their steady-state value relative to the steady-state output: $\hat{Y}_t \equiv \frac{Y_t - \bar{Y}}{\bar{Y}}$, $\hat{C}_t^j \equiv \frac{C_t^j - \bar{C}}{\bar{Y}}$ for $j = s, b$, and $\hat{F}_t \equiv \frac{F_t - \bar{F}}{\bar{Y}}$.⁸¹

Impatient agents, constrained by their borrowing limit, consume their entire disposable income:

$$(61) \quad \hat{C}_t^b = \hat{Y}_t - \hat{T}_t,$$

Patient agents (savers) follow the standard consumption Euler equation:

$$(62) \quad \hat{C}_t^s = E_t \hat{C}_{t+1}^s - \sigma(\hat{i}_t - E_t \hat{\pi}_{t+1} - \hat{r}_t^e),$$

where $\hat{i}_t$, $\hat{\pi}_t$, and $\hat{r}_t^e$ are expressed in log deviations from steady state as earlier in the paper. We focus exclusively on government spending and lump-sum taxes as fiscal instruments, with the latter now playing a central role in contrast to our previous analysis.

This extension leads to a major change in the IS equation relative to the canonical new Keynesian model. Combining the consumption behavior of all households with aggregate resource constraints yields the IS equation (63), now with heterogeneous agents. Assuming the A2 process for the efficient rate of interest, we solve for $\hat{Y}_s$ following the same steps as in deriving equation (36):

$$(63) \quad \hat{Y}_s = -\frac{\sigma}{1-\mu}(\hat{i}_s - \mu \hat{\pi}_s - \hat{r}_s^e) - \frac{\chi}{1-\chi} \hat{T}_s + \frac{1}{1-\chi} \hat{F}_s.$$

Relative to equation (36), the form of the equations is identical apart from that now government spending is multiplied by $\frac{1}{1-\chi}$, while lump-sum taxes affect aggregate demand proportionally to $\frac{\chi}{1-\chi}$.

The AS equation—the canonical new Keynesian Phillips curve in equation (32)—remains unchanged except that $\mathcal{F}_t^{AS} = -\kappa\psi\sigma^{-1}$.⁸²

To derive the multipliers at the ZLB, we first consider the limiting case in which $\kappa \rightarrow 0$ as in the original IS–LM model, which implies $\hat{\pi}_s = 0$. Under these conditions, we recover the old Keynesian multipliers:

$$(64) \quad \frac{\Delta \hat{Y}_t}{\Delta \hat{F}_t} = \frac{1}{1-\chi}, \frac{\Delta \hat{Y}_t}{\Delta \hat{T}_t} = \frac{\chi}{1-\chi}.$$

The multipliers match those of the Keynesian model when the fraction of constrained agents (χ) equals the marginal propensity to consume (c). This equivalence is somewhat intuitive: The fraction of the total households that will spend each additional dollar is χ , just as in Keynesian theory, then in the aggregate, a fraction c of every additional dollar will be spent, which corresponds to the marginal propensity to consume, c .

⁸¹In standard log-linearization, variables would be expressed as log deviations from their steady state (or percentage deviations). With such log deviations $\tilde{Y}_t = \gamma_C(1-\chi)\tilde{C}_t^s + \gamma_C\chi\tilde{C}_t^b + \gamma_F\tilde{F}_t$, where $\gamma_C \equiv \frac{\bar{C}}{\bar{Y}}$ and $\gamma_F \equiv \frac{\bar{F}}{\bar{Y}}$ are steady-state expenditure shares. Our notation avoids these weights, since to a first order $\gamma_C\tilde{C}_t^j = \frac{\bar{C}}{\bar{Y}}\frac{C_t^j - \bar{C}}{\bar{C}} = \frac{C_t^j - \bar{C}}{\bar{Y}} = \hat{C}_t^j$.

⁸²See microfoundation in Benigno, Eggertsson, and Romei (2020).

The complete solution for the government spending multiplier, accounting for policy regimes and that interest rate can be positive, we obtain

$$(65) \quad \frac{\Delta \hat{Y}_S}{\Delta \hat{F}_S} = \begin{cases} \text{if } \hat{i}_S \geq i^{zlb}: \\ \quad A1a & \frac{1}{1-\chi} \frac{(1-\mu)(1-\beta\mu)}{(1-\mu+\sigma\phi_y)(1-\beta\mu)} \\ \quad A1b & \sigma^{-1}\psi \\ \text{if } \hat{i}_S = i^{zlb}: \\ \quad A1a \text{ or } A1b & \frac{1}{1-\chi}. \end{cases}$$

This equivalence with Keynesian analysis is particularly relevant since traditional Keynesian models often assumed fixed interest rates.⁸³ The mapping between constrained agents, the ZLB, and the marginal propensity to consume provides microfoundations for the classic Keynesian multiplier. Yet, it also clarifies that Keynesian theory was less general than was typically appreciated when first formulated: At a positive interest rate, this multiplier logic breaks down completely, assuming the inflation targeting policy regime A1b. Under A1b, inflation is on target and the multiplier can be directly gauged from the AS curve. However, the policy regime specified in A1a does increase the multiplier because the central bank does not fully offset the fiscal expansion.

Table 5 presents spending and tax multipliers computed using parameter values from Table 1, now incorporating price adjustment ($\kappa > 0$).⁸⁴

We can apply the same parameterization as in Section 3.1.4. When all fiscal instruments remain at steady state, as assumed in the parameterization of the canonical new Keynesian model, the two models become identical: Equation (63) reduces to equation (36), and as noted, the Phillips curves match. This is a mathematical result: Monetary policy has identical aggregate effects across models. While perhaps surprising, this equivalence is well documented—Kaplan, Moll, and Violante (2018), who popularized the HANK terminology, highlight this result.⁸⁵

Fiscal policy, however, differs dramatically between the models: The ZLB multipliers scale by $\frac{1}{1-\chi}$ for spending and $\frac{\chi}{1-\chi}$ for lump-sum taxes. This raises the question: What is a plausible value for χ ?

⁸³In the first formal statement of the IS–LM model, Hicks (1937) discriminated between Keynesian and the “classics” by the fact that Keynesians assume fixed interest rates, assuming money demand depended on long-term rates that had a positive lower bound—see Eggertsson and Petracchi (2021) for discussion.

⁸⁴Below is the general solution when $\kappa > 0$ At the ZLB:

$$\hat{Y}_S = \frac{\sigma(1-\beta\mu)}{\vartheta} (\hat{r}_S^c - i^{zlb}) + \frac{1}{1-\chi} \frac{(1-\mu)(1-\beta\mu) - \sigma\mu\delta\kappa(1-\chi)}{\vartheta} \hat{F}_S - \frac{\chi}{1-\chi} \frac{(1-\mu)(1-\beta\mu)}{\vartheta} \hat{T}_S^b.$$

Above the ZLB:

$$\hat{Y}_S = \frac{1}{1-\chi} \frac{(1-\mu)(1-\beta\mu) + \sigma\delta\kappa(\phi_\pi - \mu)(1-\chi)}{(1-\mu+\sigma\phi_y)(1-\beta\mu) + \sigma\kappa(\phi_\pi - \mu)} \hat{F}_S - \frac{\chi}{1-\chi} \frac{(1-\mu)(1-\beta\mu)}{(1-\mu+\sigma\phi_y)(1-\beta\mu) + \sigma\kappa(\phi_\pi - \mu)} \hat{T}_S.$$

⁸⁵They note the “net effect of a monetary shock in HANK [is] quantitatively similar to RANK” (p. 799). Similar findings appear in McKay, Nakamura, and Steinsson (2016) and Gornemann, Kuester, and Nakajima (2016), though the rapid development of this literature may yet challenge these results.

TABLE 5
MULTIPLIER WITH DIFFERENT FRACTIONS OF BORROWING
CONSTRAINED AGENTS USING CALIBRATION FROM TABLE 1

		$\chi = 0$	$\chi = 0.25$	$\chi = 0.5$
<i>Great Recession</i>				
$\frac{\Delta \hat{Y}_s}{\Delta \hat{F}_s}$	$\hat{\imath} = i^{zlb}$	1.1	1.5	2.2
	$\hat{\imath} > i^{zlb}$	0.3	0.5	0.7
$\frac{\Delta \hat{Y}_s}{\Delta \hat{T}_s}$	$\hat{\imath} = i^{zlb}$	0	-0.4	-1.1
	$\hat{\imath} > i^{zlb}$	0	-0.1	-0.3
<i>Great Depression</i>				
$\frac{\Delta \hat{Y}_s}{\Delta \hat{F}_s}$	$\hat{\imath} = i^{zlb}$	2.2	3.3	5.4
	$\hat{\imath} > i^{zlb}$	0.2	0.3	0.4
$\frac{\Delta \hat{Y}_s}{\Delta \hat{T}_s}$	$\hat{\imath} = i^{zlb}$	0	-1.1	-3.2
	$\hat{\imath} > i^{zlb}$	0	-0.1	-0.2

Empirically reasonable values of χ substantially amplify fiscal policy effects. Our reading of the literature suggests that relatively generous bounds are $0.25 \leq \chi \leq 0.5$.⁸⁶ Table 5 reports multipliers for different values of χ . The resulting range of multipliers for the Great Recession aligns with empirical estimates, including Romer and Bernstein’s (2009) proposed values.

Our extension reveals a significant amplification of income effects in HANK models in response to fiscal policy intervention, even though we assume all households earn the same income. Even with this simple form of heterogeneity we find substantial increases in fiscal multipliers. Our findings align with two distinct traditions in the HANK literature. The first tradition, which our model is built on, emphasizes theoretical traceability with borrowing and lending, which is often traced to agents’ different impatience to consume.⁸⁷ Our result also aligns with the second tradition in the literature that typically is based on more complex micro-foundations in the spirit of Bewley (1977), where heterogeneity is driven by stochastically varying idiosyncratic productivity.⁸⁸

Although several studies find larger multipliers in heterogeneous agent models, others find more modest effects.⁸⁹

⁸⁶Campbell and Mankiw (1989), who first introduced “rule-of-thumb” consumers consuming their entire income, estimated $\chi = 0.5$ using time series data. The lower bound reflects more values of various papers and justifies them based on the Survey of Consumer Finances, with typical values in the range of 0.25 and 0.4.

⁸⁷Our extension builds directly on Eggertsson and Krugman (2012) and Benigno, Eggertsson, and Romei (2020), but a similar model is also studied in Bilbiie (2025) using simple form heterogeneity like we do, see also Cúrdia and Woodford (2010).

⁸⁸In the context of the ZLB, this approach was pioneered by Guerrieri and Lorenzoni (2017), who emphasize the link between heterogeneity in amplifying the size of multipliers.

⁸⁹For some recent examples from the literature see Bayer, Born, and Luetticke (2024) and Auclert, Rognlie, and Straub (2024), who find that heterogeneity amplifies the effect of fiscal spending via the income effect. Others identify conditions that produce smaller effects, such as Hagedorn, Manovskii, and Mitman (2019). Ravn and Sterk (2021) show multipliers can go either way without taking a strong position on the overall effect. McKay and Reis (2016), who gave us an early HANK

Based on our analysis of mechanisms and evidence in Section 3.2.2, we tentatively conclude that the traditional income effect increases fiscal policy impact while acknowledging that this question remains open in a rapidly evolving literature.⁹⁰

3.2.6 Assessing Robustness: Beyond the Stochastic Two-State Markov Process

So far, we have focused on a special two-state Markov process for the efficient rate of interest, which has allowed us to express the solution in closed form. Here, we show that the results are robust to assuming an autoregressive process of order 1 (AR(1)), which is common in the literature.

While replacing two-state Markov with an AR(1) process produces the same key conclusion, we uncover additional insights into the transmission mechanism, building on Erceg and Lindé (2014). We focus not on re-parameterizing the model with Bayesian methods as in previous examples, but on demonstrating that our key findings are robust to this alternative stochastic process and highlighting a few new insights.

We analyze fiscal multipliers in the canonical new Keynesian model under the standard monetary and fiscal policy regime, now adopting an AR(1) process for $\hat{r}_t^e$ (setting aside the borrowing constraints from the previous section).⁹¹ We specify the efficient rate as

$$(66) \quad \hat{r}_t^e = \rho \hat{r}_{t-1}^e + \varepsilon_t^r,$$

and assume an example of the fiscal shock follows the same persistence parameter:

$$(67) \quad \hat{F}_t = \rho \hat{F}_{t-1} + \epsilon_t^F.$$

At time zero, there is a shock to the efficient rate of interest, ε_t^r , and fiscal spending, ϵ_t^F . By choosing the same persistence parameters, ρ , we maintain our key assumption from the two-state Markov case under the standard monetary and fiscal policy regime: Fiscal spending reacts directly to exogenous shocks.

We maintain the standard policy regime (A1a) but now explicitly incorporate the natural rate of interest and output in the Taylor rule, a feature we could ignore in our previous analysis due to the discrete jump in the shock. We write the Taylor rule as

$$(68) \quad \hat{i}_t = \max\left(\hat{i}^{zlb}, \hat{r}_t^n + \phi_\pi \hat{\pi}_t + \phi_y (\hat{Y}_t - \hat{Y}_t^n)\right),$$

where we express the natural rates of interest and output as

$$(69) \quad \hat{r}_t^n = \hat{r}_t^e + \frac{\sigma^{-1}\omega}{\sigma^{-1} + \omega}(\hat{F}_t - \hat{F}_{t+1}) \quad \text{and} \quad \hat{Y}_t^n = \frac{\sigma^{-1}}{\sigma^{-1} + \omega}\hat{F}_t.$$

model, emphasized automatic stabilizers, while McKay, Nakamura, and Steinsson (2016), with another early HANK model, briefly noted that heterogeneity could amplify fiscal spending effects.

⁹⁰For an alternative treatment of fiscal multipliers that covers heterogeneity (and the open economy dimension, which we do not address here), we direct readers to a review by Farhi and Werning (2016).

⁹¹We set $\sigma = 2.35$, $\kappa = 0.05$, $\rho = 0.9$. For future research, we will use a unified estimation approach to compare the two stochastic processes.

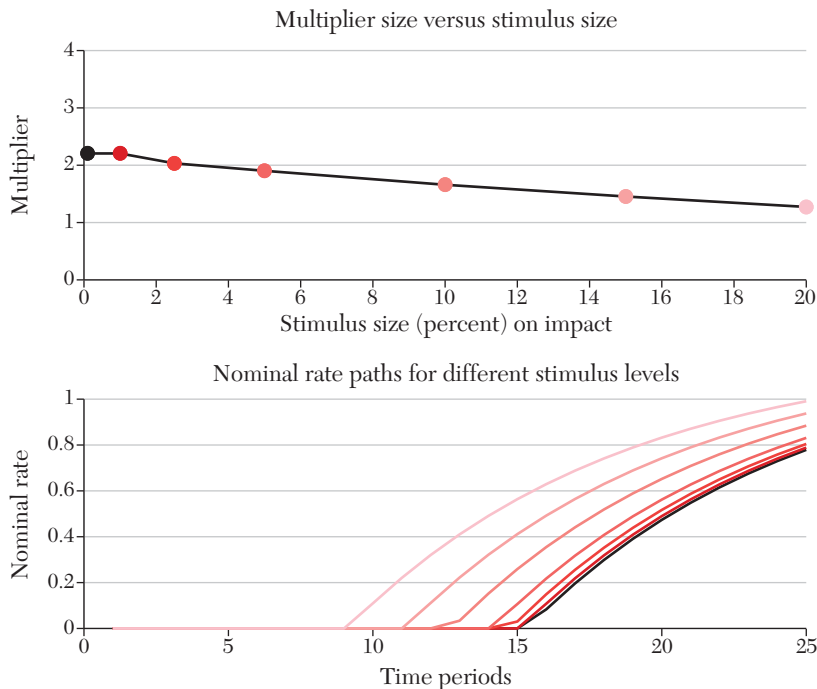


Figure 17. Multipliers and Nominal Rate Paths for Different Sizes of Fiscal Spending

The ZLB ceases to bind when $\hat{r}_t^n > i^{zlb}$. We illustrate these dynamics in Figure 17. As the lower panel shows, under the standard policy regime, the timing of lift-off from the ZLB now responds endogenously to the path of fiscal policy.⁹² This endogenous response does not reflect forward guidance, but is a mechanical consequence of the policy rule: Fiscal policy increases the natural interest rate, thereby hastening lift-off from the ZLB.

This endogenous dynamic does not dramatically alter the multiplier's value. In Figure 17, we present results under the standard policy regime with an AR(1) process. The top panel shows how the multiplier varies across different fiscal spending paths, which we index by the initial fiscal shock size, $\hat{F}_1$, ranging from $\epsilon > 0$ to 20 percent of GDP. In our illustrative example, a fiscal package of $\hat{F}_1 = 1$ percent of GDP yields a multiplier of 2.2—matching our calibration to the Great Depression. As we increase the size of the fiscal package, the multiplier decreases, although modestly. Following Erceg and Lindé (2014), this decline reveals diminishing returns: Each additional dollar of fiscal stimulus generates a smaller impact. Yet the decline is gradual, with the multiplier staying above 1 for realistic stimulus sizes.

In the bottom panel, we illustrate this result by showing nominal interest rate paths under the standard policy regime for each fiscal package examined. Larger fiscal packages shorten the ZLB duration. The mechanism is intuitive: Higher fiscal spending increases the natural

⁹²This continuous adjustment contrasts with our analysis assuming the Markov stochastic process. Conditional on the ZLB being binding in the last period, exit occurred with probability $1 - \mu$, when the natural rate of interest reversed to an absorbing steady state. We restricted attention to fiscal interventions that were too small to trigger an immediate exit.

rate of interest (see equation (69)), pushing it closer to i^{zlb} . As the fiscal stimulus grows, the time the economy spends at the ZLB decreases. This dynamic reduces the effectiveness of additional spending as the fiscal package grows. With a sufficiently large fiscal package, the economy avoids the ZLB entirely.

Our results suggest that the large multipliers under the standard policy regime are not special to the baseline two-state Markov stochastic process. The multipliers remain large when we adopt alternative shock structures and allow for endogenous lift-off timing that varies with fiscal spending paths. All paradoxes and key findings from our previous analysis carry over to this alternative stochastic process. This result aligns with Christiano, Eichenbaum, and Rebelo (2011), who examine multipliers in a more complex setting than ours, a medium-scale DSGE model with autoregressive shock structure, and standard quantitative features, such as capital accumulation and adjustment costs.

3.2.7 Eliminating the ZLB: From Gesell's Tax to Digital Currency

Theoretically, the government can, with a rich enough set of policy tools, either in the sphere of taxation or via reforming the payment system, eliminate the ZLB problem. One central challenge is political. Another is robustness; a policy may solve the ZLB in one model but become counterproductive in another. While it seems likely that these constraints can be overcome at some future date, if ZLB is sufficiently destabilizing there is little to indicate that, as of yet, central banks or governments are working toward a practical solution.

Gesell's Tax on Money, Digital Currency, and Negative Central Bank Policy Rates.—The earliest proposal for using tax policy to circumvent the ZLB is due to Gesell (1916). His idea was to impose a tax on currency. While theoretically sound, this proposal has never been implemented. Instead, several central banks experimented with negative policy rates during the Great Recession. However, both theory and empirical evidence provide mixed results regarding the effectiveness of negative policy rates.

As we discussed in Section 2.2, the ZLB is due to paper currency that has a fixed nominal value. Gesell's proposal, more than 100 years before the ZLB had ever posed a problem, was that the government could impose a tax, or carrying cost, on holding paper currency. He suggested implementing it by requiring those holding currency to obtain periodic stamps on their paper currency for a fee for it to retain its value. If you pay a fee for carrying paper currency, you will be willing to lend one dollar, even if you obtain less than one dollar back, provided the borrower pays the fee for the money to retain its value. This is equivalent to a negative nominal interest rate on loans.

Gesell's idea was revived by Goodfriend (2000) and Buiter and Panigirtzoglou (2003). These authors emphasize that while taxing reserves is feasible, taxing currency would involve significant administrative costs. Rogoff (2017) advocates eliminating most paper currency, particularly high-denomination bills. The most straightforward way to do this is if all currency is electronic. In this case, the ZLB can be violated; negative interest rates can be charged on all electronic currency, eliminating the problem. Yet, more complicated alternatives have been suggested, which allow for a simultaneous trading of paper currency and digital currency.⁹³

⁹³ Agarwal and Kimball (2015) propose a monetary system in which the value of physical cash and electronic money changes over time, creating an exchange rate between the two. This arrangement can implement a negative nominal

While these ideas have yet to retain major traction by policymakers, several central banks (in Switzerland, Japan, the eurozone, Denmark, and Sweden) have experimented with negative policy rates after the Great Recession. Here, we use the term “policy rates” to refer to the interest rates paid on the reserves of commercial banks. Private banks hold reserves at the central bank to settle transactions with one another (interbank transactions). Banks are willing to pay for these services. This makes it feasible for the central bank to charge negative interest rates.

However, the central banks’ ability to lower interest on reserves is limited, since banks can exchange reserves for paper cash. For sufficiently negative interest rates, banks can settle interbank transactions outside the central bank. This generates a bound on how a negative interest rate can be applied on reserves, as shown in Eggertsson et al. (2024), or what they call the *policy rate lower bound* on the interest rate.

The typical transmission mechanism of monetary policy works as follows: The central bank cuts its policy rate, which passes through to commercial banks’ financing costs (largely funded via deposits), which in turn gets transmitted to various loan interest rates offered to bank customers. These lower interest rates induce customers to spend more, stimulating the economy. However, Eggertsson et al. (2024) show that this relationship breaks down both empirically and theoretically with negative policy rates, as commercial banks appear unwilling to impose negative rates on depositors, fearing customers would withdraw balances and hold paper currency instead. This gives rise to a *deposit lower bound*. Meanwhile, negative policy rates do get transmitted to highly liquid assets, such as government bonds, which banks hold to insure against liquidity risk. As these assets carry negative rates, they essentially act as a tax on banks’ net worth. Taken together, this implies that negative policy rates may not be transmitted to customer loans. Indeed, in Sweden, Eggertsson et al. (2024) find that negative policy rates did not lower mortgage rates (unlike typical policy rate cuts) but modestly *increased* them.

Abadi, Brunnermeier, and Koby (2023) identify a *reversal rate* below which rate cuts are no longer expansionary due to their effect on banks’ balance sheets, estimated at -0.8 percent. In contrast, Eggertsson et al.’s (2024) model implies a 0 percent reversal rate. In summary, the literature suggests that in the presence of paper currency, there is a lower bound at which policy rate cuts are no longer effective, even if a consensus has not been reached on the exact value of this reversal rate.

To summarize, central banks can, in principle, eliminate the ZLB through technological innovation. The most apparent option is abandoning paper currency and introducing digital currency with flexible interest rates that can turn negative. One can imagine, for example, a central bank offering a “cash balance” account for every individual with a social security number, with a similar implementation for legally registered firms. In the United States, in practice, this is equivalent to extending the type of reserve accounts commercial banks already hold at the Federal Reserve to all citizens and legitimate businesses. The US government started issuing noninterest-bearing paper currency in 1861. In modern societies, no technological hurdle prevents governments from issuing digital currency. They already do so with reserve accounts, which exist only as accounting entries in computer systems and can carry either negative or positive interest rates.

interest rate on paper currency. Although this approach is theoretically feasible, communication and operational challenges have prevented central banks from experimenting.

The main hurdles to implementing such drastic changes to the monetary system lie in political constraints and the public's demand for anonymity in financial transactions. Many voters strongly prefer the anonymity afforded by cash payments instead of making payments with any form of currency that is traceable.⁹⁴ Paper currency may also have advantages in extreme circumstances, for example, if computer systems or the electric grid collapses. The trade-offs between implementing radical changes to the payment system to eliminate the ZLB are primarily determined by how frequently ZLB imposes a meaningful constraint on policy in the future. There is no consensus on the answer to that question today. Our prediction, however, is that sooner than later paper currencies will be replaced by other forms of payment. In designing such payment systems, it is urgent that policymakers provide sufficient flexibility so that interest rates can turn negative.

Using Multiple Tax Instruments Simultaneously.—Using a combination of separate tax instruments can circumvent the liquidity trap. This policy faces two challenges: The first is institutional constraints, and the second is that while this type of proposal may work in one plausible model, it may backfire in another equally plausible one.

Feldstein (2002) is the first to suggest that Japan suspend its sales tax (τ_t^s) and gradually lift it to its previous level, making up revenue loss by raising income taxes (τ_t^I) (Feldstein 2001). Eggertsson and Woodford (2004) formalize Feldstein's proposal as

$$(70) \quad \hat{\tau}_S^s = \frac{(\chi^s)^{-1}}{1 - \mu} \hat{r}_S^e < 0,$$

$$(71) \quad \hat{\tau}_S^I = -\frac{\chi^s}{\chi^I} \hat{\tau}_S^s > 0.$$

Adjusting the two tax instruments completely offsets the shock and prevents output contraction in the model we have considered under A1–A5. Intuitively, the commitment to raise consumption taxes works via the exact mechanism of increasing inflation expectations at constant interest rates. Correia et al. (2013) extend this result to a more general setting, including a model with capital. With a rich enough tax structure, negative natural rate shocks can typically be offset.

Eggertsson (2004) examines whether fiscal policy can substitute for monetary policy in a monetary union (which may or may not be subject to the ZLB) through a simple rule: having fiscal spending follow each country's specific natural interest rate. If national interest rates differ across countries, for example, due to imperfect financial integration as in Eggertsson et al. (2016), then the monetary union cannot replicate each country's natural rate.

However, local fiscal authorities can raise the domestic natural rate of interest using fiscal instruments to match the union-wide natural rate, thus stabilizing domestic inflation

⁹⁴While paper currency provides anonymous transactions, digital currencies create traceable records. Central banks' attempts to limit paper currency, even partially, have often faced strong opposition. The ECB's May 2016 decision to discontinue producing €500 notes (while maintaining existing ones as legal tender) proved highly controversial despite citing illegal activity concerns. Similar policies met varying resistance elsewhere: The Bank of Canada faced limited opposition when discontinuing \$1,000 notes in 2000, while the US decision to stop issuing \$500, \$1,000, \$5,000, and \$10,000 bills in 1969 was largely uncontroversial. These notes remain legal tender but are rarely seen and never reissued.

and output.⁹⁵ This insight is corroborated and shown to apply in a richer setting by Farhi, Gopinath, and Itskhoki (2014), who refer to this ability of fiscal authorities to make up for lack of ability to react to shocks with a common monetary policy as fiscal devaluations.

Tax policy offers powerful tools for addressing the ZLB, but important caveats limit Feldstein's insight in practice. First, as Eggertsson and Woodford (2004) emphasize, implementation may require negative sales taxes. Second, implementation details depend on the interactions between firm pricing strategies and tax policies. If the price the firm sets includes the tax in the final price, this has different implications than if the price the firm sets does not include the tax. Eggertsson and Woodford (2004) argue that the former assumption is more appropriate for value-added taxes common in Europe (where prices posted typically include the tax). At the same time, the second is more suitable for sales taxes as in the United States (where prices are quoted commonly excluding the sales tax).

Third, and most fundamentally, the effectiveness of tax policies at the ZLB depends on model specifications. Lancaster (2017) challenges Feldstein's core proposal by showing that introducing liquidity-constrained agents fundamentally alters the results: In an overlapping generations model with nominal frictions and borrowing constraints, the policies in equations (70) and (71) deepen the recession rather than circumventing the ZLB. While reforming the tax system to respond to ZLB-inducing shocks could yield substantial benefits, institutional constraints and incomplete understanding of the current economic impacts limit practical implementation.

3.2.8 *Resolving Contradictions in the Literature*

The literature on fiscal policy at the ZLB in new Keynesian models begins with Eggertsson (2001a, b) and Christiano (2004), who first demonstrate the significant role of fiscal policy.⁹⁶ Our analytical results in Section 3.2 build primarily on Eggertsson (2010) and Denes, Eggertsson, and Gilbukh (2013). Christiano, Eichenbaum, and Rebelo (2011) extend the analysis to a medium-scale quantitative model, finding large fiscal multipliers. Woodford (2011) provides a comprehensive overview, including multiplier determinants at positive interest rates—a topic beyond our current focus. Farhi and Werning (2016) further broaden the analysis to include open economies and non-Ricardian agents, thus incorporating similar effects as in the HANK model.

Some papers find smaller multipliers, notably Cogan et al. (2010) and Drautzburg and Uhlig (2015). These differences stem from their assumptions about the timing of spending increases and future offsetting distortionary taxes. This analysis relates closely to our discussion in Section 3.2.4, which highlighted that the effect of short-run fiscal policy is highly contingent on its impact on the budget deficit and, if negative, how this would affect expectations about future taxation, or what we term confidence at the ZLB in Section 3.2.4. Once we account for these timing differences, the common message emerges: Clear long-run fiscal frameworks are crucial to complement short-run interventions.⁹⁷

⁹⁵ See discussion in section 4 in Eggertsson (2004) that elaborates on this.

⁹⁶ These citations refer to the working paper version of this work, which later found its way into several papers by these authors discussed in detail in this review.

⁹⁷ For example, permanent spending increases that violate A3 affect only the AS equation, having no impact on the IS equation (29). Coenen et al. (2012) demonstrate this distinction across several models, finding large multipliers for temporary interventions but small ones for permanent increases.

Several papers challenge the findings about large multipliers and the paradoxes presented in this section. Subsequent research over the past decade has resolved the contradictory findings.

Boneva, Braun, and Waki (2016) argue that the results we review under the standard monetary and fiscal policy regime are artifacts of log-linearization in new Keynesian models by solving a model with Rotemberg pricing using nonlinear methods (Rotemberg 1982). However, Eggertsson and Singh (2019) demonstrate the robustness of these results using a closed-form nonlinear model with Calvo pricing or, alternatively, assuming Rotemberg pricing once implausibly high costs of changing prices are removed from the resource constraint.⁹⁸

Kiley (2016) suggests that sticky information models overturn the results and that they are an artifact of the nature of the assumed price setting. Eggertsson and Garga (2019) demonstrate that this difference emerges because Kiley's ZLB scenario differs from more standard assumptions in the literature: the standard monetary and fiscal policy regime and conditions analogous to A1–A5. They show that under A1–A5, sticky information models predict even higher multipliers and stronger paradoxes than sticky price models.

Finally, Cochrane (2017) attributes large multipliers and related paradoxes to arbitrary “equilibrium selection” and proposes alternative equilibria he views as more plausible in which there are no paradoxes and the multiplier of government spending is low. Section 4.2 shows that his preferred equilibrium selection implies a regime change—to the one that closely approximates optimal policy under commitment (detailed in Section 4 with implications for multipliers and paradoxes discussed in Section 4.2), rather than the standard monetary and fiscal policy regime, which is the focus of this section. We discuss the empirical plausibility of alternative policy regimes during the Great Recession in Section 4.2.2.

3.3 *Supply-Side Policies at the ZLB within a Fixed Policy Regime: The Paradox of Toil*

We now turn from aggregate demand to supply. The paradox of toil, a startling result of modern liquidity trap literature, suggests that *positive* supply shocks *contract* output at the ZLB, a prediction supported by recent empirical evidence. The paradox implies that structural reforms proposed in Europe during the Great Recession might have been counterproductive at the ZLB. At the same time, the much-maligned National Industrial Recovery Act of 1933, implemented by President Roosevelt as part of the New Deal, can be cast more positively, despite virtually uniform condemnation from economists across the spectrum. Notably, Keynes himself wrote an op-ed in the *New York Times* on December 31, 1933, warning against the policy. Keynes's argument was that demand policies, not supply restrictions, were the key to recovery and that to think otherwise was “a technical fallacy” related to “the part played in the recovery by rising prices.” Keynes's logic will be recognized by a modern reader as a basic IS–LM argument, which by construction ignored any effect expectations might play, which in turn will play a central role in this section.⁹⁹

⁹⁸Eggertsson and Singh (2019) show that Boneva, Braun, and Waki's (2016) different results stem directly from their assumption of implausibly large quadratic price adjustment costs in their nonlinear model. Once these costs are removed from the aggregate resource constraint, their model produces nearly identical results to the nonlinear Calvo model.

⁹⁹See Eggertsson (2012a).

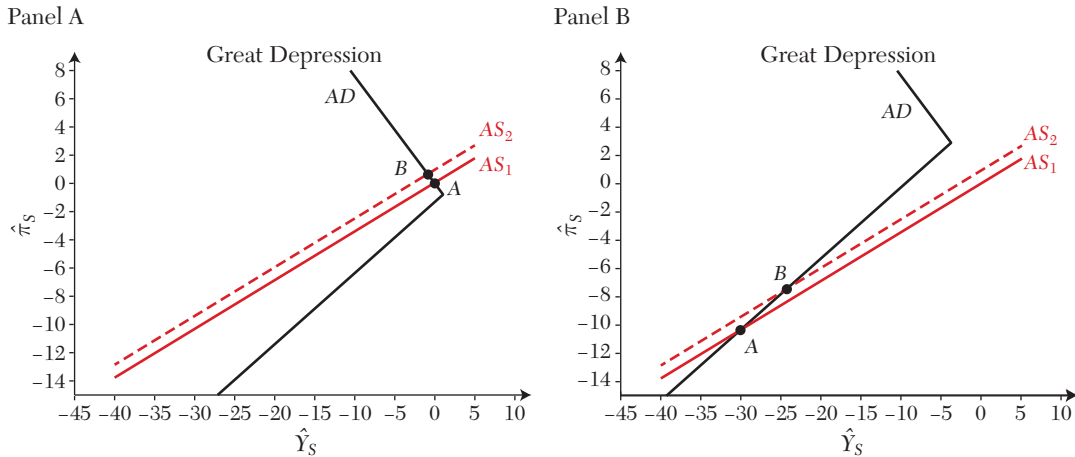


Figure 18. The Effect of a Negative Supply Shock at a Positive Interest Rate (Left) and the ZLB (Right)

3.3.1 The Paradox of Toil: Theory

We start by stating the paradox of toil via the following thought experiment:

THE PARADOX OF TOIL: Imagine you wake up one day and decide to work more. You start looking for extra jobs and, if lucky, find some (partial equilibrium). Now imagine everyone waking up with the same idea. At first, it may seem that everyone will work at least as much as before. However, the paradox of toil shows that *at the ZLB*, everyone wanting to work more may actually result in less work in general equilibrium.

This paradox is a classic fallacy of composition, just like Keynes's paradox of thrift. The fallacy of composition is the presumption that what is true for a single individual can be extrapolated to the population. For example, if one person stands up at a concert, he or she can see better, but if everyone stands up, nobody's view improves.¹⁰⁰ Our depiction of the paradox is a simplification, for it depends in a subtle way on expectations, which requires elaboration.

The left part of Figure 18 shows that normally in response to a negative supply shock (imagine, for example, an increase in oil prices) a central bank raises rates and reduces aggregate demand. At the ZLB, however, the right part of the figure shows the central bank will not raise the rate in response to inflationary pressures stemming from a negative supply shock. A negative supply shock at the ZLB triggers an increase in inflation and, more importantly, an increase in inflation expectations, which reduces the real interest rate and increases demand. With inflation below target, this is exactly what the central bank wants.

¹⁰⁰We do not state the paradox of thrift here, which is well known, as it involves introducing capital, which we abstract from here; see Eggertsson (2010) for this extension.

To characterize the paradox, we adopt the notation $\hat{\omega}_t \equiv \psi \chi^I \hat{\tau}_t^I$ to emphasize the interpretation that $\hat{\tau}_t^I$ is a stand-in for a large range of supply disturbances, commonly known as the labor wedge. It is straightforward to solve equations (37) and (38) together so that A1–A5 lead to

$$(72) \quad \frac{\Delta \hat{Y}_S}{\Delta \hat{\omega}_S} = \begin{cases} \text{if } \hat{i}_S \geq i^{zlb}: \\ \quad \text{A1a} & -\frac{\sigma(\phi_\pi - \mu)}{(1 - \mu + \sigma\phi_y)(1 - \beta\mu) + \sigma\kappa(\phi_\pi - \mu)} < 0 \\ \quad \text{A1b} & -1 < 0 \\ \text{if } \hat{i}_S = i^{zlb}: \\ \quad \text{A1a or A1b} & \frac{\sigma\mu\kappa}{(1 - \mu)(1 - \beta\mu) - \sigma\mu\kappa} > 0. \end{cases}$$

3.3.2 The Paradox of Toil: Empirical Evidence

Recent empirical evidence suggests that the effects of oil price changes on key aggregate variables switch signs once the ZLB is binding, as predicted by the paradox of toil.

Datta et al. (2021a) show that the correlation between daily oil and equity prices flips from negative to positive at the ZLB, as shown in Figure 19.¹⁰¹ They also find that oil and equity returns become more responsive to macroeconomic news surprises at the ZLB, in contrast to historical experience. This is consistent with the idea that the ZLB prevents the central bank from offsetting the shock.

Miyamoto, Nguyen, and Sergeyev (2024a) use high-frequency data to identify the effect of oil prices on industrial production and unemployment. Following Känzig (2021), they identify the shocks using the change in oil futures prices in a tight window around Organization of the Petroleum Exporting Countries (OPEC) production announcements. The series of future oil price changes then becomes an external instrument in a VAR, allowing the authors to tightly estimate the impact of structural oil shocks on industrial production and unemployment. As shown in Figure 20, the result of this analysis suggests that the response of unemployment and industrial production to oil shocks changes sign in both the United States and Japan. Positive supply shocks are expansionary at a positive interest rate but negative at the ZLB. This change is statistically significant.

These papers build on Wieland (2019), which proposes a test for the existence of the paradox empirically. First, he uses the Great East Earthquake in Japan as a natural experiment to test the paradox. Yet, this experiment is not ideal for testing the paradox, since it is a one time shock, not a sequence of deflationary shocks as we have considered in constructing the paradox, which triggers deflationary expectations over some horizon in the future. Second, he considers the effect of oil prices on output, which is a better-designed test. While Wieland (2019) point estimates for oil shocks at the ZLB have the opposite sign of Miyamoto, Nguyen, and Sergeyev's (2024a), the results are not inconsistent with each other. Wieland (2019) reports that the difference between the impulse responses estimated at the ZLB and positive interest rates is not statistically significant. A possible interpretation is that Miyamoto, Nguyen, and Sergeyev's (2024a)

¹⁰¹The data used in this figure is digitized.

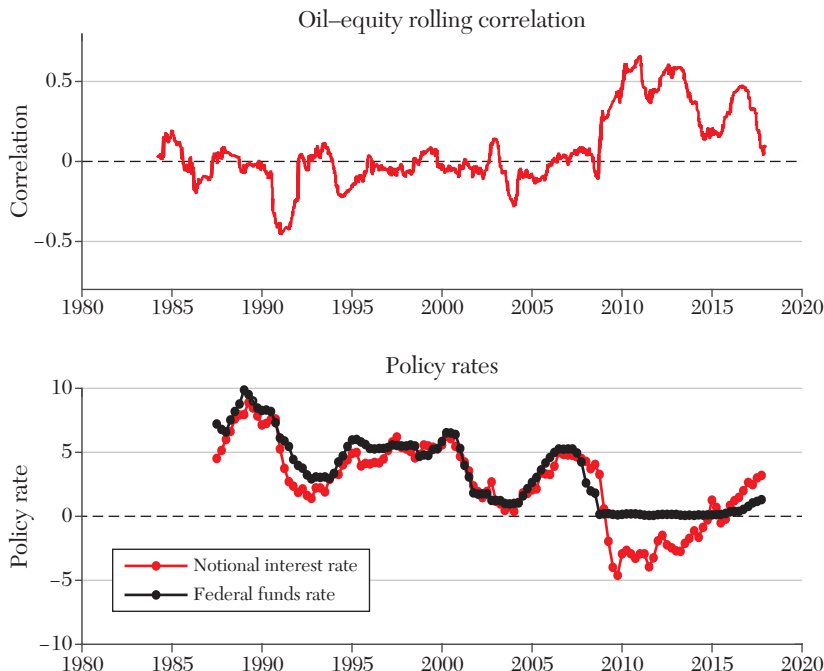


Figure 19. Daily Oil–Equity Correlation and Policy Rates

Sources: Datta et al. 2021a, b; Board of Governors of the Federal Reserve System 2025a.

high-frequency identification of oil shocks gives them more statistical power, as they are able to establish a statistical difference between the impulse response functions.

One of the central issues about whether one should expect the paradox to arise is the extent to which the underlying shock is permanent or temporary. This will be particularly transparent in the next subsection in the context of structural reforms in Europe that were discussed during the Great Recession. A central issue is if the public expects the policy to be temporary or permanent.

3.3.3 *The Great Recession and the Debate on Structural Reforms in Europe*

In this section we explore the implications of the paradox of structural reforms.

During the Great Recession, the European Central Bank, like the Federal Reserve, was constrained by the ZLB. The main policy advice for southern eurozone countries lacking fiscal space to stimulate demand was structural reforms. However, the paradox of oil suggests that these policies could have resulted in counterproductive short-run effects.

We define structural reforms as policies that increase production capacity, or alternatively, policies that reduce monopoly power of firms or worker unions. While these policies are clearly growth enhancing under normal circumstances, the paradox of oil suggests they may trigger deflationary effects at the ZLB, which, if sufficiently strong, suggest the policy may backfire.

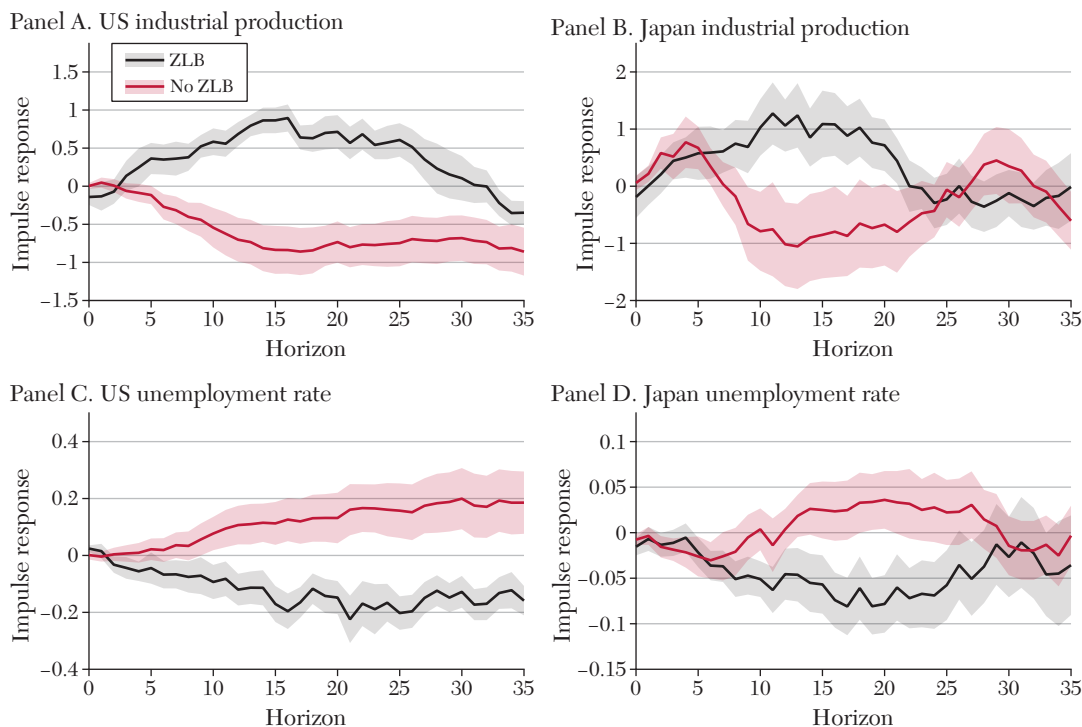


Figure 20. Estimated Impulse Response Functions of Industrial Production and Unemployment in the United States and Japan to Positive Oil Price Shocks

Notes: Shaded areas show 68 percent confidence intervals.

Source: Miyamoto, Nguyen, and Sergeyev (2024b).

Following the Great Recession, the scope for fiscal or monetary expansion was challenging, especially in the southern part of Europe. Austerity measures had to a large extent failed, while the short-term interest rates remained at zero. With limited fiscal space, the European Commission started suggesting to these countries that the main avenue for growth was structural reforms. These reforms aimed to increase competition in product and labor markets, as evidence suggests less competition in the service sector in the southern periphery than at the core, as discussed in Eggertsson, Ferrero, and Raffo (2014).¹⁰²

Structural reforms have positive long-term effects on growth and demand but also short-term deflationary effects. Consider a supply shock with effects in both the short and long run, that

¹⁰²For instance, in his closing remarks following the 2012 State of the Union, J.M. Barroso, the president of the European Commission at the time, stated, “The biggest problem we have for growth in Europe is the problem of lack of competitiveness that has been accumulated in some of our Member States, and we need to make the reforms for that competitiveness. ... to get out of this situation requires ... structural reforms, because there is an underlying problem of lack of competitiveness in some of our Member States” (Eggertsson, Ferrero, and Raffo 2014).

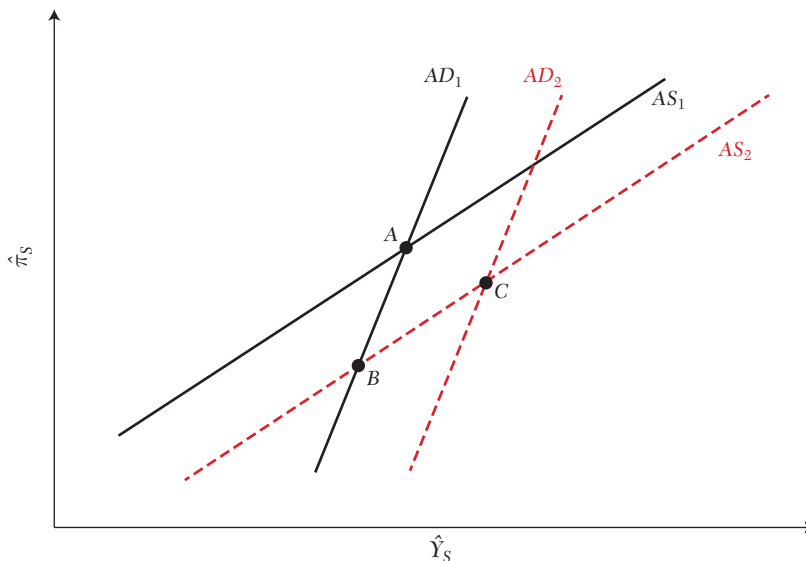


Figure 21. Short-Run Equilibrium at the ZLB under Permanent Structural Reforms

is, $\hat{\omega}_S < 0$ and $\hat{\omega}_L < 0$. Assuming the ZLB is not binding and policy regime A1b holds, so that $\hat{\pi}_t = 0$, we have

$$(73) \quad \hat{Y}_S = -\hat{\omega}_S \text{ and } \hat{Y}_L = -\hat{\omega}_L.$$

At the ZLB, consider the long-run effect of structural reforms:

$$(74) \quad \text{AD: } \hat{Y}_S = \hat{Y}_L + \frac{\sigma^{-1}\mu}{1-\mu}\hat{\pi}_S + \frac{\sigma^{-1}}{1-\mu}\hat{r}_S^e$$

$$(75) \quad \text{AS: } \hat{\pi}_S = \frac{\kappa}{1-\mu\beta}\hat{Y}_S + \frac{\kappa}{1-\mu\beta}\hat{\omega}_S.$$

In Figure 21, the short-run equilibrium at the ZLB is shown at point A. Reforms may be contractionary or expansionary in the short run: The outcome depends on the relative strength of the permanent-income effect (**AD** shift, obtained by substituting $\hat{Y}_L = -\hat{\omega}_L$ into the AD equation) and the deflationary effect (**AS** shift, given by $\frac{\kappa}{1-\mu\beta}\hat{\omega}_S$). If there is no income effect, the equilibrium is at point B. More realistically, credible structural reforms will have an income effect, which, if strong enough, leads to an increase in output as illustrated in point C. Eggertsson, Ferrero, and Raffo (2014) find that a 10 percent reduction in markups in the service sector of the periphery leads to a 5 percent increase in long-run output but a 1 percentage point additional contraction at the ZLB, thus the equilibrium is between point A and C. Fernández-Villaverde, Guerrón-Quintana, and Rubio-Ramírez (2014) show that supply-side policies with no short-run effects but positive long-run effects on productivity and income are always expansionary.

3.3.4 *The Great Depression and the US National Industrial Recovery Act of 1933*

This section studies one of the most criticized economic policies in US history, which was implemented at the height of the Great Depression: The National Industrial Recovery Act (NIRA) of 1933. One of its architects, Rexford Guy Tugwell, said about it “for the economic philosophy which it represents there are no defenders at all.”

The paradox of toil casts a fresh light on this policy. As we will see, the objective of the legislation is consistent with its predicted effect in our framework, given the assumed policy regime.

The NIRA was a key component of the New Deal. It aimed to curb the deflationary spiral during the Great Depression by encouraging firms to raise prices. To do so, it used a blunt instrument: suspending antitrust laws so firms could collude on price increases and encouraging workers to unionize to support higher wages. The paradox of toil provides a theoretical rationale for why such a policy can turn out to be expansionary *at the ZLB* after all.

An interesting aspect of the NIRA was that it was explicitly designed to be *state contingent*. This suggests that the architects of the policy recognized that the policy was not desirable in the long run, but could be helpful to curb deflation. The NIRA legislation stated: “A national emergency productive of widespread unemployment and disorganization of industry [...] is hereby declared to exist.” It then went on to specify that “this title shall cease to be in effect and any agencies established hereunder shall cease to exist at the expiration of two years after the date of enactment of this Act, or sooner if the President shall by proclamation or the Congress shall by joint resolution declare that the emergency recognized by section 1 has ended.” Hence, a reasonable assumption is that the policy was expected to be an emergency measure and to last only as long as the deflationary emergency. It follows that we can use exactly the same **AD** and **AS** apparatus as in Figure 18.

As we already noted, the economics profession, ranging from Keynes to Friedman, universally condemned NIRA. Influential quantitative work by Cole and Ohanian (2004) argued that the New Deal slowed the recovery from the Great Depression. It was only later that modern theory caught up with policymakers’ intentions, with De Long and Summers (1986) emphasizing NIRA’s role in making wages and prices more rigid and Eggertsson (2012a) highlighting it as a direct tool for propping up prices via the paradox of toil. Eggertsson (2012a) provides narrative evidence suggesting that the actual objective of NIRA—propping up prices—was indeed similar to the role it plays analytically in the model.¹⁰³ This curious history of thought flips a famous quote from Keynes’s *General Theory* on its head: “Practical men, who believe themselves to be quite exempt from any intellectual influences, are usually the slaves of some defunct economist. Madmen in authority, who hear voices in the air, are distilling their frenzy from some academic scribbler of a few years back.”

In retrospect, FDR himself viewed the policy as a failure, and it was struck by the Supreme Court in 1935. We will discuss the Great Depression in greater detail in Section 4.3. Cohen-Setton, Hausman, and Wieland (2017) provide evidence from France in the 1930s, which was even more ambitious than NIRA. Overall, they find that these policies led to output stagnation.

¹⁰³The *overall* legacy of NIRA, on the other hand, is mixed at best, as it brought about a large bureaucracy with associated inefficiencies.

One challenge in interpreting the evidence from France is that in contrast to the parallel element of the New Deal, the policies in France promoted a *permanent* change. As shown in equation (73), a permanent change of this kind is *always* contractionary. In contrast, NIRA was an “emergency” legislation that was installed temporarily to reflate the price level.

3.4 *Unconventional Monetary Policy within a Fixed Standard Policy Regime: Credit Policy and Central Bank Lending Facilities*

So far in Section 3, we have considered policy interventions in the absence of a regime change in monetary policy. Yet, Section 2 suggest that financial market frictions can be a central source of a temporary drop in the efficient interest rate. Can the government directly intervene to offset these market shocks? This opens the door for policy beyond typical supply and demand analysis considered so far raising a crucial question: Can direct government intervention in financial markets prevent the drop in the natural rate of interest? Bhattacharai and Neely (2022) provides a comprehensive review in this journal of this class of policies, typically referred to as unconventional monetary policy. This section is a more targeted overview of these interventions and their relationship with the broader themes of this review.

3.4.1 *Crisis Response Tools: QE1 and Liquidity Facilities as an Extension of the Discount Window*

The light blue area in Figure 22 shows the Federal Reserve’s direct credit extension to financial markets through various liquidity facilities during the Great Recession. One branch of the literature interprets this as the government directly substituting bank lending via direct credit extensions. Another branch of the literature rationalizes these policies as the government taking on banks’ illiquid assets in exchange for perfectly liquid government paper, which is more similar to the function of the Federal Reserve’s discount window. Either interpretation captures elements of the policies of the Federal Reserve via the so-called liquidity facilities (a wide range of policies aimed at providing additional credit to the financial sector and the broader economy).

Financial distress, as discussed in Section 2, is a plausible explanation for a temporary reduction in the natural rate of interest, triggering both the Great Depression and Great Recession. Policies aimed at easing financial conditions may, therefore, directly influence the natural rate of interest. During the initial phase of the Great Recession, the Federal Reserve expanded its balance sheet through quantitative easing (QE1) by approximately \$1.5 trillion, roughly 10 percent of GDP, as further interest rate cuts were no longer feasible. This expansion reached unprecedented levels following the COVID-19 crisis, with QE4 pushing the balance sheet to nearly 40 percent of GDP. The empirical literature, particularly Swanson (2021), documents substantial effects of QE, especially on targeted asset prices at the ZLB.

Theoretically, it is not obvious that these policies have meaningful effects. This was the message of the irrelevance results stated in Section 2.3. However, these irrelevance results can be invalidated if certain assumptions are violated, potentially giving QE meaningful potency. The irrelevance results specifically rely on two key assumptions: (i) Financial markets where (a) markets are complete with no limits to arbitrage, and (b) interventions have no effect on the state-contingent payoffs of the assets purchased; and (ii) there is no change in expectations about future interest rate policy.

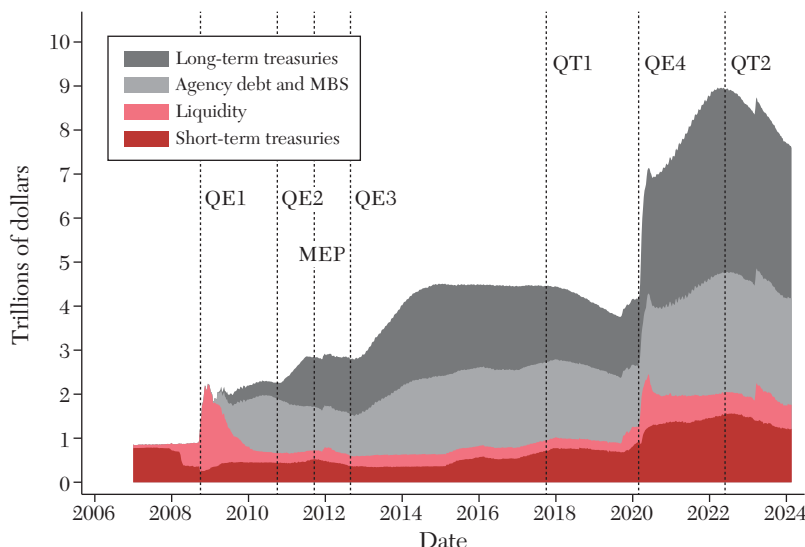


Figure 22. Evolution of the Federal Reserve Balance Sheet since the 2008 Financial Crisis

Notes: Vertical lines denote QE, the Maturity Extension Program (MEP), and quantitative tightening (QT). Data from the Federal Reserve Bank of Cleveland, “Credit Easing” (Federal Reserve Bank of Cleveland 2024).

QE1 fundamentally differed from subsequent quantitative interventions in both the target and the approach. While QE1 provided direct liquidity to the private sector through various facilities, later QEs primarily targeted government debt and debt of government-sponsored enterprises. As Figure 22 shows, long-term Treasury securities became the dominant component of these purchases. Additionally, the Fed made substantial purchases of agency mortgage-backed securities (MBS) from government-sponsored enterprises such as Fannie Mae, Freddie Mac, and Ginnie Mae, while also acquiring bonds these institutions issued directly to fund mortgage activities.

The literature generally finds significant effects for QE1 while results for later QEs appear to be more mixed. Distinguishing direct effects from signaling effects on future interest rates poses a particular challenge for later QEs, as intervention announcements often convey multiple policy signals simultaneously.

Gertler and Karadi (2011) provide an influential account of QE1. The key assumption, relative to the ideal assumption listed in the irrelevance result, is that the intervention relaxes the assumption of complete markets and that the intervention can directly affect the viability of the underlying asset that is bought via various liquidity facilities. Their starting assumption is a negative shock to bank capital. This directly reduces lending through commercial banks. QE counters this at a somewhat abstract level, relative to what happened in practice. The government is directly extending credit to the private sector. This policy creates an interesting trade-off: The Federal Reserve can restore credit flow, but it operates less efficiently than private intermediaries. Overall, however, the framework effectively captures both QE1’s role in addressing credit market disruptions and various government lending

facilities. However, it less readily accommodates the later QE phases' focus on long-dated government bond purchases.

An important aspect of QE1's liquidity facilities was that they effectively extended the Federal Reserve's discount window beyond its traditional scope. The role of the discount window is to serve as a lender of last resort to banks that accept demand deposits. The discount window serves as a crucial safeguard against bank runs in which simultaneous customer withdrawals force banks to liquidate long-term investments at a discount, potentially collapsing even fundamentally sound institutions. This socially inefficient outcome prompted the Federal Reserve's establishment in 1913 and the discount window's creation in 1914, allowing eligible financial institutions, primarily those serving small depositors, to borrow at a discount rate using long-term assets as collateral. This mechanism reduces both the need for fire sales and the risk of bank runs. During the Great Depression, the Banking Act of 1933 further strengthened the system by establishing the Federal Deposit Insurance Corporation, creating additional guardrails for banks serving small depositors and helping prevent banking panics.

Diamond and Dybvig (1983) formalize the economics of bank runs, demonstrating that they can occur even when all agents are perfectly rational. Bank runs require two conditions: that banks hold on their balance sheet assets that are illiquid in the short run (so they will need to sell them at a discount if a large enough number of depositors withdraw money) and that depositors are served in sequence (late withdrawers might get less than the first). However, Diamond and Dybvig (1983) did more than simply formalize the existing wisdom of policymakers. When establishing the discount window (1914) and Federal Deposit Insurance Corporation (1933), policymakers had assumed that bank runs stemmed from small customers' irrationality, rumor chasing, and herd behavior. They viewed large, sophisticated investors as naturally immune to such erratic behavior, requiring minimal regulation.

However, Diamond and Dybvig (1983) clarify a crucial insight: Inefficient bank runs can occur even among large, sophisticated, and fully rational investors. Moreover, the logic of bank runs extends beyond traditional banks to any firm engaging in bank-like behavior—borrowing short term and investing in long-term illiquid assets, making them vulnerable to liquidity risk. Hedge funds and investment banks exemplify these vulnerable institutions, forming part of what Paul McCulley called *shadow banking* in 2007: Shadow banks are financial institutions that operate like banks but without the same regulations, safeguards, and discount window protections. These institutions face the very same vulnerabilities that policymakers once thought were limited to banks serving small, irrational, herd-like, and low-information depositors.

Modern macroeconomic theory has formalized how the Federal Reserve's liquidity facilities, during the Great Recession, effectively extended the traditional role of the discount window to the shadow banking system. Del Negro et al. (2017b) develop this idea in an extension of the canonical new Keynesian model, incorporating investment opportunities and imperfect asset resellability. Their model shows how the Federal Reserve expanded discount window services, targeting specific frozen markets through liquidity facilities. The Great Recession, in their framework, triggered a sudden reduction in private asset liquidity, leading to a sharp drop in available credit and investment. This decline in investment directly reduced the natural interest rate. Combined with the ZLB and nominal frictions, a negative natural rate could generate a full-blown economic crisis of the order of the Great Depression—if not for the liquidity facilities, hence their paper's title: "The Great Escape?" The model demonstrates

how credit intervention helps: Liquidity facilities, like the traditional discount window, replace illiquid assets with perfectly liquid government-issued paper. Gertler, Kiyotaki, and Prestipino (2020) provide complementary theoretical support by incorporating Diamond and Dybvig (1983) into a dynamic general equilibrium model—while not their focus, their framework naturally rationalizes the Fed’s liquidity facilities as well.

3.4.2 *Later QE Programs: Transmission Mechanisms and Effectiveness*

The subsequent QE programs of the Federal Reserve, marked by yellow and green areas in Figure 22, represented a major expansion into two asset classes: mortgage markets and government securities. These interventions, known as large-scale asset purchases (LSAP), directly challenged assumption (1 ii) of the irrelevance result in Eggertsson and Woodford (2003), which suggested that central bank purchases would not change the state-contingent payoffs of the securities in question. Through trillion-dollar interventions, the Fed clearly did change these payoffs by purchasing agency MBS from Fannie Mae, Freddie Mac, and Ginnie Mae, while also acquiring these institutions’ bonds. These actions effectively extended government guarantees to these institutions’ bonds, thus largely eliminating the risk premiums that these securities carried relative to government bonds and enhancing market liquidity. The result was a direct lowering of mortgage costs for American home buyers.¹⁰⁴

Explaining QE’s effectiveness through risk-premium reduction presents a theoretical challenge when considering long-term Treasury purchases, since these securities presumably carry the same default risk as reserves. Chen, Cúrdia, and Ferrero (2012) addresses this challenge by modeling long-term government bond purchases through preferred-habitat theory, following Vayanos and Vila (2021). In analyzing QE2 and subsequent programs, Chen et al. relax A1a from the irrelevance results in Section 2.3 by incorporating market segmentation and transaction costs. Their key finding, however, is that these market imperfection mechanisms alone generate only modest effects *unless* the purchases also alter expectations about future interest rates, effectively signaling a change in the monetary policy regime—thus also violating A2.

Bhattarai, Eggertsson, and Gafarov (2023) explores how QE violates the second assumption of the irrelevance results in Section 2.3—that QE does not change expectations of future monetary policy. When the Federal Reserve purchases long-term government bonds with short-term liabilities, it effectively shortens the maturity of consolidated government debt, analogous to a homeowner switching from a 30-year fixed mortgage to a floating rate. Bhattarai, Eggertsson, and Gafarov (2023) demonstrate that in a Markov perfect equilibrium, this maturity shortening makes interest rate increases more costly for the government through higher fiscal costs. Equivalently, from the central bank’s perspective, raising rates becomes more costly due to potential capital losses on its balance sheet. These losses may have consequences beyond what is implied by social welfare considerations, as they could—if large enough—threaten central bank independence from a political standpoint. This concern has historical precedent: Paul Samuelson claimed that the Federal Reserve was “the prisoner of its

¹⁰⁴For additional transmission channels, see Vissing-Jorgensen and Krishnamurthy (2011) and the comprehensive review in Bhattarai and Neely (2022).

own independence” during the early phase of the Great Depression, as fear of balance sheet losses led to reluctance in fighting deflation, thereby exacerbating the economic slump.¹⁰⁵

A common argument before the Great Recession was that monetary policy affected the economy beyond just influencing expectations of future policy. Clouse et al. (2003), for example, argued that institutions often hold specific securities for regulatory or habitual reasons, suggesting that large interventions could lower the interest rates on long-term government bonds beyond what expected future short-term rates would imply. Milton Friedman (see Friedman 2000) stressed the importance of the positive *long-term* interest rates on government bonds in a 2000 Bank of Canada keynote address:

Now, the Bank of Japan’s argument is, “Oh well, we’ve got the interest rate down to zero; what more can we do?” It’s very simple. They can buy long-term government securities, and they can keep buying them and providing high-powered money until the high-powered money starts getting the economy in an expansion. What Japan needs is a more expansive domestic monetary policy.

In direct contrast, Eggertsson and Woodford’s (2003) irrelevance result argues that long-term government bond purchases are ineffective if they leave expectations about future interest rate policy unchanged. This is because bond prices reflect expected future short rates along with risk and term premiums, neither of which are affected by the size of the intervention in their model.¹⁰⁶ As in the case of Krugman (1998), this represents a pure model-based prediction that challenged the conventional wisdom of the time.

Some institutions may have idiosyncratic preferences for holding long-term government bonds, enabling the central bank to lower these rates through asset purchases beyond what open market forces would justify. However, this does not necessarily mean the intervention stimulates the broader economy—it may simply reduce the government’s financing costs. Eggertsson and Woodford (2003) predict that if only a small number of institutions hold these bonds at unfavorable rates for idiosyncratic reasons, lower government yields become disconnected from broader private-sector borrowing costs. This disconnect separates government bond yields from other asset prices, limiting the transmission of monetary policy to the real economy.

Evidence from the Great Recession supports Eggertsson and Woodford’s (2003) prediction. Although central banks successfully lowered government bond rates through large purchases, this had little impact on *private* borrowing costs except when these purchases affected expectations about future short-term interest rates. Lucca and Wright’s (2024) analysis of Australia’s explicit targeting of three-year government bond yields provides perhaps the clearest demonstration of this disconnect. Despite successfully maintaining its yield target, the Australian central bank ultimately had to purchase most bond issuance at this maturity to sustain the peg. This intervention eventually caused Australian government yields to diverge significantly from both current and expected short-term interest rates, as evidenced by the overnight index swap curve, and even from state government yields—precisely the market segmentation that Eggertsson and Woodford (2003) predicted.

There is also limited evidence that central bank asset purchases of long-dated securities—in contrast to the yield-targeting approach of the Reserve Bank of Australia—spilled over to private lending conditions beyond the expectation channel of lower future short rates. Studies

¹⁰⁵See Mayer (1993), p. 6.

¹⁰⁶See detailed discussion in Eggertsson and Woodford (2003) in the section titled “Discussion,” pp. 159–65.

such as those by Swanson (2011) and Krishnamurthy and Vissing-Jorgensen (2013), along with further references in Lucca and Wright (2024), consistently find minimal transmission from government bond yield reductions to broader private-sector borrowing costs when controlling for signaling effects about future monetary policy.

4. *Regime Changes and Optimal Policy Design: Understanding the US Great Recession and the Great Depression*

In Section 3 we assumed a standard monetary and fiscal policy regime, precluding discussion of key strategies used in the Great Recession and Great Depression that emphasized changing expectations about future policy rates and inflation. In Section 4 we depart from this policy regime. We start by reviewing two polar cases for optimal monetary policy: full commitment and no credible commitment (discretion). These polar cases provide useful interpretations of US monetary policy during the Great Recession. We also review a joint monetary-fiscal regime under optimal discretion with constraints like, for instance, the gold standard, potentially abandoned by subsequent policymakers, triggering regime change. We argue that the radical regime change in 1933 implemented by Franklin Delano Roosevelt ended the Great Depression.

4.1 *The Great Recession through the Lens of Monetary Policy Regimes: Forward Guidance and Deflation Bias*

This section reviews optimal monetary policy under full commitment or no commitment (discretion), which sheds light on the Federal Reserve's use of forward guidance about future interest rates during the Great Recession. Forward guidance implies a regime change relative to the standard policy regime assumed in Section 3. We argue that US policy fell short of the full optimal commitment, partially explained by credibility problems and the *deflation bias* of discretionary policies. We provide empirical and narrative evidence from policy makers demonstrating that imperfect credibility was a major constraint on policy. Yet, we also review empirical evidence establishing that forward guidance had a significant effect, which cannot be explained by a policy regime that lacks all credibility. We propose a sketch of a generalized policy regime for the United States during the Great Recession informed by the theoretical results that is consistent with the empirical evidence reviewed.

4.1.1 *The Optimal Monetary Policy Regime under Commitment: The Theoretical Ideal of Perfect Forward Guidance*

This section reviews optimal monetary policy when the central bank can fully commit to future policy in all states of the world. Like a strategic chess player thinking several moves ahead, a central bank successfully implementing the optimal monetary policy regime under commitment thinks several moves ahead, taking into account every possible contingency that may emerge, and computes optimal action in all states of the world. Critically, this plan is fully credible and its implementation is reflected in market expectations.

If the central bank can optimally commit to future policy, it can largely undo the destabilizing effect of the ZLB. To do this, it can promise to keep the interest rate low even *after* the economic conditions improve, but this date is, under A2, uncertain. Such promises are

a powerful tool that can greatly mitigate downturns at the ZLB. The objective of the central bank is to maximize welfare given by household utility functions, which can be approximated to a second order by¹⁰⁷

$$(76) \quad -\frac{1}{2}E_0 \sum_{t=0}^{\infty} \beta^t (\hat{\pi}_t^2 + \lambda_y \hat{Y}_t^2).$$

The problem of the central bank can then be solved by formulating the Lagrangian:

$$\begin{aligned} \mathcal{L}_0 = E_0 \sum_{t=0}^{\infty} \beta^t \Big\{ & \frac{1}{2} [\hat{\pi}_t^2 + \lambda_y \hat{Y}_t^2] + \phi_{1t} [\hat{Y}_t - E_t \hat{Y}_{t+1} + \sigma(\hat{i}_t - E_t \pi_{t+1} - \hat{r}_t^e) - \mathcal{F}_t^{IS}] \\ & + \phi_{2t} [\hat{\pi}_t - \kappa \hat{Y}_t - \beta E \hat{\pi}_{t+1} - \mathcal{F}_t^{AS}] + \phi_{3t} [\hat{i}_t - i^{zlb}] \Big\}. \end{aligned}$$

The first-order conditions and complementary slackness conditions for the optimal policy problem are

$$\begin{aligned} (77) \quad \partial \hat{\pi}_t : \quad & \hat{\pi}_t + \sigma \beta^{-1} \phi_{1,t-1} + \phi_{2,t} - \phi_{2,t-1} = 0, \\ \partial \hat{Y}_t : \quad & \lambda_y \hat{Y}_t + \phi_{1,t} - \beta^{-1} \phi_{1,t-1} - \kappa \phi_{2,t} = 0, \\ \partial \hat{i}_t : \quad & -\sigma \phi_{1,t} + \phi_{3t} = 0, \\ \text{CS :} \quad & \hat{i}_t \geq i^{zlb}, \phi_{3t} \geq 0, \phi_{3t}(\hat{i}_t - i^{zlb}) = 0. \end{aligned}$$

The Lagrange multipliers $\phi_{j,t}$ for $j = 1, 2, 3$ refer to equations IS (29), AS (32), and ZLB (31) respectively. The equilibrium can now be defined.¹⁰⁸

The optimal monetary policy regime under commitment implies keeping the real interest rate low *after* the shock reverts to a steady state. Figure 23 shows the optimal policy for the Great Recession calibration. The central bank maintains the policy rate at the ZLB beyond the duration of the shock—in contrast with the standard policy regime. The figure shows a single realization of the shock, corresponding to the stochastic contingency when it reverts back to the steady state in period 5, that is, the realization of $t_L = 5$. Of course we could alternatively show this figure for any realization of t_L but choose 5, as it represents well the general property of a typical contingency.¹⁰⁹

Figure 23 shows that optimal monetary policy regime under commitment mitigates the drop in output and inflation relative to the standard monetary and fiscal policy regime (where fiscal policy is held fixed) substantially. The successful mitigation of the recession is due to a number

¹⁰⁷See Woodford (2003).

¹⁰⁸The equilibrium is a set of stochastic processes for $\{\hat{\pi}_t, \hat{Y}_t, \hat{i}_t, \phi_{1,t}, \phi_{2,t}, \phi_{3,t}\}$ that solve equations (29), (32), and (77), given $\{\hat{r}_t^e, \mathcal{F}_t^{IS}, \mathcal{F}_t^{AS}\}$, as for now we will take fiscal policy as exogenous given and keep both $\mathcal{F}_t^{IS} = \mathcal{F}_t^{AS} = 0$ unless otherwise noted. The complementary slackness (CS) condition complicates the solution, so that standard method cannot be applied. The stochastic duration of the ZLB policy now depends both on the realization of the shock as well as the policy choice of the government. Eggertsson and Woodford (2003) suggest a guess-and-verify algorithm, which is automated for a more general setting in Eggertsson et al. (2021).

¹⁰⁹While the earlier literature is cast in discrete time, Werning (2011) is the first characterization of the optimal monetary policy problem in continuous time, a solution method that has found applicability in a number of papers that have followed. His findings are largely consistent with those reported herein.

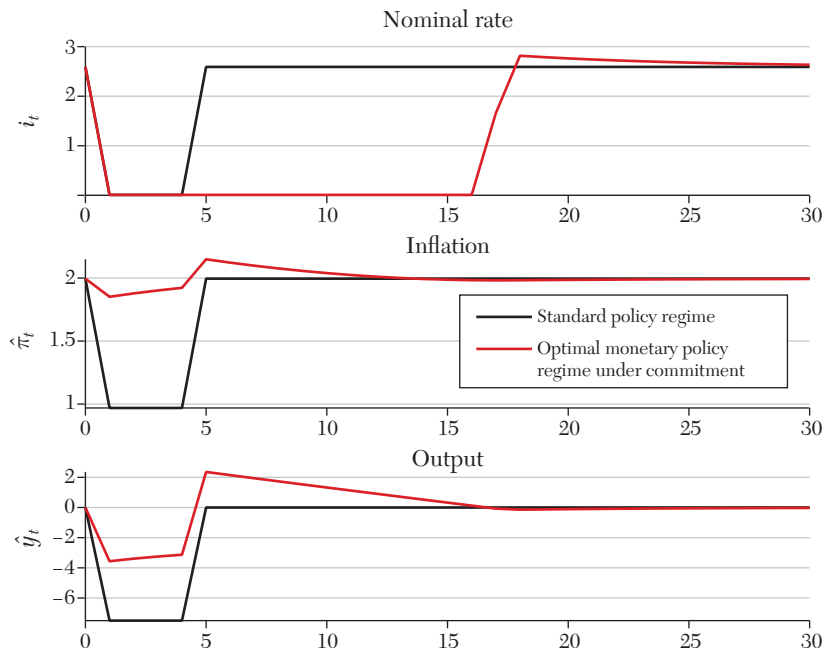


Figure 23. The Standard Monetary and Fiscal Policy Regime and the Optimal Monetary Policy Regime under Commitment When the Shock Lasts Five Periods

of mechanisms. The commitment to lower the *future* nominal interest rate, together with a commitment to higher inflation, reduces the real interest rate, which stimulates demand. Moreover, the optimal monetary policy regime under commitment generates expectations of higher future output relative to the standard monetary and fiscal policy regime, which also increases spending via the permanent income hypothesis, as evident in IS equation (29).

In the optimal monetary policy regime under commitment, the duration of zero interest rates beyond the point when the efficient rate returns to positive territory depends on how long the efficient rate has been negative. The longer the efficient rate remains negative, the longer the central bank maintains the additional zero interest rate policy, as illustrated in Figure 24. Importantly, what matters is not the extended duration of low *nominal interest rates* but rather the lower *real interest rates*, which can in principle also be achieved via higher inflation. However, our simulations indicate that the optimal policy commitment generally takes the form of extended duration of the policy rates at zero.¹¹⁰

Under the standard monetary and fiscal policy regime, the duration of the zero interest rate policy is exactly equal to the duration of the negative natural rate episode. It is the *expected* duration of the negative natural rate episode that is important for the size of the output and

¹¹⁰This observation becomes relevant when considering suboptimal policy regimes, such as those in Cochrane (2017). He arbitrarily fixes the policy rate at zero through the same duration as the efficient rate. Cochrane illustrates policy regimes that closely mimic the optimal regime, but with policy rates fixed across the regimes, monetary expansion is instead expressed via higher inflation expectations. See further discussion of his results in Section 4.2.

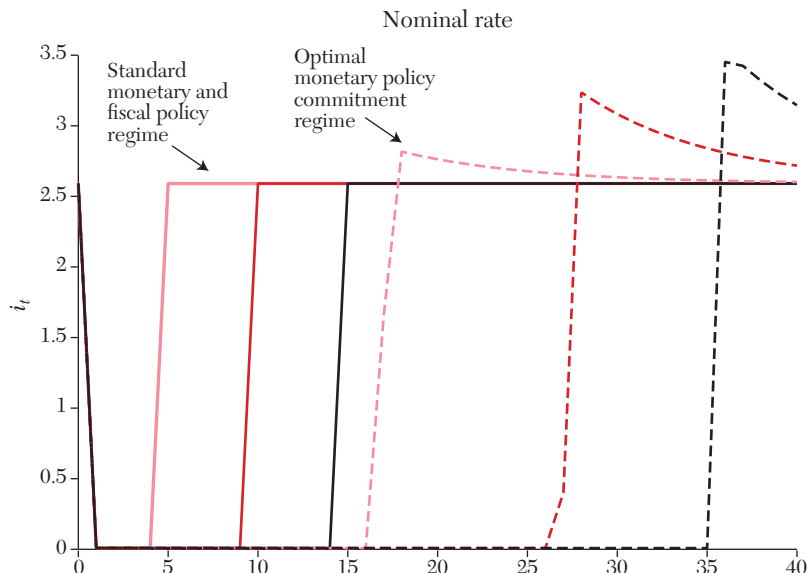


Figure 24. Path of Policy Rates for Alternative Realization of the Shock

Notes: The interest rate assuming the standard monetary and fiscal policy regime and optimal monetary policy regime under commitment under Great Recession calibration.

inflation deviations from the steady state in the short run. $\hat{Y}_S$ and $\hat{\pi}_S$ depend upon μ but not upon a particular realization of the shock duration. For example, a shock that is expected to be persistent will lead to a bigger decline in $\hat{Y}_S$ and $\hat{\pi}_S$ even if ex post realized duration turns out to be short. Vice versa, a shock that is expected to subside soon will lead to a smaller decline in $\hat{Y}_S$ and $\hat{\pi}_S$ even if ex post realized duration turns out to be much longer.

The optimal monetary policy regime under commitment is different because the duration of the zero interest rate policy is not equal to the duration of the negative natural interest rate episode, but instead changes with its realized stochastic duration. For each realized duration of the episode, there's an appropriate policy response with an additional zero interest rate policy after the shock has already subsided. This is shown in Figure 24.

The Optimal monetary policy regime under commitment was derived independently by Eggertsson and Woodford (2003) and Jung, Teranishi, and Watanabe (2005).¹¹¹ Two important papers predate these results. Rotemberg and Woodford (1997) account for the ZLB indirectly: by penalizing interest rate variation in the central bank loss function. They conclude that the problem implied by the ZLB can be substantially mitigated if interest rates move slowly in response to shocks.¹¹² Reifschneider and Williams (2000) simulate the FRB/US model assuming a Taylor rule augmented by the ZLB and find significant output losses associated with ZLB episodes. They suggest modifying the Taylor rule to keep track of when it is violated due to the

¹¹¹While the former considers a two-state Markov process for the natural rate, the latter considers a deterministic AR(1) process. Adam and Billi (2006) generalize Jung, Teranishi, and Watanabe (2005) to a stochastic AR(1) process.

¹¹²See the discussion in footnote 65.

ZLB. Their modified policy rule says that if the policy rate is constrained by the ZLB, future interest rates are to be kept low. This captures one of the key elements of the optimal monetary policy regime under commitment, as we have seen.¹¹³

4.1.2 *Implementing Optimal Commitment in Practice: From Theory to Operational Frameworks*

This section shows that the optimal policy commitment analyzed in the previous section can be communicated in a surprisingly simple form: The central bank simply defines appropriate thresholds that depend on a combination of the price level and output, which, once met, signal lift-off from the ZLB. This policy shares important commonalities with the so-called Evans rule adopted by the Federal Reserve in 2012, named after the President of the Federal Reserve Bank of Chicago, Charles Evans, who was instrumental in its design.

Even in our simple example where shocks follow a stochastic two-state Markov process, the optimal path for the policy rate is a maze of contingencies depending on how long the recession shock lasts. The complexity of the optimal commitment is already apparent in Figure 24, which displays only a few possible realizations for the two-state Markov Process in A2.

Yet, this seemingly impractical theory can be implemented in a surprisingly straightforward way, as we review below. The policy protocol implements the optimal policy regime under commitment, even allowing for arbitrary stochastic processes in both the IS equation ($r_t^e, \mathcal{F}_t^{IS}$) and the AS equation ($\mathcal{F}_t^{AS}$).

Bernanke, Reinhart, and Sack (2004) distinguish between unconditional (calendar-based) and conditional (economic-based) forward guidance at the ZLB. Most major central banks used both during the Great Recession, but relied more on calendar-based commitments. This is illustrated for the Federal Reserve in Table 6 adopted from Swanson (2021), which documents forward guidance from the start of the Great Recession until the Fed began raising rates in 2015.

Despite the complexities of stochastic shock duration, unobserved efficient rates, and hard-to-measure AS equation shocks, Eggertsson and Woodford (2003) show that the optimal monetary policy regime under commitment can be communicated simply and only on basis of publicly observed data as long as there are estimates for κ , σ , β , and λ_y , which can be made public as well.¹¹⁴

They propose that the central bank announce a threshold, formulated using the output adjusted and price level (in its most simple formulation, nominal GDP) that must be met before raising rates. The output-adjusted price-level target is

$$(78) \quad \tilde{p}_t = p_t^*,$$

¹¹³Specifically, Reifschneider and Williams (2000) consider the rule $\hat{i}_t = \max(\hat{i}_t^{Taylor} - \alpha Z_t, i^{zlb})$, where $\alpha \in (0, 1]$, $\hat{i}_t^{Taylor}$ is the nominal rate implied by an unconstrained Taylor rule, $d_t = \hat{i}_t - \hat{i}_t^{Taylor}$ measures the difference between the policy rate and the Taylor rule rate, and Z_t is the sum of past d_t values. This rule implies that if policy is constrained by the ZLB, it will “make up” for it by remaining lower in future periods when the ZLB no longer binds, replicating key aspects of the optimal monetary policy regime under commitment. Eggertsson et al. (2021) show this remains true in the new Keynesian model simulations.

¹¹⁴The implementation is independent of the stochastic properties of the underlying shock.

TABLE 6
MAJOR UNCONVENTIONAL MONETARY POLICY ANNOUNCEMENTS BY THE FEDERAL OPEN MARKET
COMMITTEE (FOMC), 2009–15

Date	Announcement
March 18, 2009	FOMC announces it expects to keep the federal funds rate between 0 and 25 basis points (bp) for “an extended period,” and it will purchase \$750B of mortgage-backed securities, \$300B of longer-term Treasuries, and \$100B of agency debt (a.k.a. QE1).
November 3, 2010	FOMC announces it will purchase an additional \$600B of longer-term Treasuries (a.k.a. QE2).
August 9, 2011	FOMC announces it expects to keep the federal funds rate between 0 and 25 bp “at least through mid-2013.”
September 21, 2011	FOMC announces it will sell \$400B of short-term Treasuries and use the proceeds to buy \$400B of long-term Treasuries (a.k.a. Operation Twist).
January 25, 2012	FOMC announces it expects to keep the federal funds rate between 0 and 25 bp “at least through late 2014.”
September 13, 2012	FOMC announces it expects to keep the federal funds rate between 0 and 25 bp “at least through mid-2015” and it will purchase \$40B of mortgage-backed securities per month for the indefinite future.
December 12, 2012	FOMC announces it will purchase \$45B of longer-term Treasuries per month for the indefinite future and it expects to keep the federal funds rate between 0 and 25 bp at least as long as unemployment remains above 6.5 percent and inflation expectations remain subdued.
December 18, 2013	FOMC announces it will start to taper its purchases of longer-term Treasuries and mortgage-backed securities to \$40B and \$35B per month, respectively.
December 17, 2014	FOMC announces that “it can be patient in beginning to normalize the stance of monetary policy.”
March 18, 2015	FOMC announces that “an increase in the target range for the federal funds rate remains unlikely at the April FOMC meeting.”

Source: Adapted from Swanson (2021).

where $\tilde{p}_t \equiv p_t + \frac{\kappa}{\lambda} \hat{Y}_t$ and p_t^* is computed as

$$(79) \quad p_{t+1}^* = p_t^* + \beta^{-1}(1 + \kappa\sigma)\Delta_t - \beta^{-1}\Delta_{t-1},$$

with $\Delta_t \equiv p_t^* - \tilde{p}_t$ measuring the ZLB-induced target miss.

Figure 25 illustrates the optimal monetary commitment policy for a five-period shock realization analogous to Figure 23. If the output-adjusted price-level target (p_t^* in equation (78)), determined at time $t - 1$, is missed at time t , the central bank uses equation (79) to determine the next period's price level target. Eggertsson and Woodford (2003) show that satisfying this criterion implements the optimal monetary policy regime under commitment.¹¹⁵

¹¹⁵The analysis of Eggertsson and Woodford (2003) is anticipated by Wolman (2005), who suggested replacing inflation in the Taylor rule with the price level and shows that this approach substantially reduces the cost of the ZLB in a model with rigid prices.

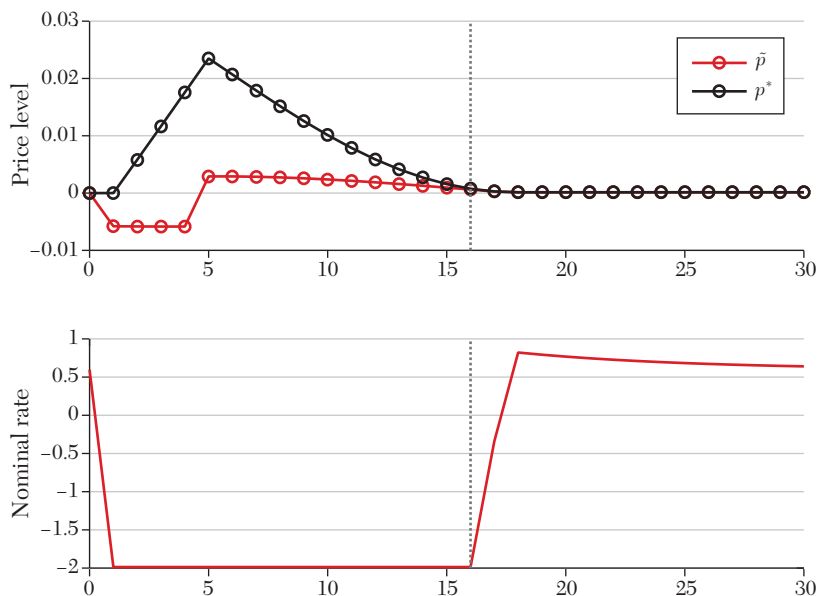


Figure 25. Price Level, Actual and Target, under the Commitment Policy Regime

This conditional commitment approach, as Bernanke, Reinhart, and Sack (2004) term it, allows market expectations to adjust automatically to new shocks. When correctly specified, monetary policy becomes more accommodative in response to negative shocks, as markets understand that these shocks push the economy further from the threshold. This key advantage distinguishes it from unconditional commitments, where the central bank needs to be constantly adjusting its stance based upon incoming shocks. Figure 24 compares interest rate paths under the optimal monetary policy regime with commitment versus the standard monetary and fiscal policy regime across different shock realizations. As the figure clearly shows, the duration of the ZLB depends on the shock duration under commitment, unlike in the standard monetary policy regime.

The Federal Reserve initially used unconditional announcements during the Great Recession (Table 6) and thus tied its commitment to a time frame instead of outcomes. The 2012 Evans rule, however, specified conditions for rate increases, thus representing a shift toward the conditional approach described above. While it didn't fully implement the new Keynesian commitment solution, it featured a threshold rather than a deterministic time horizon—the key idea of the Eggertsson–Woodford proposal—allowing markets to forecast the ZLB duration themselves. While an improvement in communication strategy, its effectiveness may have been dulled by the fact that the Federal Reserve specified a relatively high unemployment threshold.

4.1.3 The Deflation Bias: When Forward Guidance Lacks Credibility

What happens when a central bank faces a credibility crisis, unable to convince markets it will keep future promises? In this section we show how the optimal monetary policy regime in a Markov perfect equilibrium captures this problem—where the central bank must navigate

the present without the power to shape expectations about future policy rates. Instead, policy-makers can only react to current conditions.

This policy regime stands as the polar opposite of the commitment solution we just examined, as forward guidance becomes completely ineffective in the absence of endogenous state variables that can constrain the actions of the central bank to implement the announced commitment.¹¹⁶ This is absent in the canonical new Keynesian model. It is perfectly forward looking, so it has no endogenous state variables.

This property is useful and allows us to establish an equivalence result: The resulting equilibrium path for inflation output and interest rates is equivalent if the central bank follows the optimal monetary policy regime in a Markov perfect equilibrium or the standard monetary policy regime considered in Section 3.¹¹⁷ This gives key insight into the properties of the standard monetary policy regime, but also points to the plausibility of this characterization of policy, since the standard policy regime is, indeed, standard in the literature.

The optimal monetary policy regime in a Markov perfect equilibrium suffers from a credibility problem, leading to what Eggertsson (2001a) coined as the deflation bias.¹¹⁸ In the Markov perfect equilibrium, actions and expectations depend only on observed state variables. The only state variables in the model are $\{\hat{r}_t^e, \mathcal{F}_t^{AS}, \mathcal{F}_t^{IS}\}$, which the central bank takes as exogenous. For now, we set $\mathcal{F}_t^{AS}, \mathcal{F}_t^{IS} = 0$. In this setting, expectations of future inflation and output are the functions $E_t \hat{\pi}_{t+1} = \pi^e(\hat{r}_t^e)$ and $E_t \hat{Y}_{t+1} = Y^e(\hat{r}_t^e)$, which the central bank takes as given.

The central bank's optimization problem becomes static and is solved by the Lagrangian method using the same notation for multipliers as under commitment, but now taking expectations as exogenous variables. The first-order conditions are

$$\begin{aligned}
 (80) \quad \partial \hat{\pi}_t : \quad & \hat{\pi}_t + \phi_{2t} = 0, \\
 \partial \hat{Y}_t : \quad & \lambda_y \hat{Y}_t + \phi_{1t} - \kappa \phi_{2t} = 0, \\
 \partial \hat{i}_t : \quad & -\sigma \phi_{1t} + \phi_{3t} = 0, \\
 CS : \quad & \hat{i}_t \geq i^{zlb}, \phi_{3t} \geq 0, \phi_{3t}(\hat{i}_t - i^{zlb}) = 0.
 \end{aligned}$$

The equilibrium can now be defined as before.¹¹⁹

We can walk through the same steps as when we derived the solution under the standard monetary policy regime. In the long run ($t \geq t_L$), if the ZLB is not binding, $\phi_{3t} = \phi_{2t} = 0$. The unique bounded solution is $\hat{\pi}_L = \hat{Y}_L = 0$. As according to the standard monetary and

¹¹⁶We will see examples of these type of endogenous state variables later.

¹¹⁷With the exception of the case when we added debt in the analysis, for that then becomes an additional state variable in the policy game.

¹¹⁸Krugman recognized this credibility problem, stating that monetary policy can be effective in a liquidity trap if the central bank can credibly promise to be irresponsible and seek a higher future price level. Eggertsson (2001a) formalized this idea, terming it deflation bias using the statistical process in A2, while Adam and Billi (2007) show it using global methods for a general first-order autoregressive process.

¹¹⁹The equilibrium is a set of stochastic processes for $\{\hat{\pi}_t, \hat{Y}_t, \hat{i}_t, \phi_{1,t}, \phi_{2,t}, \phi_{3,t}\}$ that solve equations (29), (32), and (80), given $\{\hat{r}_t^e, \mathcal{F}_t^{IS}, \mathcal{F}_t^{AS}\}$, as for now we will take fiscal policy as exogenous given and keep both $\mathcal{F}_t^{IS} = \mathcal{F}_t^{AS} = 0$ unless otherwise noted. The same solution method can be applied as for the commitment problem discussed in footnote 109.

fiscal policy regime, this requires $\hat{i}_t = \hat{r}_t^e = i^{zlb}$. In the short run ($t < t_L$) this violates the ZLB, suggesting $\phi_{3t} > 0$. If the ZLB binds, the solution then satisfies exactly the same set of equations as the standard monetary and fiscal policy regime, resulting in the same solution. Hence, the Markov perfect equilibrium is a natural game theoretic *foundation* for the standard monetary and fiscal policy regime. Figure 23 shows that the optimal monetary policy regime under commitment delivers better results than the optimal monetary policy regime in a Markov perfect equilibrium, with output and inflation falling less due to expectations of lower interest rates after the shock and expectations of higher future output.

To see why the optimal monetary policy regime under commitment is not credible and thus does not satisfy the requirement of Markov perfect equilibrium, consider the solution (in isolation) from period five onward. From the perspective of period $t \geq 5$, the central bank is creating excess inflation and output above the target when no shocks hit the economy. Clearly, from the perspective of period $t \geq 5$ onward, the central bank can achieve the best possible outcome, for instance, both the inflation and the output targets: $\hat{\pi}_t = \hat{Y}_t = 0$. Indeed, independent observers of the central bank at time five would feel justified in asking a policymaker following the optimal commitment plan, Why are you generating inflation and output boom today when economic conditions that we observe give you no justification for doing it? This is at the heart of the credibility problem at the ZLB.

One important implication of the Markov perfect equilibrium is that forward guidance by the central bank has no effect. Below, we briefly summarize empirical evidence from the United States about the impact of forward guidance. This evidence paints a more optimistic picture of its impact than an analysis of a Markov perfect equilibrium.

4.1.4 *Forward Guidance in Practice: Empirical Evidence of Market Reaction to the Communications of the Federal Reserve*

Not all promises are created equal—especially when made by central banks. When the Federal Reserve issued forward guidance in August 2011, markets reacted negatively, while a seemingly similar forward guidance in September 2012 triggered a rally. In this section we will explore in more detail the nature of the forward guidance of the Federal Reserve and why they did not always generate the same reaction. Swanson (2021) documents the effect of forward guidance more broadly: Before August 2011, forward guidance typically lowered medium- and long-term Treasury yields by 10–20 basis points, while large-scale asset purchases reduced them by 15–45 basis points. The evidence shows that forward guidance significantly impacted financial markets during the Great Recession, contradicting the prediction that such promises would have no effect, as in a Markov perfect equilibrium. But did the Federal Reserve's actions fully embody the theoretical ideal of full commitment? Although statistical evidence confirms that forward guidance mattered, FOMC minutes and economic data tell a story of a central bank caught between theoretical ideals and practical limitations, where the problem of credibility appears to have been given considerable weight.

The literature identifies the impact of forward guidance by considering market reactions within short event windows, for instance, 30 minutes, typically around FOMC meetings, but more recently also speeches by FOMC members (Swanson and Jayawickrema 2023).¹²⁰

¹²⁰ Kuttner (2001) and Gürkaynak, Sack, and Swanson (2005) are two influential papers that started this rapidly growing literature. For an overview of different branches of the literature see Swanson (2024).

The broad conclusion of this empirical work is that forward guidance was an effective monetary policy tool at the ZLB during the Great Recession. Table 6 from Swanson (2021) shows examples of forward guidance events, such as the FOMC's January 25, 2012 announcement to keep the federal funds rate between 0 and 25 basis points "at least through late 2014." A number of recent empirical studies find a significant effect of forward guidance on financial markets and the economy.¹²¹

Central banks often face unpredictable market interpretations when signaling. This creates a communication challenge for monetary policy. Campbell et al. (2012) distinguish between "Delphic" and "Odyssean" forward guidance. Delphic guidance signals weaker economic conditions than previously expected, while Odyssean guidance indicates the Federal Reserve's willingness to keep interest rates lower for longer than anticipated, even at the risk of higher inflation. Odyssean forward guidance aligns with the optimal monetary policy regime under commitment, as it involves pledges to future policy actions. Conversely, in a Markov perfect equilibrium framework, Odyssean guidance becomes irrelevant, since the central bank cannot make credible long-term promises.

Market reactions to seemingly similar forward guidance statements reveal the crucial distinction between Delphic and Odyssean interpretations. Del Negro, Giannoni, and Patterson (2023) document two contrasting FOMC announcements that exemplify this difference. The August 9, 2011, statement—which the authors classify as Delphic forward guidance—indicated that "economic conditions...are likely to warrant exceptionally low levels for the federal funds rate at least through mid-2013." Markets interpreted this as reflecting pessimistic economic projections rather than a policy shift. Conversely, the September 12, 2012, announcement acknowledged only "moderate" growth and "subdued" inflation while promising accommodative policy "for a considerable time after the economy strengthens." The authors suggest the strong positive market reaction to this announcement reflected Odyssean optimal monetary policy under commitment: maintaining rates at the ZLB even after the economy has recovered. They estimate that while the first announcement lowered expected GDP growth without affecting inflation expectations, the latter increased both inflation expectations and anticipated growth.

Empirical evidence of yield responses to macroeconomic news provides insights into monetary policy effectiveness at the ZLB. Swanson and Williams (2014) analyze how Treasury bond yields react to economic news to determine whether the ZLB constrained Federal Reserve policy. Their reasoning: If the Fed successfully implemented the optimal monetary policy regime under commitment, longer-term yields would respond to news similarly whether at the ZLB or not. We highlight three key findings relevant to our review. First, yields up to six months ahead responded differently than normal to news—an intuitive result since markets would typically adjust near-term rate expectations based on news, but wouldn't if they expected the ZLB to bind for at least two quarters. Second, one- to two-year yields remained responsive to news during 2008–10, though this responsiveness ceased in late 2011. The most plausible explanation, among many offered by the authors, is that until approximately August 2011, markets expected the ZLB to constrain policy for only a few quarters, minimizing effects

¹²¹ For some recent work see, e.g., see Del Negro, Giannoni, and Patterson (2023), Swanson and Williams (2014), and Swanson and Jayawickrema (2023). Earlier work by Gertler and Karadi (2015) and Bauer and Swanson (2023) is also consistent with forward guidance having significant effects, although these latter studies measure monetary surprises as a combination of federal funds rate surprises and forward guidance news and strictly speaking do not separate the two.

TABLE 7
SUMMARY OF ECONOMIC PROJECTIONS BY FEDERAL RESERVE BOARD MEMBERS AND FEDERAL RESERVE BANK
PRESIDENTS, DECEMBER 2015

Variable	Median					Central tendency				
	2015	2016	2017	2018	Long run	2015	2016	2017	2018	Long run
GDP growth	2.1	2.4	2.2	2.0	2.0	2.0–2.3	2.3–2.5	2.0–2.3	1.8–2.2	1.8–2.3
Unemployment	5.0	4.7	4.7	4.7	4.9	5.0–5.1	4.6–4.8	4.6–4.8	4.5–5.0	4.7–5.8
PCE ^a inflation	0.4	1.6	1.9	2.0	2.0	0.3–0.5	1.2–1.7	1.8–2.0	1.9–2.0	2.0
Policy rate	0.4	1.4	2.4	3.3	3.5	0.1–0.6	0.9–1.4	1.9–3.0	2.9–3.5	2.9–3.9

Sources: Board of Governors of the Federal Reserve System (US); Summary of Economic Projections, December 2015 (Board of Governors of the Federal Reserve System 2015).

^aPersonal consumption expenditures.

on medium- and long-term yields. This is broadly consistent with surveys of primary dealers, conducted by the Federal Reserve, at the time. An alternative explanation—that the Fed was actually following the optimal monetary policy regime under commitment and thus endogenously offsetting shocks by adjusting the duration of its low-rate commitment—seems less compelling, which is an interpretation we turn to next.

4.1.5 Assessing Federal Reserve Policy: How Close to the Optimal Commitment Benchmark?

While there is statistically significant empirical evidence that forward guidance impacted markets, contradicting the prediction of the optimal monetary policy regime in a Markov perfect equilibrium, this section reviews data suggesting that Federal Reserve policy fell short of the full optimal monetary policy regime under commitment.

We also provide narrative evidence from policymakers, indicating that this shortfall is partially explained by the deflation bias of the optimal monetary policy regime in a Markov perfect equilibrium.

The signature of the optimal monetary policy regime under commitment is that the Federal Reserve keeps real interest rates low even after the economy has fully recovered and inflation can be stabilized at its target, overshooting both parts of its dual mandate. This, however, is not consistent with the Federal Reserve's behavior as it started projecting a full recovery and raising rates. The record shows that the FOMC aimed to have inflation slowly approach its 2 percent target *from below* rather than replicating the key property of the optimal monetary policy regime under commitment (see Figure 26). Table 7 reports the projections of the 19 FOMC members when they started raising rates in December 2015, with inflation at 0.4 percent. FOMC members projected inflation would reach its target only in 2018. The FOMC started tightening policy in 2015, despite inflation being below target, because they estimated the natural rate of unemployment to be 4.9 percent (Table 7, Long run). With unemployment at 5 percent, the FOMC projected it to go slightly below its natural rate, pushing inflation back to target slowly. The interest rate increase was a preemptive attempt to avoid inflation overshooting, which is more consistent with the optimal monetary policy regime in a Markov perfect equilibrium.

Why did the Federal Reserve fall short of implementing the optimal monetary policy regime under commitment? In a 2019 speech, Chair Jerome Powell (2019) reflected on this question,

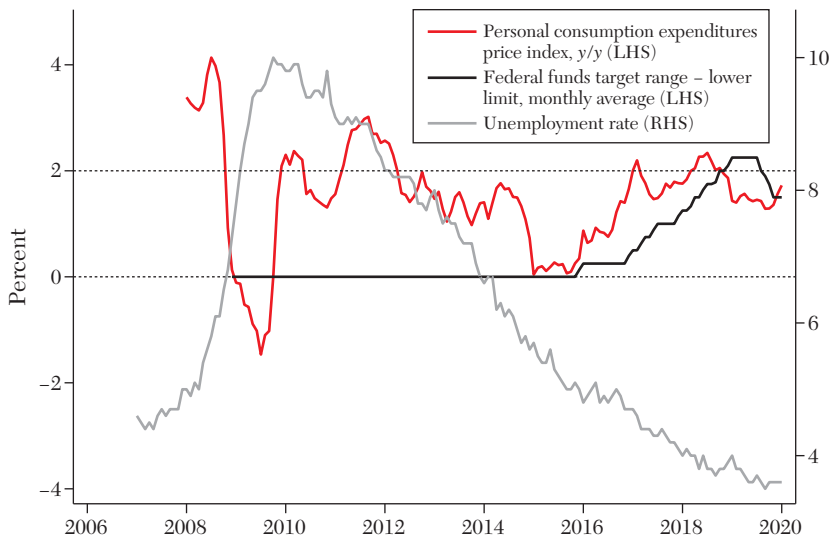


Figure 26. Federal Reserve Interest Rate Policy, Unemployment, and Inflation, 2008–19

Sources: Data from FRED (Board of Governors of the Federal Reserve System 2025b; US Bureau of Labor Statistics 2025; US Bureau of Economic Analysis 2025b).

which includes allowing for excess inflation to make up for the fall in the price level during the ZLB episode, what Powell (2019) terms a *makeup strategy*, see Figure 25:

For makeup strategies to achieve their stabilizing benefits, households and businesses must be quite confident that the “makeup stimulus” is really coming. This confidence is what prompts them to raise spending and investment in the midst of a downturn. In models, confidence in the policy is merely an assumption. In practice, when policymakers considered these policies in the wake of the crisis, they had major questions about whether a central bank’s promise of good times to come would have moved the hearts, minds, and pocketbooks of the public. Part of the problem is that when the time comes to deliver the inflationary stimulus, that policy is likely to be unpopular—what is known as the time consistency problem in economics.

Powell cites FOMC minutes suggesting that members shared similar concerns about the Optimal monetary policy regime under commitment. This statement can be supported by reading declassified minutes. One example: John Williams, vice Chair of the FOMC and president of the Federal Reserve Bank of New York, noted in FOMC minutes from 2011: “In the jargon of academics, our commitment technology is very limited. It is simply impossible for us to set a predetermined course of policy that will bind future Committees.” The dynamic inconsistency problem, the deflation bias, is, therefore, one of the reasons policymakers did not implement a policy consistent with the optimal monetary policy regime under commitment. Recognition of this bias led the Federal Reserve to change its monetary policy framework in 2020.¹²²

¹²²It includes a reformulation of Fed’s objectives in a way that makes implementation of makeup strategies more credible. See Eggertsson and Kohn (2023) on the 2020 Fed framework’s inflation impact. It revised the 2012 strategy with dual changes: targeting above-2 percent inflation after below-target periods and reducing concern about above-maximum

During the Great Recession, the Federal Reserve faced the same credibility problem as policymakers did during the Great Depression. Section 4.3 examines how FDR's 1933 regime change credibly committed to reflation of the economy, confronting similar issues as the Federal Reserve in the Great Recession.

Before we analyze the Great Depression, the next section considers the importance of the assumed policy regime for the theoretical predictions of the new Keynesian model for the fiscal multipliers and the paradoxes we document in Section 3. We then collect the lessons from the theoretical results and the empirical evidence to conjecture what a general policy regime between the two polar extremes of full commitment and none may look like, and how the general regime captures the main empirical evidence we reviewed in Sections 3 and 4.

4.2 *Regime-Dependent Policy Effect: How Optimal Commitment Changes Predictions about Fiscal Multipliers and Paradox*

4.2.1 *Predictions at the ZLB under the Optimal Commitment Policy Regime*

The analysis in this section is based on Eggertsson, Egiev, and Roulleau-Pasdeloup (2024). This section shows how optimal monetary policy under commitment fundamentally changes several key results we derived under the standard monetary and fiscal policy regime. Government spending multipliers become small, and the paradoxes from Section 3 disappear. In a nutshell, what happens under optimal monetary policy regime under commitment is that the central bank tightens its policy, that is, raises expected future real interest rates, in response to, for example, higher government spending. Monetary tightening, in response to a fiscal expansion, is exactly what happens at *positive* interest rates.

One challenge to interpreting the United States as having adopted the optimal monetary policy regime under commitment during the Great Recession is that it is inconsistent with the evidence we presented on the impact of government spending and the paradox of toil. We have also presented evidence that the Federal Reserve policy regime did not leave the classic signature of optimal commitment policy in the data. Moreover, policymakers openly discussed why policy fell short of the optimal commitment.

The government spending multiplier when monetary policy follows the optimal monetary policy regime under commitment can be analyzed in the same way as in Section 3 by assuming A3.¹²³ The result, shown in Table 8, reveals that the multiplier drops considerably and closely approximates the government spending multiplier at positive interest rates.

Figure 27 compares the realization of each of the variables under the assumption the shock reverts back to the steady state in period 5. The left panel in each pair of figures shows the results under the standard monetary and fiscal policy regime. The right side shows the optimal monetary policy regime. Importantly, when modeling the optimal monetary regime, we treat the fiscal impulse as being exogenous and the monetary policy authority reacts optimally to it. This way, the endogenous response of the variables in each policy regime reflects the response

employment following 2015's premature tightening. The September 2020 guidance—requiring both maximum employment **and** above-2 percent inflation before rate hikes—likely contributed to delayed tightening until March 2022 despite approaching a double-digit CPI, when the Fed's primary labor market metrics (unemployment and prime-age employment/population ratio) had returned to pre-pandemic levels.

¹²³This is because the monetary policy regime treats the path of government spending, $\hat{F}_t$, as exogenous.

TABLE 8 FISCAL MULTIPLIERS				
Multiplier	Great Recession		Great Depression	
	Standard monetary and fiscal policy regime	Optimal monetary policy commitment regime	Standard monetary and fiscal policy regime	Optimal monetary policy commitment regime
$M_F = \frac{E_0 \sum_{t=0}^{\infty} \beta^t \Delta \hat{Y}_t}{E_0 \sum_{t=0}^{\infty} \beta^t \Delta \hat{F}_t}$	1.1	0.4	2.2	0.5

Notes: Government spending multipliers contrasting standard monetary and fiscal policy regime (equivalent to optimal monetary policy regime in a Markov perfect equilibrium) from Section 3 to the optimal monetary policy regime under commitment.

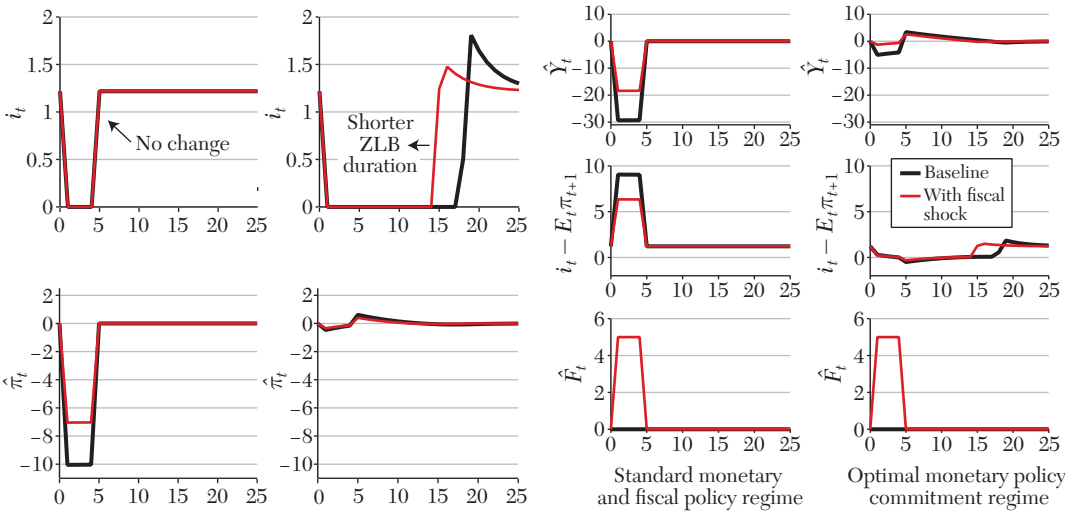


Figure 27. Effect of a 5 Percent of GDP Increase in Government Spending That Lasts Five Periods Using the Great Depression Calibration

Notes: Under the standard monetary and fiscal policy regime from Section 3 and the optimal monetary policy commitment regime from Section 4.

to the same shocks: A temporarily negative efficient interest rate and a fiscal policy expansion that is in direct response to it.

We use the Great Depression calibration in Figure 27. The increase in government spending is 5 percent of steady-state output, of a similar order of magnitude to that in the New Deal. The figure on the upper right-hand side shows that this reduces the output drop to 19 percent instead of 30 percent under the standard monetary and fiscal policy regime as suggested by the multiplier of 2.2. In contrast, the right panel shows a modest increase in output under the optimal monetary policy regime under commitment. This is consistent with the multiplier being more than four times smaller, as shown in Table 8. We also observe that the drop in output without the intervention was large in the first place.

The top panel to the left shows why the multiplier is smaller under the optimal monetary policy regime under commitment. In response, we see by the red line, that monetary policy tightens considerably in response to the fiscal expansion. Meanwhile, under the standard monetary and fiscal policy regime, there is no change in the monetary policy instrument. Under monetary commitment the impact of fiscal expansion most closely resembles its effect under positive interest rates, as a tightening of monetary policy counters the fiscal expansion.

The paradox of toil and the paradox of flexibility no longer apply under the optimal monetary policy regime under commitment. These paradoxes arise when monetary policy cannot offset deflationary forces and welcomes inflationary shocks associated with fiscal spending or oil shocks, since inflation is below target. Under the optimal monetary policy regime under commitment, the central bank has full control of the price level through its ability to commit to future interest rate policy, so for an inflationary shock, it will reduce its commitment to expansionary monetary policy.

Cochrane (2017) emphasizes that the results reviewed in Section 3 are “sensitive to equilibrium choice.” Recall that those results can be derived assuming either the standard monetary and fiscal policy regime or the optimal monetary policy regime in a Markov perfect equilibrium. Instead of the policy regimes we have already considered, Cochrane (2017) suggests “backward stable” and “no jump” equilibria. In both alternatives the equilibria are free of paradoxes. In our view, these alternative equilibria should be interpreted as alternative policy regimes and their plausibility should be judged in that light. Moreover, these policy regimes are very close to being identical to the optimal monetary policy under commitment regime, and can for practical purposes be considered one and the same—even if they are not given the firm logical foundations of the optimal monetary policy under commitment regime.¹²⁴

4.2.2 *Characterizing the Actual Policy Regime: The Middle Ground between the Polar Cases of Commitment and Discretion*

A fundamental open question is what the best way to describe the monetary policy regime in the United States during the Great Recession is. Empirical evidence from high-frequency movements in oil prices and the response of the economy to government spending are consistent with the standard monetary and fiscal policy regime. At the same time, empirical evidence suggests

¹²⁴To see why they are close to identical, observe that Cochrane’s suggested “backward stable” and “no jump” equilibria closely resemble the flexible price equilibrium. The optimal monetary policy regime under commitment is designed to replicate the flexible price equilibrium. Accordingly, Cochrane’s (2017) policy regimes inherit the very same properties we have considered in this section for the optimal commitment equilibrium and he obtains the same results we get when considering that equilibrium. We consider the question the impact of fiscal policy on output as not being a question of equilibration choice. Instead, we think the best way of framing it is to say that the effect of fiscal policy depends upon the assumed policy regime—which is one of the two organizing themes of this review. The next section will focus on what we consider a plausible empirical description of a policy regime in the United States during the Great Recession. A possible source of confusion is that Cochrane’s policy regimes are all associated with a path for the policy rate that is identical, and follows the efficient interest rate one-to-one, and thus readers may suspect equilibrium choice is what matters. However, as first shown by Sargent and Wallace (1975), an exogenous path for the nominal interest rate is consistent with an infinite number of equilibria. The path of the policy rate in isolation is therefore uninformative about the nature of the policy regime or the ultimate effect of one variable on another. As we have stressed, what matters to determine the stance of monetary policy is, instead, the evolution of the real interest rate. Cochrane (2017) defines the different policy regimes he considers by imposing restrictions which do not appear to have clear economic interpretation, such as “backward stable” and “no jump” equilibria.

that forward guidance had a meaningful effect which is more consistent with the optimal monetary policy regime under commitment. These two observations suggest that the policy regime cannot be described as either polar case.

One way to reconcile the evidence is to posit a generalization of the optimal monetary policy regime in a Markov perfect equilibrium that still falls short of the optimal monetary policy regime under commitment. To identify the properties of a more general regime, it is helpful to consider contributing factors: On the one hand, those leading to large fiscal multipliers at the ZLB and the positive effect of oil shocks and, on the other hand, factors that give power to forward guidance.

Under an optimal monetary policy regime in a Markov perfect equilibrium, a large multiplier and positive effect of a negative supply shock (paradox of toil) emerge because monetary policy does not offset the inflationary impact of these shocks. Under the optimal monetary policy regime under commitment, forward guidance has power because the central bank can influence expectations about the future path of the policy rate.

The empirical evidence we review points to both large multipliers and the paradox of toil. To be fair, this empirical debate is still active, so the consensus can at best be called tentative. At the same time, the empirical literature has reached tentative consensus that forward guidance reduces the real interest rates and aggregate spending. Can they be encompassed in a single more general policy regime?

Several plausible policy regimes can account for the patterns documented by the empirical literature in a theoretically consistent way, but this topic remains largely unexplored by the theoretical literature. Here we sketch one illustrative example. The key property of the standard and Markov perfect equilibrium regimes is that interest rates are fixed throughout the ZLB episode (i.e., they do not respond to fiscal shocks). It is easy to imagine a policy regime that maintains this property while at the same time giving forward guidance some power.

Consider optimal monetary policy in a Markov perfect equilibrium augmented with a limited commitment technology of a simple form: The central bank can credibly announce a calendar date before which it commits not to raise the policy rate, even if the natural rate of interest recovers before that date arrives. If credible, this type of commitment changes the expected path of the policy rate when announced, explaining why forward guidance has a statistically significant impact in the data. Yet, this policy regime also retains the key property that generates large spending multipliers and the paradoxical behavior we covered in Section 3: The interest rate does not endogenously respond to fiscal policy expansion or oil shocks at the ZLB. Consider two possible cases. If the natural rate reverts *before* the calendar date to which the central bank commits to keep rates low, the ZLB remains binding, explaining the power of forward guidance. If the natural interest rate reverts to the steady state *after* the binding forward calendar date, the policy follows the optimal monetary policy regime in a Markov perfect equilibrium, and thus once again predicts large fiscal multipliers at the ZLB and the paradox of toil.

4.3 *The Great Depression: Explaining the 1933 Credible and Radical Regime Change that Ended the US Great Depression*

This subsection reviews the optimal monetary and fiscal policy regime in a Markov perfect equilibrium subject to additional institutional commitments. We call this regime, which now

includes possible institutional constraints, the optimal joint monetary and fiscal policy regime in a Markov perfect equilibrium. We use this framework to analyze the US Great Depression.

President Franklin Delano Roosevelt's 1933 policy regime change marked a dramatic turning point in the Great Depression, abandoning institutional constraints like the gold standard and the principle of small government and balanced budget in favor of an aggressive reflation program. This section shows how an optimal monetary and fiscal policy in a Markov perfect equilibrium can explain, at the same time, the sharp recovery that followed in 1933–37, the recession in 1937–38 (the mistake of 1937), and the subsequent recovery from 1938 onwards.

We interpret the spring of 1933, when FDR became president, as a regime change that triggers a sharp recovery, marking the *beginning* of the end of the US Great Depression, but the transition to full recovery was far from smooth. We justify the institutional commitments faced by policymakers at the time by historical records. The regime change is defined as the elimination of these institutional commitments.

The government's overarching objective in 1933 was inflating prices to pre-Depression levels, typically formulated as the price level in 1926 to be reached in three–four years. This view is broadly accepted among historians. Yet its importance has only more recently caught the attention of economists, who have traditionally placed a higher emphasis on the gold standard. This will also be part of our narrative, in addition to the evolution of the money supply and fiscal outlays.¹²⁵ The omission of this policy element in economic analysis is partially because early Keynesian studies were often conducted within the confines of IS–LM models, according to which expectations are treated as exogenous shocks, while the endogenous change in expectations as a result of regime change are center stage in this review. Moreover, in early Keynesian models, price dynamics typically played a peripheral role, remaining fixed in Hicks's (1937) first rendition of the IS–LM model.¹²⁶

The optimal joint monetary and fiscal policy regime in a Markov perfect equilibrium rationalizes, within the model, *why* the inflation program credibly shifted expectations from deflationary to inflationary. In fact, President Hoover kept promising recovery of prices and output too. The analysis answers why FDR achieved this goal while Hoover did not.

Sargent's (1982) seminal paper "The Ends of Four Big Inflations" demonstrates the importance of modeling regime changes when analyzing abrupt shifts in macroeconomic aggregates during crises. Using a rational expectations model, Sargent studies four major post-World War I inflationary episodes, emphasizing that a policy regime change's effectiveness relies on its credibility and expectations, with fiscal policy playing a central role. Thus, the success of a policy regime change depends not only on the technical aspects of reform, but also on the public belief in the government's commitment to it.

Temin and Wigmore (1990) apply Sargent's insight to analyze the Great Depression in their classic paper "The End of One Big Deflation." The interpretation of the US recovery from the

¹²⁵Following is a representative quote from Stanford historian David M. Kennedy: "And yet, amid the chaos of the Hundred Days, and indeed through the tense stand-off of the interregnum that preceded it, one threat flashed and dove like a scarlet skein shot through brocade: inflation" (Kennedy 1999). Kennedy's work is generally considered the definitive historical account of the era and was awarded the Pulitzer Prize, and as the quote reflects, he identifies inflating the price level as the central objective of FDR.

¹²⁶Once expectations and price dynamics became part of the canon of economics, however, attention had largely shifted elsewhere, with a return to the topic happening once the ZLB became an important topic of practical relevance again.

Great Depression, reviewed in this subsection, formalizes Temin and Wigmore's idea,¹²⁷ which can be applied using the standard new Keynesian model.

4.3.1 *Modeling Institutional Constraints and Policy Credibility: The Role of Policy Dogmas*

This section reviews optimal discretionary monetary and fiscal policy in an extended new Keynesian model where households derive utility from government consumption and taxation is distortionary. The social welfare function integrates the Federal Reserve's dual objective from the previous section and contains new terms for fiscal considerations.

Relative to the model in Section 3, we simplify the analysis by assuming a distortionary tax, $\hat{T}_t^d$, associated with deadweight loss as in Barro (1979b) (instead of the multiple tax instruments as in Section 3) and a limited access to lump-sum taxation.

Imagine that only a part of government spending, G_t , contributes to social welfare and enters utility additive separately to consumption via the function $g(G_t)$, which is increasing and concave in its argument. Taxation absorbs real resources via the increasing and convex function $f(T_t^d)$. This function represents tax-collection costs or "indirect misallocation costs" that create economic distortions, as proposed by Barro (1979b). Total government spending is then

$$(81) \quad F_t = G_t + f(T_t^d).$$

We assume that total taxes consist of two components:

$$(82) \quad T_t = T_t^d + T_t^L,$$

where T_t^L is a lump-sum tax. This tax carries no distortions, but the government has limited access to it. Allowing for some form of lump-sum taxation simplifies the analysis and enables us to consider the economy local to the efficient steady state. Welfare is then approximated by¹²⁸

$$(83) \quad -\frac{1}{2}E_0 \sum_{t=0}^{\infty} \beta^t \left(\underbrace{\hat{\pi}_t^2 + \lambda_x \hat{x}_t^2}_{\text{Dual Mandate of the Federal Reserve}} + \underbrace{\lambda_F \hat{F}_t^2 + \lambda_T (\hat{T}_t^d)^2}_{\text{Fiscal Policy Consideration}} \right),$$

where $x_t \equiv \hat{Y}_t - \sigma^{-1} \psi \hat{F}_t$ is the output gap, with all variables defined as in the last section and $\hat{T}_t^d \equiv \frac{T_t^d - \bar{T}^d}{\bar{Y}}$.

A policymaker's problem is to maximize the social welfare function (83), which, in addition to the dual mandate, now includes a desire to keep government spending at a constant target level and taxation smoothing. In addition to the constraints or dogmas of the government's policy problem we consider shortly, the policymaker takes into account the government budget constraint, written as in equation (51) in Section 3, but with the simplified taxation system

$$(84) \quad \hat{w}_t = \beta^{-1} \hat{w}_{t-1} + w_y \hat{t}_t - \beta^{-1} w_y \hat{\pi}_t + \beta^{-1} \hat{F}_t - \beta^{-1} \hat{T}_t^d.$$

¹²⁷They use a narrative approach instead of providing an explicit model.

¹²⁸See Bhattacharai, Eggertsson, and Gafarov (2023) for a derivation of this objective.

Consider three additional institutional commitments, termed dogmas, based on the historical record:¹²⁹

- (85) No-deficit dogma: $\hat{w}_t = 0$,
- (86) Small-government dogma: $F_t = \bar{F}$,
- (87) Gold standard dogma: $M_t \leq \lambda_g p_g g_t^m$.

The first two policy dogmas are self-explanatory: The no-deficit dogma maintains the real value of government debt at the steady-state level, while the small-government dogma fixes government spending at the (low) steady-state level, $\hat{F}_t = 0$.

The gold standard dogma requires elaboration. M_t is the nominal money stock, g_t^m is the gold reserves of the Federal Reserve, p_g is the dollar price of gold, and λ_g measures the required gold backing of the monetary base. In the United States, the Federal Reserve needed to keep gold reserves at 40 percent of the monetary base.¹³⁰ For simplicity, g_t is exogenous.

The gold standard formulation follows Barro (1979a), except that we allow for the Federal Reserve to hold gold in excess of what was required to support the outstanding monetary base. The inequality reflects the Federal Reserve's commitment to hold *at least as much* gold as required. This asymmetry was important: The Federal Reserve "sterilized" gold inflows throughout the 1920s and much of Herbert Hoover's term, hence not increasing the money supply in proportion to gold inflows.¹³¹

The gold standard's asymmetry was crucial for understanding US monetary policy, as it implied that it did not tie the hands of the Federal Reserve to expand the supply of money. The Fed held more gold than it needed to support the outstanding money supply and could therefore expand it further if it chose to do so, which it did not. This point has been highlighted by several authors, including Friedman and Schwartz (1963), Temin (1991), Eichengreen (1992), and Meltzer (2003).¹³²

The government's problem is to maximize (83) subject to constraints imposed by the new Keynesian model but with the additional institutional commitments implied by the no-deficit dogma, the small-government dogma, and the gold standard dogma, which enter as additional constraints.

The policy regime change when FDR took power in the spring of 1933 is modeled as:

Hoover regime $\rightarrow$ Unexpected elimination of the no-deficit dogma,
the small-government dogma, and the gold standard dogma $\rightarrow$ Roosevelt regime

This historical record provides compelling evidence that President Hoover's policy regime was constrained by the policy dogmas; see discussions in Temin and Wigmore (1990) and Eggertsson (2008), with a few illustrative quotes in Table 9. What is equally clear from the

¹²⁹See Eggertsson (2008) for a detailed discussion.

¹³⁰The price of gold was 20.67 per ounce, so g_t^m is measured in ounces.

¹³¹In case of outflows, if the constraint was close to being binding, the Federal Reserve lacked this flexibility and would need to contract the monetary base, a situation many other gold standard countries faced during the Great Depression.

¹³²Meltzer notes that in February 1932, gold covered 71 percent of outstanding money supply, suggesting the constraint was far from binding.

TABLE 9
THE REGIME CHANGE IN THE SPRING OF 1933

President Hoover	President Roosevelt
<i>Small-government dogma (D1)</i> “Every additional expenditure placed upon our government in this emergency magnifies itself out of all proportion into intolerable pressures, whether it is by taxation or by loans. Either loans or taxes [...] will increase unemployment. [...] We can carry our present expenditures without jeopardy to national stability. We can carry no more without grave risks.” September 21, 1931	“Our greatest primary task is to put people to work. This is no unsolvable problem if we face it wisely and courageously. It can be accomplished in part by direct recruiting by the government itself, treating the task as we would treat the emergency of a war, but at the same time, through this employment, accomplishing greatly needed projects to stimulate and reorganize the use of our natural resources.” March 4, 1933
<i>No-deficit dogma (D2)</i> “For the Government to finance by bond issues deprives industry and agriculture of just that much capital for its own use and for employment. Prosperity cannot be restored by raids on the public Treasury.” July 18, 1930	On debt: “In the first place, government credit and government currency are really one and the same thing.” On the purpose of increasing debt: “That is why powers are being given to the Administration to provide, if necessary, for an enlargement of credit [...] These powers will be used when, as, and if it may be necessary to accomplish the purpose [i.e., increasing inflation].” April 19, 1933
<i>Gold-standard dogma (D3)</i> During the fall of 1932 presidential campaign, both President Hoover and FDR were strong supporters of the gold standard.	FDR formally suspended the gold standard on April 20, 1933.

Notes: Adapted from Eggertsson (2008).

record is that a central part of FDR’s policy when he assumed the presidency was to eliminate these policy dogmas, which was largely unanticipated (as it violated FDR’s election platform from the fall of 1932). In response to the radical policy changes, several senior administration officials resigned in protest, with one, Lewis Douglas, the Director of the Budget, declaring: “This is the end of Western Civilization.”¹³³

Considering the equilibrium free of the institutional constraints imposed by the policy dogma, we observe that the real value of government debt, $\hat{w}_{t-1}$, is an endogenous state variable. In a Markov perfect equilibrium, agents’ expectations depend only on the minimum set of state variables. We consider a solution where the expectation functions are

(88)

$$\begin{aligned} E_{t,j} \hat{Y}_{t+1} &= Y_r^J \hat{r}_t^e + Y_w^J \hat{w}_t, \\ E_{t,j} \pi_{t+1} &= \pi_r^J \hat{r}_t^e + \pi_w^J \hat{w}_t, \\ E_{t,j} \hat{F}_{t+1} &= F_r^J \hat{r}_t^e + F_w^J \hat{w}_t, \end{aligned}$$

¹³³See Eggertsson (2008).

with unknown constants $Y_r^J, Y_w^J, F_r^J, F_w^J, \pi_r^J$, and π_w^J solved using the method of undetermined coefficients, and J being S for short run and L for long run. The government's problem is characterized by the Bellman equation:

$$(89) \quad V(\hat{w}_{t-1}, \hat{r}_t^e) \\ = \max_{\hat{T}_t^d, \hat{F}_t, \hat{i}_t} - \left\{ \pi_t^2 + \lambda_y (\hat{Y}_t - \delta \hat{F}_t)^2 + \lambda_F (\hat{F}_t)^2 + \lambda_T \hat{T}_t^d + \beta E_t V(\hat{w}_t, \hat{r}_{t+1}^e) \right\},$$

subject to the IS equation (29), ZLB (31), AS equation (32), and the budget constraint (84), with expectations in every equation by the relevant expectation variable in equation (88). In addition, the no-deficit dogma, the small-government dogma, and the gold-standard dogma apply in the Hoover regime. However, the gold standard constraint is assumed not to be binding under President Hoover, even if its elimination is critical for the credibility of the FDR inflationary regime change, since the inflation program implied it *could* be binding in the future (Eggertsson 2008).

4.3.2 *The Hoover Regime and the Economic Collapse 1929–33: Policy Dogmas of Gold Standard, Balanced Budget, and Small Government*

This section shows the Hoover regime, with its strong deflationary bias, yields results identical to the standard monetary policy regime in Section 3 and the optimal monetary policy regime in a Markov perfect equilibrium reviewed in Section 4.1.3

Hoover's adherence to three rigid policy dogmas—balanced budgets, small government, and the gold standard—effectively paralyzed both fiscal and monetary policy during the crucial early years of the Great Depression. This section illustrates how these constraints on the Hoover regime (as long as the gold standard constraint is not binding), when incorporated into our framework, generate an equivalent equilibrium to the one derived in Section 4.1 for optimal *monetary* regime in a Markov perfect equilibrium, thereby accounting for both the massive output collapse and severe deflation of 1929–33. Recall, that we had already established an equivalence between that policy regime and the standard monetary policy regime.¹³⁴

Suppose government spending that contributes to utility is fixed at $G_t = \bar{G}$. Real debt is constant by the no-deficit dogma, and we assume lump-sum taxes adjust to cover variations in interest rate costs.¹³⁵

¹³⁴ See Eggertsson (2008) for a more detailed discussion of the role of this constraint under the Hoover regime.

¹³⁵ Taking account of how monetary policy affects these interest costs has trivial quantitative effects, since our approximation is local to the efficient steady state. To be more concrete, we assume total government spending and the welfare-contributing part are fixed at $G_t = \bar{G}$. With real debt constant at $\bar{w}$, the budget constraint implies taxes are determined by

$$\hat{T}_t = (1 + \bar{i})w_y \hat{i}_t - w_y \hat{\pi}_t.$$

Taxes must adjust to offset changes in real interest rate payments given fixed government spending, affecting the government's tax-smoothing incentive. While we will fix lump-sum taxes at their steady state, we will make the exception of letting them vary to cover interest cost to abstract from this effect that leads to unnecessary complications:

$$\hat{T}_t^L = (1 + \bar{i})w_y \hat{i}_t - w_y \hat{\pi}_t,$$

where $\bar{T}^d$ is distortionary taxation at steady state and T_t^L is lump-sum taxation adjusting to offset variations in debt payment costs.

Consider now social welfare shown in (76), which includes the fiscal variable F_t and T_t^d . The no-deficit says real debt is constant. The small-government dogmas indicate constant spending. An implication is that the central bank's objective now boils down to the dual mandate. As long as the gold standard constraint is not binding, which it was not in the United States, the government's policy problem yields equations shown in (80) once again.

The Hoover regime delivers the solution depicted in Figures 9 and 23, which correspond to the equilibrium under the standard monetary and fiscal policy regime and the optimal monetary policy regime in a Markov perfect equilibrium.

4.3.3 *Roosevelt's Radical Regime Change in 1933: How It Triggered a Recovery*

Inflation was the overriding policy objective of the regime change when FDR took office. FDR employed all available tools, via trial and error, to reverse the 10 percent annual deflation and increase the price level. This section rationalizes FDR's inflation program as the optimal joint fiscal and monetary regime under discretion, where a regime change is rationalized by the elimination of Hoover's policy dogmas.

Breaking dramatically not only with Hoover's policies but also with his own Democratic presidential campaign platform from the fall 1932, President Roosevelt was inaugurated March 4th, 1933, with one overriding objective: to reverse the devastating 10 percent annual deflation and restore prices to pre-Depression levels using every available policy tool. FDR defined the pre-Depression price level as that prevailing in 1926. While the administration's target horizon of 3 to 4 years and exact price metrics remained somewhat ambiguous, FDR's resolve was unmistakable: "If we can't do it one way, we will do it another. But do it we will," he declared in the fall of 1933, when the policy's credibility was being questioned.

This subsection rationalizes FDR's inflation program as the optimal monetary and fiscal policy in a Markov perfect equilibrium once the government breaks free from Hoover's institutional constraints. There are other fruitful interpretations of the complex nature of the regime change in 1933 and subsequent recovery, such as Jacobson, Leeper, and Preston (2019).¹³⁶ We expect future research to combine elements of both the optimal commitment and optimal Markov perfect equilibrium regimes when modeling policy during the Great Depression, similar to our discussion of the Great Recession in Section 4.2.2. An important element of a more general framework would be modeling both the varying credibility of the program and the evolution of policymakers' thinking as they gathered experience and experimented with different policy strategies.

Under this regime, the government uses both monetary $\hat{t}_t$ and fiscal policy, (real government spending, $\hat{F}_t$, and distortionary taxes $\hat{T}_t^d$) to solve the dynamic programming problem given by the policy problem (89), but now free of policy dogmas. This means that the real value of debt ($\hat{w}_t$) is an endogenous state variable. Moreover, the government now has a blunt tool in its arsenal that it can directly employ to increase spending, that is, government spending $\hat{F}_t$. Expectations are a linear function of the state variable but with an undetermined coefficients, Y_w , π_w , and F_w , which can be found using methods of undetermined coefficients. The first-order conditions are computed by formulating a Lagrangian for the government's problem in any period t , with

¹³⁶Eggertsson and Pugsley (2006), for example, model it as a reduced-form commitment to inflation.

Lagrange multipliers ϕ_{1t} , ϕ_{2t} , ϕ_{3t} , and ϕ_{4t} for the IS equation (29), AS equation (32), budget constraint (84), and ZLB (31). The first-order conditions are¹³⁷

$$\begin{aligned}
 (90) \quad \partial \hat{\pi}_t: \quad & \hat{\pi}_t + \phi_{2t} + \beta^{-1} w_y \phi_{4t} = 0, \\
 \partial \hat{Y}_t: \quad & \lambda_x (\hat{Y}_t - \sigma^{-1} \psi \hat{F}_t) + \phi_{1t} - \kappa \phi_{2t} = 0, \\
 \partial \hat{F}_t: \quad & \lambda_F \hat{F}_t - \phi_{1t} + \kappa \psi \sigma^{-1} \phi_{2t} = 0, \\
 \partial \hat{T}_t^d: \quad & \lambda_T \hat{T}_t^d + \phi_{4t} = 0, \\
 \partial \hat{w}_t: \quad & -E_t \phi_{4t+1} - \phi_{1t} (\hat{Y}_w + \sigma \pi_w - F_w) - \beta \pi_w \phi_{2t} + \phi_{4t} = 0, \\
 \partial \hat{i}_t: \quad & -\sigma \phi_{1t} + \phi_{3t} = 0, \\
 CS: \quad & \hat{i}_t \geq i^{zlb}, \phi_{3t} \geq 0, \phi_{3t} (\hat{i}_t - i^{zlb}) = 0.
 \end{aligned}$$

Monetary and fiscal policy coordination improves the outcome through two mechanisms. First, free from the small-government dogma, the government is able to increase spending. This increases aggregate demand by direct actions today, which is not subject to the credibility problem associated with a promise of some future policy.

Second, the key element of the regime change was that FDR promised to inflate the price level to pre-Depression levels. Free from the no-deficit dogma, the government had a direct tool to back this up and make the inflation pledge credible: It could finance spending by running deficits. As illustrated in Table 9, this is how FDR himself rationalized how the goal of inflation could be reached, stating, “That is why powers are being given to the Administration to provide, if necessary, for an enlargement of credit. [...] These powers will be used when, as, and if it may be necessary to accomplish the purpose.”

By most measures, FDR backed up the rhetoric with action. The federal government’s consumption and investment, for example, was 90 percent higher in 1934 (Roosevelt’s first full calendar year in office) than in 1932 (Hoover’s last), but the fiscal year was defined by the middle of the year during this period. The deficit during Roosevelt’s first fiscal year was one of the highest in US history outside of wartime and 2020 (during the COVID-19 epidemic), increasing by 66 percent in FDR’s first fiscal year (from June 1933 to June 1934) and reaching 9 percent of GDP. Unlike Hoover, who tried and failed to balance the budget, FDR deliberately pursued deficit financing and explicitly said he would use it to bring up inflation.¹³⁸ This made a permanent monetary expansion credible, firming up inflation expectations and making the target of the price level of 1926 more credible. Fiscal expansion made the reflation

¹³⁷ Here, we have substituted out the derivative of the value function $E_t V(\hat{w}_t, \hat{r}_{t+1}^e)$ using the Lagrange multiplier ϕ_{3t} and an envelope condition, typically referred to as the Benveniste–Scheinkman condition in economics:

$$V_w(\hat{w}_{t-1}, \hat{r}_t^e) = -\beta^{-1} \phi_{3t}.$$

By forwarding this condition once and taking conditional expectations at time t , we can substitute the value function in the above first-order condition.

¹³⁸ See Eggertsson (2008) for more conservative estimates. Relative to government spending, the deficit was larger than in 2020, due to the smaller government spending and taxing capacity during this period.

objective also a critical part of the governments fiscal policy strategy, as it had every incentive to satisfy the commitment to minimize distortionary taxation. Reneging on the commitment would increase both the real value of government debt as well as interest rate payments.¹³⁹

FDR recognized the problem that the Federal Reserve might not be pliable to his fiscal needs and fully on board with his inflation program. This was one reason for the “credibility crisis” in the fall of 1933, which historians have referred to as mini recession. To address this issue, he rewrote the Federal Reserve Act to centralize power in Washington, DC, and appointed Marriner Eccles, who favored active and coordinated fiscal and monetary response, as chair. See Eggertsson and Schüle (2024), who argue that this was as a significant boost to the credibility of the reflation regime in 1934.

Figure 28 compares the simulated model path to the data, using the calibration from Eggertsson (2008). The only difference is the assumption of external habits in consumption and labor, replacing output with the quasi-growth rate, $\tilde{Y}_t = \hat{Y}_t - \rho \hat{Y}_{t-1}$ in the AS and AD equations.¹⁴⁰ The vertical line denotes the regime change when FDR takes power and the dashed green line represents the data. Panel A shows that under the Hoover regime the output would have continued collapsing, while the regime change generated the blue-circled line, closely matching the recovery in the data. The turning point was not driven by interest rate cuts (panel E) or money supply changes (Temin and Wigmore 1990). Instead, the regime change can be explained by eliminating the three policy dogmas in the optimal joint monetary and fiscal policy regime in a Markov perfect equilibrium. Panels F and G show substantial increases in real and deficit spending,¹⁴¹ triggering a shift in inflation and inflation expectations.

Panel B shows the model-generated increase in inflation, matching the data. Panel D shows the drop in ex ante real interest rates, driven by the change in expected inflation, measured by Cecchetti (1992) based on time series forecasts using data available to market participants at the time, and Hamilton (1992) that uses commodity futures traded during this period to estimate the shift in expectations. Jalil and Rua (2015) provide compelling narrative evidence that a major change in inflation expectation occurred during the regime shift using three sources: (i) new data series on inflation news coverage, (ii) the reports of contemporary observers contained in the historical news accounts, and (iii) the forecasts of contemporary business analysts. Moreover, Eggertsson (2008) documents major shifts in various variables that are sensitive to changes in expectations about future policy and economic conditions, including individual commodity prices, stocks, and investment, all of which abruptly started surging following the regime change. Taken together, the overall empirical evidence is strongly supportive of major shifts in inflation expectation in the spring of 1933.

Although the gold standard was not binding, its elimination played a crucial role. Even with sufficient gold reserves in 1933 for monetary expansion, there was no guarantee the constraint would not become binding if the administration succeeded in restoring the 1926 price level. Eliminating the gold standard may therefore have been central to coordinating expectations toward an inflationary regime, which is broadly consistent with its central role in Eichengreen (1992). As shown in Eggertsson and Krugman (2012), which models debtors and creditors in Section 2.1, the redistribution aspect of inflation can have a significant effect, since most debt

¹³⁹Table 9 shows a quote from FDR indicating that he considered increasing deficits as a tool to increase inflation.

¹⁴⁰This implies a more gradual fall in output.

¹⁴¹As a fraction of GDP.

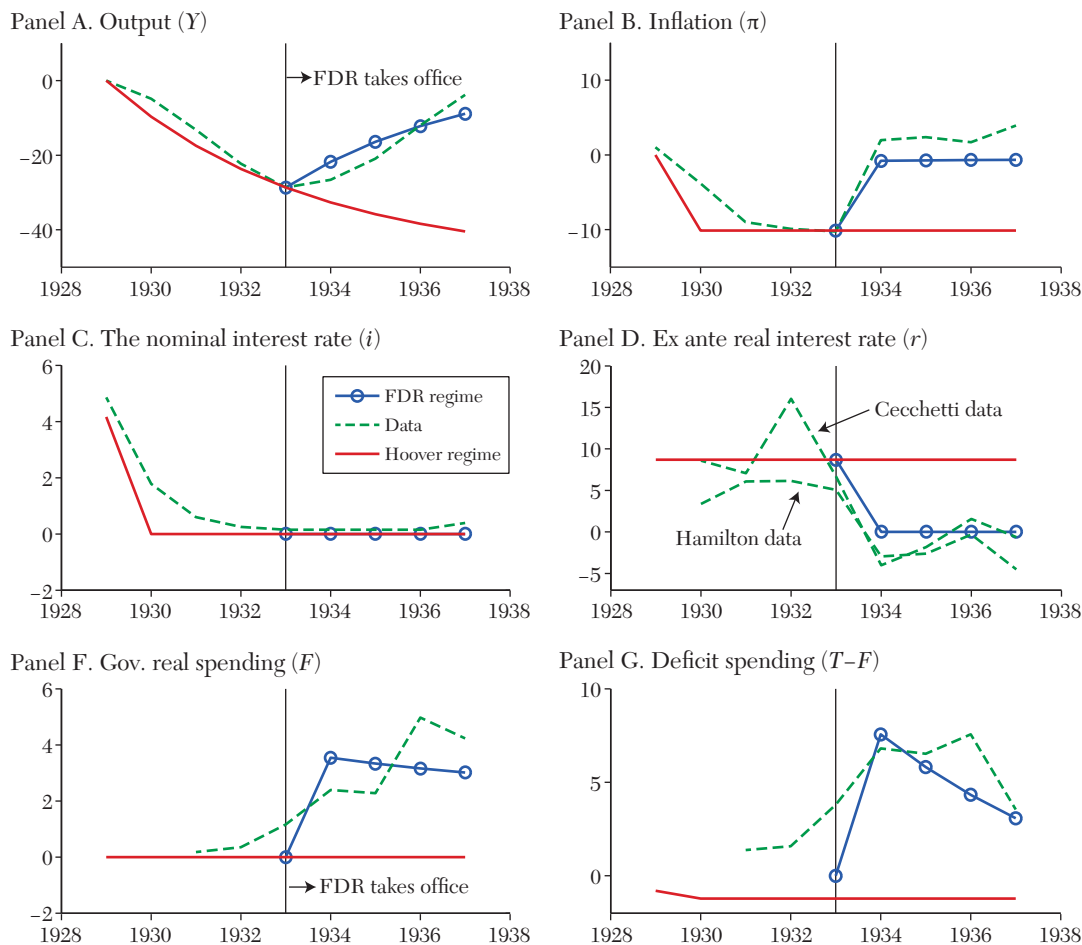


Figure 28. Data and the Hoover Regime Change versus the FDR Regime Change

Notes: The FDR regime change implies a collapse in real interest rates and a robust recovery in prices and output. The regime change can account for 67 percent of the recovery in prices and 79 percent of the recovery in output in the period 1933 to 1937.

contracts are nominal. Hausman, Rhode, and Wieland (2021) provide interesting evidence that this mechanism may have played an important role in the recovery of 1933.

4.3.4 The Mistake of 1937 and the Recovery of 1938

This section reviews evidence suggesting that the 1937 recession, the second phase of the Great Depression, was partially triggered by doubts about the credibility of FDR's inflationary commitment reviewed in the previous section, with the markets anticipating a reversal to the

previous regime. In contrast, we review evidence that the 1938 recovery is interpreted as the administration recommitting to its 1933 inflationary pledge, triggering a sharp rebound.

While defense spending picked up following the fall of France in the Q2 of 1940 (Gordon and Krenn 2014), the US economy had already made significant headway. It is a caricature to claim World War II ended the Great Depression, a common assertion, ignoring the impact of FDR's inflationary policies in 1933–37 and the resumption of recovery in 1938 before the major boost in defense spending.

We consider our general account of the recovery from the Great Depression to be consistent with the broader literature. Bernanke (1983) and Friedman and Schwartz (1963) emphasize the role of disruption in financial intermediation as an important source of the Great Depression. As we saw in Section 2, this narrative gives a natural interpretation for the reduction in the natural rate of interest triggering the Great Depression. Similarly, Romer (1992) stresses the role of monetary policy in generating the recovery from 1933, recognizing that with the interest rate at zero, “the main way that the monetary expansion could stimulate the economy was by generating expectations of inflation and thus causing a reduction in real interest rates,” thus highlighting what became the central theme of the modern literature on the liquidity trap and is at the core of this review.

The analytic framework we outlined in the last subsection can be used to cast light on the second phase of the Great Depression. One interpretation is that it was caused by an erosion of trust in FDR's inflationary commitment, with markets anticipating restoration of the previous regime. The 1938 recovery can be interpreted in the model as the administration recommitting to its 1933 inflationary pledge, triggering a sharp rebound.

Eggertsson and Pugsley (2006) term the turning point in 1937 as “the mistake of 1937” and the second “the Reversal of 1938.” By 1937, the price level was far from the 1926 target despite industrial production reaching pre-Depression levels (Figure 29). The administration weakened its commitment to the price level target, as evidenced by the narrative examples in Table 10 and reinforced by the Federal Reserve raising reserve requirements and the federal government attempting to balance the budget. Market expectations shifted, believing the administration had reverted to the Hoover regime. Eggertsson and Pugsley (2006) show that a small change in beliefs about inflationary regime can generate large output movements, as observed in the data. The drop in industrial production following the mistake of 1937 was the sharpest contraction in US history. Recovery resumed in 1938, when the administration recommitted to reinflating prices.

5. *A Permanently Negative Natural Rate of Interest and the Secular Stagnation Hypothesis: Understanding Japan's Long-Run Recession*

Sections 3 and 4 analyzed a liquidity trap driven by fast-moving forces generating a temporary drop in the efficient rate of interest. However, as discussed in Section 2, a branch of the literature contemplates slow-moving forces generating a permanently negative efficient rate of interest, the second organizing principle of this review, often referred to as the secular stagnation hypothesis. We first review the origin of the hypothesis. We then move on to its modern formalization, which suggests this possibility has significant policy implications by elevating the importance of robust fiscal response and weakening the power of monetary policy. We use the framework to interpret Japan's Long Recession and conclude with future research directions combining fast and slow-moving forces.

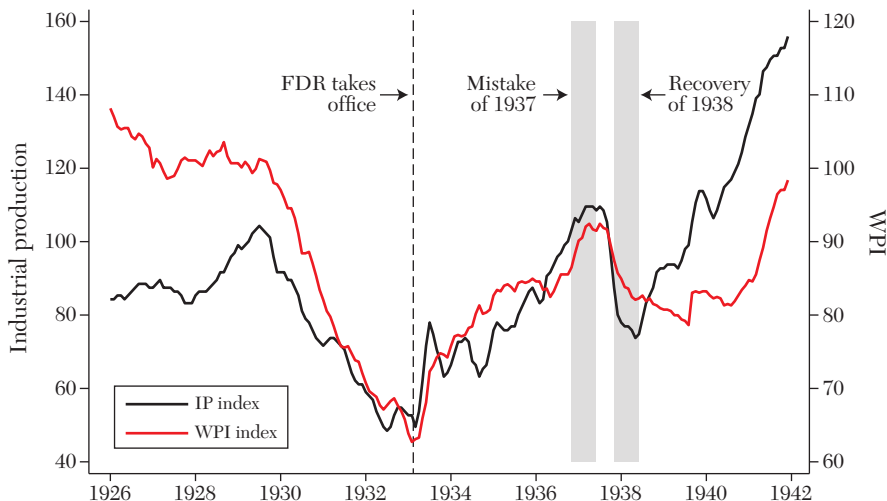


Figure 29. Industrial Production and Price Level in the Great Depression

Notes: Both wholesale prices (WPI) and industrial production (IP) collapsed in 1929–33 but abruptly started to recover in March 1933, when FDR took office and implemented a regime change along with New Deal policies. The second phase of the Great Depression occurred after the mistake of 1937, when the Supreme Court struck down parts of the New Deal and the administration appeared to back away from its commitment to reflate the price level to pre-Depression levels. Data on IP and WPI is from Eggertsson (2012b).

5.1 *The Origin of the Secular Stagnation Hypothesis: From Hansen to Summers*

The idea that the natural rate of interest can be permanently negative can be traced back to the secular stagnation hypothesis developed during the Great Depression.

During the closing phase of the Great Depression, President of the American Economic Association, Alvin Hansen, delivered a disturbing message in his Presidential Address to the association (Hansen 1939). He suggested that the Great Depression might start a new era of ongoing unemployment and economic stagnation without any natural force pulling the economy toward full employment. This idea was termed the secular stagnation hypothesis. According to Hansen, some of the main driving forces of secular stagnation was a decline in the population birth rate, oversupply of savings, and lack of investment opportunities, suppressing aggregate demand. Soon after Hansen’s address, the Second World War led to a massive increase in government spending, effectively ending any concern about insufficient demand. Moreover, the baby boom following World War II drastically changed the population dynamics in the United States, thus effectively erasing the problem of excess savings of an aging population.

Hansen, often called the “American Keynes,” based his analysis on Keynesian foundations.¹⁴² He implicitly assumed a classic Keynesian consumption function, $C_t = C^a + c(Y_t - T_t)$,

¹⁴²The following equations represent our interpretation of Hansen’s (1939) verbal exposition. We believe his exposition is clear enough, however, so that there is little doubt about the interpretation in the context of the modeling tradition of his time.

TABLE 10
THE REGIME CHANGES IN 1937 AND 1938

The Mistake of 1937	The Reversal of 1938
February 18, 1937. Marriner Eccles, Chair of the Board of Governors, speaks in Senate hearings: "The short-term rates are excessively low, and there may be a tendency for rates near the vanishing point to increase" (<i>Wall Street Journal</i> , February 19, pg. 1).	February 15, 1938. President Roosevelt holds a press conference: "At his press conference today, the President said that he believes now, as he did in 1933, that achievement of permanent prosperity depends on raising general price levels to those prevailing in 1926" (<i>Chicago Daily Tribune</i> , February 16, 1938, pg. 1).
March 15, 1937. Marriner Eccles, Chair of the Board of Governors, gives a statement: "The upward spiral of wages and prices into inflationary levels can be as disastrous as the downwards spiral of deflation." (<i>Chicago Daily Tribune</i> , March 16, pg. 1).	February 18, 1938. President Roosevelt releases a written statement at a press conference that was prepared by Henry Morgenthau Jr., Secretary of the Treasury; Henry A. Wallace, Secretary of Agriculture; Frances Perkins, Secretary of Labor; Marriner Eccles, Chair of the Board of Governors of the Federal Reserve System; and economists of various executive departments: "It is clear that in the present situation, a moderate rise in the general price level is desirable [...]. Our program seeks a balanced system of prices that will promote a balanced expansion in production. Our goal is a constantly increasing national income through increasing production and employment. This is the way to increase the real income of consumers."
March 17, 1937. Commerce Secretary Daniel C. Roper and Secretary of Agriculture Henry A. Wallace hold press conferences. Both secretaries warn against excessive inflation. (<i>Wall Street Journal</i> , March 18, 1937, pg. 8)	April 14, 1938. FDR addresses Congress, announcing that the reserve requirement will be abandoned: "The measures for expanding excess reserves which were announced on Thursday by President Roosevelt will re-create the bases for a great credit inflation [...]. Monetary management, after having been directed for some time towards guarding against a possible inflationary boom, has turned, under the pressure of the business depression, toward the other extreme" (<i>New York Times</i> , April 17, 1938).
March 24, 1937. Marriner Eccles, Chair of the Board of Governors, outlines five steps to avert "dangerous inflation" in <i>Forbes</i> magazine: (i) reserve requirement "to eliminate excess reserves," (ii) fiscal policy that balances the budget, (iii) reduction in the gold price of the dollar, (iv) increase in the labor share of national income, (v) antitrust legislation. (<i>Christian Science Monitor</i> , March 25, 1937)	
April 2, 1937. FDR holds a press conference: "I am concerned—we are all concerned—over the price rise in certain materials."	
August 3, 1937. FDR's views on price-level targeting are revealed. Senator Elmer Thomas publishes a letter from FDR to himself rejecting his proposal that the Federal Reserve formally target the 1926 price level. (<i>Wall Street Journal</i> , August 4, 1937, pg. 6)	

Note: Adapted from Eggertsson and Pugsley (2006).

where $C_a > 0$ and $0 < c < 1$. His analysis is consistent with a typical early Keynesian investment function, $I_t = I^a - b i_t$, where $I^a > 0, b > 0$. The variables C^a and I^a are autonomous consumption and investment spending, while i_t is the interest rate.¹⁴³ Combining this with the resource constraint $Y_t = C_t + I_t + F_t$ we obtain:

$$(91) \quad Y_t = \frac{C^a}{1-c} + \frac{F_t}{1-c} - \frac{c}{1-c} T_t + \frac{I^a}{1-c} - \frac{b}{1-c} i_t.$$

Hansen (1939) proposed that the declining growth of population exerted downward pressure on autonomous investment spending, I^a , through various mechanisms. He doubted that future innovation would generate enough new investment opportunities to sustain the necessary investment and compensate for this reduced spending. In fact, he suggested that introducing new production technology would lead to falling production costs over time, a concept modern economists refer to as the decrease in the relative price of investment. This, according to Hansen, would lead to a long-term decline in investment spending in the absence of technological innovation.

With short-term nominal interest rates near zero and a very limited understanding of price dynamics at the time, it was natural for Hansen, like John Maynard Keynes, to have little confidence in interest rate policy.

The situation in 2013 strongly resembled that of 1938 when Hansen delivered his speech. Although governments had largely contained a meltdown on the order of the Great Depression following the 2008 financial crisis, growth remained anemic, investment weak, population and productivity growth low, prompting people to seek alternative explanations. In this environment, Lawrence Summers gave an influential speech at the International Monetary Fund in the Fall of 2013 (see Summers 2014). He revived Hansen's secular stagnation hypothesis, but proposed a novel and provocative reformulation that tied it directly to the modern ZLB literature. Summers posed the following question: How do policy prescriptions change if we assume a *permanently* negative natural interest rate instead of a *temporarily* negative one?

We have already discussed in Section 2.1.4 how to model this in an endowment economy. However, we have not explored the implications for output, inflation, and policy. We covered these implications for *temporary* shocks in sections 3 and 4. We now examine them for *permanent* shocks.

Given the maturity of the ZLB literature, readers might find it peculiar that previous literature had not contemplated the possibility of permanently negative natural rates. The reason for this is twofold. First, what is needed is to abandon the standard IS equation common in the new Keynesian literature to be able to allow for the possibility of permanently negative real interest rates. The second (and, perhaps, the most important) reason is that to pursue this line of thought one needs to sacrifice one of the most sacred cows of modern macroeconomics: that there is no permanent trade-off between output and inflation. That is, in the long run, output is not represented as a vertical line in the output–inflation space. Yet, the judgment about the absence of permanent trade-off was reached during the heyday of high inflation in the 1970s,

¹⁴³The distinction between real and nominal was not always clear in early Keynesian literature, since the Hicksian IS–LM model assumed fixed prices. Moreover, the relevant interest rate was typically assumed to be the long-term interest rate.

and generally based upon studies of countries suffering from persistently high inflation. No such empirical consensus was reached at low inflation.¹⁴⁴

Hansen's interest in *permanent demand recessions* rendered post-1970s models inherently inadequate, as they excluded by assumption the very possibility of what he—and later Summers—sought to explore. Addressing even the theoretical possibility of a demand recession of arbitrary duration required not only an open-heart surgery on the new Keynesian IS curve but, more radically, the replacement of the new Keynesian Phillips curve with a production framework that permits meaningful permanent trade-offs between output and inflation, thereby opening the door to demand recessions of arbitrary length.

5.2 A Modern Model of Secular Stagnation: Modifying the New Keynesian Framework

Here, we sketch out a simplified version of the first formal model of secular stagnation, that of Eggertsson and Mehrotra (2014). Relative to Section 2, we now introduce price level determination and a production side. The presence of money leads to a consumption Euler equation that prices one-period risk-free debt, yielding a nominal interest rate controlled by the monetary authority. The pricing of one-period risk-free debt is

$$(92) \quad \frac{1}{C_t^m} = \beta E_t \frac{1}{C_{t+1}^o} (1 + i_t) \frac{P_t}{P_{t+1}},$$

using the notation of Section 2. We obtain the standard Fisher equation if we assume perfect foresight and then combine the middle equation in (15) and equation (92):

$$(93) \quad 1 + r_t = \frac{1 + i_t}{\Pi_{t+1}}.$$

Monetary policy is set according to a Taylor rule:

$$(94) \quad 1 + i_t = \max \left(1, (1 + i^*) \left(\frac{\Pi_t}{\Pi^*} \right)^{\phi_\pi} \right),$$

where ϕ_π and Π^* and i^* are parameters of the policy rule taken as given. The rule states that the central bank stabilizes inflation around the target Π^* associated with the steady-state interest rate i^* unless it is constrained by the ZLB.

In steady state all variables have constant values. The derivation that led us to equation (18) in Section 3 remains unchanged. However, we no longer treat Y as an endowment but as an endogenously determined aggregate demand that has to equal aggregate supply in equilibrium. In steady state this yields

$$(95) \quad 1 + r = \frac{1 + \beta(1 + g)\tilde{D}}{\beta(Y^d - \tilde{D})},$$

where Y^d now represents aggregate demand instead of an income endowment.

¹⁴⁴See Eggertsson, Lancastre, and Summers (2019) for further discussion and citations on this point.

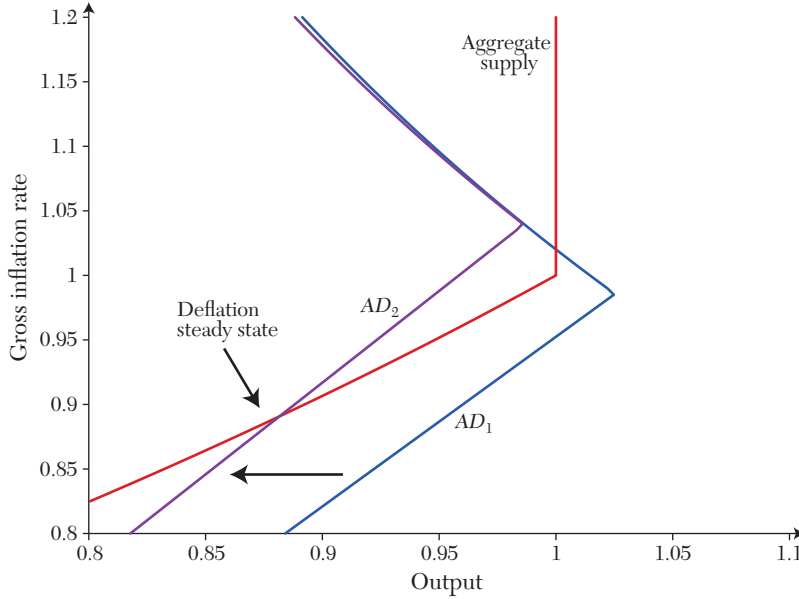


Figure 30. Steady-State Aggregate Demand and Aggregate Supply Curves

Collecting the starred component of the monetary stance at positive interest as $m^* \equiv (\Pi^*)^{\phi_\pi} / (1 + i^*)$ we can now rearrange equation (95) in terms of output demanded, substitute for the policy rule in the Fisher equation, and finally replace the real rate using the Fisher equation (93) combined with the policy rule to arrive at:

$$(96) \quad Y^d = \begin{cases} \tilde{D} \left(1 + \frac{1 + \beta}{\beta} (1 + g) m^* \frac{1}{\Pi^{\phi_\pi - 1}} \right) & \text{for } i > 0, \\ \tilde{D} + \frac{(1 + \beta)(1 + g)\tilde{D}}{\beta} \Pi & \text{for } i = 0. \end{cases}$$

This AD equation has the same form as the AD considered for a temporary shock in Figure 11, as can be seen in Figure 30. The fundamental difference is that it corresponds to a steady state. For example, this relationship can be *permanently* shifted by a slowdown in population growth ($1 + g$) and the change in the debt limit, $\tilde{D}$. More generally, any force changing the relative demand for savings and investment, as discussed in Section 2, shifts it. Such forces include rising inequality, increasing life expectancy, or growing government debt.

We close the model by specifying the production side. Economists like Hansen, following Hicks's original IS–LM model, would simplify the problem by depicting the red line as a straight horizontal one, reflecting fixed prices. This horizontal line, when placed below the kink of the AD curve, would illustrate the futility of interest rate cuts.

Ever since Phillips's (1958) work, economists have enriched the IS–LM model with a theory of inflation determination. Today's economists seem to agree that if inflation remains

persistently high, expectations about future inflation will eventually adjust, eliminating any long-run trade-off between inflation and unemployment. The model incorporates this idea with a vertical AS curve *when inflation is sufficiently high* (see the top half of the AS curve in Figure 30).

In contrast, economists never reached a similar consensus on the long-run neutrality of inflation *when it's low*. Tobin (1972) argued that during the Great Depression, firms were reluctant to cut nominal wages despite high unemployment, suggesting a permanent inflation–unemployment trade-off at low inflation. Later Akerlof et al. (1996) showed that the assumption of downward wages implies a long-run trade-off.¹⁴⁵ We incorporate this idea with an upward-sloping Phillips curve at low inflation (see the bottom half of the AS curve in Figure 30).

Here, we sketch a parsimonious model that captures the gist of the idea. Households elastically supply labor. They reject work paying below the prevailing, downward rigid wage rate. In contrast, they happily work for higher wages. Yet the labor endowment is not infinite and capped at full employment, that is, L^j . A representative firm maximizes profits using the following technology:

$$(97) \quad Y_t = L_t^\alpha.$$

Here, $1 > \alpha > 0$ and L_t is the supply of labor by the middle-aged generation. The firm hires workers to decide production, taking the price level P_t and wage rate W_t as given, yielding the first-order condition:

$$(98) \quad \frac{W_t}{P_t} = L_t^{\alpha-1}.$$

Nominal wages are downward rigid:

$$(99) \quad W_t = \max\{\tilde{W}_t, W_t^{flex}\}.$$

Here, $\tilde{W}_t$ is the wage norm and W_t^{flex} is the wage rate if wages are flexible. This assumption says that households will not supply labor unless the salaries are at least equal to a social norm given by $\tilde{W}_t$. However, they are perfectly willing to work for higher wages; hence, if the market-clearing salary is higher than the wage norm, the firm will bid up the wages to the market-clearing level.

Keynes's original idea was that wages are rigid downward—that is, workers would not accept a nominal wage rate lower than last year (Keynes 1937). It is one example of a wage norm. Here, we posit a norm that generalizes Keynes's idea modestly so that one could vary the rigidity parameter, γ , with perfectly downward rigid prices at $\gamma = 1$ and perfectly flexible at $\gamma = 0$:

$$(100) \quad \tilde{W}_t = W_{t-1}^\gamma (W_t^{flex})^{1-\gamma}.$$

¹⁴⁵The idea of downward rigid nominal wages dates back at least to Malthus, who noted that “it very rarely happens that the nominal price of labor universally falls” (Malthus 1798). Bewley (1999) interviewed corporate executives, documenting their reluctance to cut nominal wages. More recently, researchers have discovered substantial nominal wage rigidity using US administrative data (Fallick, Lettau, and Wascher 2011), worker surveys (Barattieri, Basu, and Gottschalk 2014) and cross-country data (Schmitt-Grohé and Uribe 2016). Downward wages were also at the core of the original Phillips curve.

Imposing a steady state and combining equations (97)–(100) yields the AS relationship

$$(101) \quad Y^s = \begin{cases} Y^f & \text{for } \Pi \geq 1 \\ Y^f \Pi^{\frac{1-\gamma}{1-\alpha}} & \text{for } \Pi < 1 \end{cases},$$

where Y^f denotes the level of output when the entire labor endowment has been utilized. Figure 30 plots this relationship along with the AS curve, which stands vertical at positive inflation ($\Pi > 1$).¹⁴⁶ With inflation below target (including deflation), real wages exceed the market-clearing level and firms do not hire all workers. The figure shows two AS–AD intersections: the upper one, where the natural rate of interest is positive (AD_1 and AS cross, shown with a positive inflation target), and the lower one, where the natural rate is sufficiently negative, shifting the AD curve to AD_2 , and the interest rate is zero. The latter case features deflation and unemployment. This deflationary steady state (AD_2) is interesting because it represents *a permanent recession without any natural force to pull the economy to full employment*. Here, the price and wage market mechanisms do not generate a recovery. More flexible wages exacerbate the recession, another manifestation of the price-flexibility paradox from Section 3. Eggertsson, Mehrotra, and Robbins (2019) show that the model can generate output and inflation dynamics observed for the Long Recession in Japan via a permanent reduction in the natural interest rate.

5.3 Policy Implications of Secular Stagnation: The Limitations of Conventional Approaches

Secular stagnation fundamentally reshapes policy effectiveness compared to standard new Keynesian models with temporarily negative natural rates. When the natural rate remains persistently negative, several traditional policy approaches fail. First, waiting for the shock to subside proves futile—a strategy that likely dominated initial responses in the euro area. Second, forward guidance loses effectiveness when markets expect the ZLB to persist indefinitely. Third, higher inflation targets, often portrayed as a panacea in the literature, face what Krugman (1999) calls the “law of the excluded middle.” Figure 31 illustrates this problem: If the announced target is too low (see AD_2 in the left panel), it remains unattainable despite public confidence in the central bank’s commitment, as the upper branch of the AD curve never intersects the AS curve. Only sufficiently high targets shift the AD curve upward enough (see AD_3) to permit full employment.

Fourth, even appropriate inflation targets face a crucial challenge: Their mere announcement does not eliminate the original secular stagnation equilibrium, leaving expectation coordination uncertain. Finally, fiscal policy retains and even amplifies its power under secular stagnation. Effective fiscal policy shifts the AD curve rightward in Figure 31, creating a *unique* full-employment equilibrium at point D. Unlike higher inflation targets, sufficiently aggressive fiscal policy eliminates secular stagnation entirely. However, in overlapping-generations models, fiscal policy operates differently than in standard new Keynesian frameworks, with distributional consequences becoming crucial (Eggertsson, Mehrotra, and Robbins 2019).

¹⁴⁶In this case the wage norm is not binding, wages are flexible, and the entire labor endowment is employed.

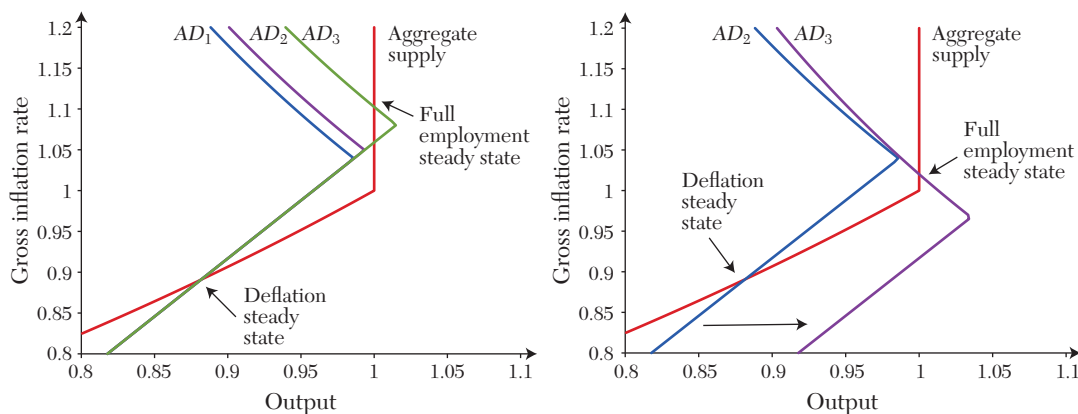


Figure 31. Monetary and Fiscal Policy Responses

We already cited Caballero and Farhi (2018), who propose an alternative explanation for negative natural interest rates based on a shortage of safe assets. Their approach complements ours by emphasizing excess demand for safe assets, driving up a “safety premium,” which reduces the return on safe relative to risky assets. Using the AD–AS framework from Eggertsson and Mehrotra (2014), they derive the same basic insights. A key new policy implication is that increasing the supply of safe assets—either through encouraging private issuance or via public guarantees—offers another policy dimension to address economic stagnation.

5.4 Japan’s Long Recession and Abenomics: Three Decades at the Zero Bound

Shinzo Abe was elected Prime Minister of Japan on December 26, 2012, and served until 2020, when he resigned due to health reasons. His platform, Abenomics, consisted of “three arrows”: expansionary monetary policy, aggressive fiscal policy, and structural reforms. This agenda shared similarities with FDR’s 1933 regime change and drew inspiration from it.

Abe threatened to revise the law that granted the Bank of Japan independence if it did not agree to increase its inflation target and engage in “unlimited easing” to achieve that goal, a threat reminiscent of FDR ultimately rewriting the Federal Reserve Act in his first term to centralize power in Washington and fulfill his pledge of reflation the price level.¹⁴⁷ The Japanese Parliament also passed legislation for a 2 percent increase in government spending. However, the actual increase was only about half of that.¹⁴⁸

While the Abenomics experiment is relatively recent, and thus, a full assessment is only tentative, some observations can be made based on our preceding analysis of secular stagnation, considering Japan has been at or near the ZLB for about 30 years, with only a glimmer of hope of escape briefly in 2006, and also as of current writing in 2025.

Early evaluations of Abenomics, such as Hausman and Wieland (2014, 2015) and Romer (2014), were relatively positive based on its initial success in 2013. However, the following years yielded mixed results, creating fertile ground for future research.

¹⁴⁷ See Hausman and Wieland (2014).

¹⁴⁸ *ibid.*

During the Abenomics period (2013–20), inflation averaged 0.87 percent. This rate exceeded the near-zero inflation from 1996 to 2008, but fell well short of the 2 percent target promised by the Bank of Japan's governor Haruhiko Kuroda in 2013. Kuroda pledged to achieve this target in two years through “open-ended asset purchases.” GDP growth during Abenomics averaged 1.04 percent, slightly lower than the 1.18 percent average from 1996 to 2008.¹⁴⁹

Hausman and Wieland (2015) suggest the lack of success of the regime change is partly explained by the imperfect credibility of the inflation target of the Bank of Japan, arguing that the BoJ would reduce the monetary stimulus before inflation reached the target just as it did in 2006, as we documented in Section 2.

Our analysis of secular stagnation offers a different perspective, as shown in Figure 31. If the natural interest rate were below -2 percent, no equilibrium would be consistent with achieving this inflation target. It must be large enough to accommodate the natural interest rate; for example, if it is -5 percent, the target must be 5 percent or more.

At a G7 meeting in Japan in 2016, one of the authors of this paper discussed a critical policy dilemma with a senior Japanese policymaker regarding the uncertain natural interest rate. The Bank of Japan had failed to reach its 2 percent inflation target. When presented with the argument that 2 percent might be too low, given the law of the excluded middle, and that a higher target was necessary, the policymaker responded (paraphrased): “We promised 2 percent inflation and could not deliver. How seriously would the public take us if we announced an even higher target when we cannot even reach 2 percent?” One interpretation of this exchange suggests that central banks should initially set higher inflation targets under such circumstances, since failing to achieve a lower target can damage their credibility and make it harder to raise targets later. Another interpretation is that the announced target should have similar properties as the implementation of the optimal policy commitment suggested by Eggertsson and Woodford (2003). Announce a target for prices, and if this is not reached, then increase the target further in the next period to provide additional stimulus.

Comparing Abenomics to FDR's regime change is instructive. In the spring of 1933, FDR committed to increasing the price level to its 1926 levels. This implied a 30–40 percent increase in the price level. Despite the administration's somewhat vague timeline, the general understanding was to achieve this within three–four years or less. That would imply inflation of around 10 percent per annum. Such monetary transformation, unlike Abenomics, could thus respond to a substantially negative natural rate of interest. In addition, FDR implemented a significantly more aggressive fiscal expansion and aimed his supply-side policies primarily at raising prices instead of addressing longer-term structural issues, which as we have seen in Section 3.3.3, carry the danger of triggering deflationary forces in the short run.

5.5 *Integrating Temporary and Permanent Forces: Future Research Direction after the 2020s Inflation Surge*

Integrating temporary and persistent causes of liquidity traps represents a critical frontier for economic research. The literature we have reviewed shows that an economy can enter a liquidity trap either due to the forces that are sharp and temporary or due to forces that are

¹⁴⁹This lower growth partly resulted from unfortunate sales tax increases on April 1, 2014 (from 5 to 8 percent), and October 1, 2018 (from 8 to 10 percent), policies determined before Abe took office and severely contradicting his objective to an extent that is important for future research to evaluate.

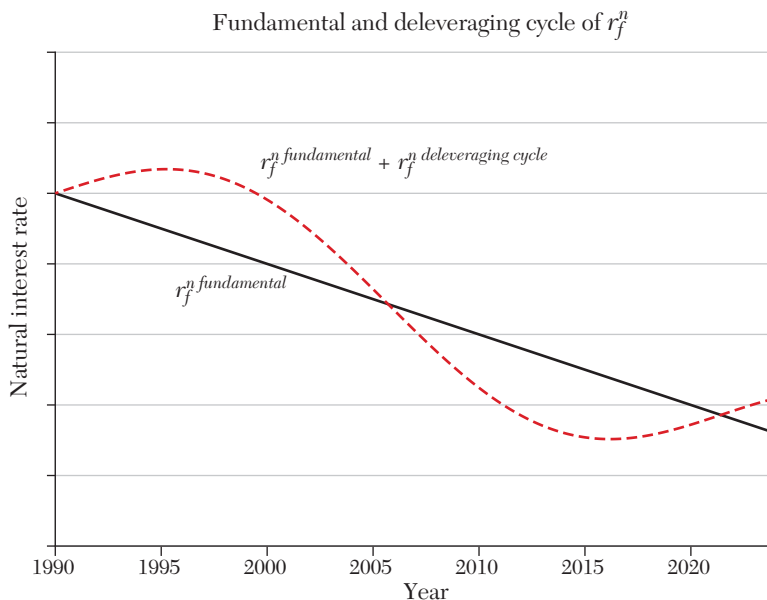


Figure 32. Combining a Debt Deleveraging and a Secular Stagnation Narrative

persistent and slow. The literature has not yet integrated the two perspectives. Here, we offer some speculative remarks as to how such integration might look. We argue that this line of research is of central importance to the current policy strategy. We also offer highly speculative comments on the current state of affairs in the United States as of writing.

Figure 32 provides a graphical illustration of how such an integration might look. The figure highlights four points. First, temporary phenomena can mask secular trends in interest rates, as is evident in the data we showed in Figure 5. A temporary relief from the ZLB may soon be undone. Second, temporary phenomena can lead to a sharper decline in interest rates than slow-moving forces would suggest if acting alone. Third, in this theoretical framework the real interest rate should return to the lower trend level once temporary factors disappear.

Fourth, an inflationary surge emerged in the 2020s following the COVID-19 pandemic, which at the time of writing appears to be receding, though this remains uncertain. If inflation is successfully controlled after this surge, and the secular stagnation hypothesis holds, long-term interest rates will likely resume their downward trend, forcing the Federal Reserve to confront the ZLB in coming decades. The long-term trends driving the secular decline in the natural rate assuming the secular stagnation framework remain firmly in place: inequality and demographic patterns, for example, have not fundamentally changed during the 2020s inflation surge and are unlikely to shift radically in the foreseeable future.

However, multiple variables could upend these predictions. Recent AI innovations arguably represent the most important technological development of recent years, potentially altering the downward trend in the natural interest rate through accelerated productivity growth. At the time of writing, whether these impressive AI technologies will translate into productivity gains strong enough to pull the economy out of secular stagnation remains uncertain, but plausible.

Less optimistically, a growing productivity divergence between the United States and Europe has emerged, particularly in the AI sector, both in innovation and adoption. Consequently, the United States possesses at least one discernible candidate technology that could theoretically increase the natural rate of interest. In Europe a possible catalyst is a widely discussed military buildup in response to the Russia–Ukraine war and the United States decreasing involvement in the North Atlantic Treaty Organization.

The regime change in 2025 in the United States introduces additional uncertainty through several proposed policies of a scale rarely seen in US post–World War II history, which may imply a number of unintended consequences: major tariff increases, stronger enforcement of immigration laws against undocumented immigrants, and possible pressure on the Federal Reserve to lower rates. Tariffs could raise prices while reducing the workforce, via tightened enforcement of illegal immigration, increasing wage pressures. Lower interest rates in this inflationary environment would further accelerate price increases. At sufficiently high inflation rates, the ZLB does not constrain policy.

While budget deficit trajectories remain unclear, increasing public debt would push the natural interest rate upward. Public promise of major cuts in government spending pushes in the opposite direction. Tax policy favoring the top 1 percent would likely increase savings and exert downward pressure on interest rates. Finally, the long-term treasuries have traditionally been considered a “safe haven” for investors.

We have already cited work by Caballero and Farhi (2018). They trace low interest rates to shortage of safe assets. If the US regime change in 2025 implies that treasuries no longer maintain their “safe haven status,” Caballero and Farhi’s (2018) theory predicts that this would intensify the shortage of safe assets and push the natural rate of interest down further.

Together, these factors create unprecedented uncertainty regarding how the ultimate policy mix will affect the investment–savings balance going forward, and thus the natural interest rate.

Looking ahead, Hansen’s gloomy post–World War II prediction appears possible for both Europe and the United States and other industrialized economies in coming decades. It is difficult to judge if Japan’s Long Recession has at long last come robust end.

The new normal natural rate of interest remains one of the great unknowns in macroeconomics.

6. Conclusion

This survey has proposed a unified theoretical framework for understanding three major economic crises—the Great Depression, the Great Recession, and Japan’s Long Recession. It identifies both fast-moving forces, such as financial crises, and slow-moving forces, such as demographic changes, as potential triggers for liquidity traps. We have analyzed how effectively various monetary and fiscal policies performed in these contexts.

A key takeaway is that sufficiently flexible and credible monetary policy can largely offset temporary shocks leading to liquidity traps; however, fiscal support becomes essential when central bank credibility is lacking or shocks are permanent. The US experience during the Great Depression vividly illustrates the power of coordinated monetary and fiscal expansion.

Looking ahead, a critical question is whether the United States and global economies will settle into a “new normal” of persistently low interest rates, similar to Japan’s experience over the past 30 years, if inflation pressures from the 2020s recede—a highly uncertain proposition,

given the considerable political uncertainty. The secular drivers identified in the literature, such as rising inequality and declining population growth, clearly suggest the possibility of persistently low rates becoming the new normal.

Given these prospects, episodes in which the ZLB constrains monetary policy may become increasingly frequent and prolonged in the decades ahead. Fiscal policy would then need to play a more active and ongoing role in macroeconomic stabilization. Monetary policy frameworks might also need to adapt significantly, including serious consideration of major changes in payment technology that could eliminate the ZLB. Our current system of US government-issued paper currency dates back to 1861. Much has changed since then, and various technical solutions could be implemented to improve the current state of affairs.

While the literature surveyed here has substantially advanced our understanding of liquidity traps, important questions remain. Key priorities include integrating fast- and slow-moving forces into a unified framework and better characterizing the credibility challenges faced by central banks when interest rates are constrained. Ultimately, it is essential to develop a robust, empirically grounded theory of the *policy regime* prevailing during episodes such as the Great Recession.

In our view, the central and overarching lesson—a theme with which we opened this paper—is that the economics profession largely ignores the prospect of major and persistent contractions to the peril of free democratic societies. Poorly managed macroeconomic crises can cast a long shadow. When assessing the importance of investigating a scientific question, one should consider not only the precision of the answers obtained but also the marginal benefit to society from incremental improvements in our understanding. Few areas offer marginal benefits as obvious as improving our understanding of major economic crises—not only because of their immense costs, but also because policymakers who manage responses are often directly informed by existing and ongoing research.

The Great Depression was not an aberration, but a warning of what can happen absent vigorous and well-informed policy responses. During the Great Recession, the United States had the rare good fortune of having, at the helm of the Federal Reserve, a leader who had spent his career studying the Great Depression, along with economic advisors leading the two key advisory groups in the White House who had similarly devoted significant portions of their academic careers to studying that episode. It seems exceedingly unlikely we will have such luck next time, leading to worse economic outcomes. There is considerable consensus, for example, that the tariff war initiated by the passage of the Smoot–Hawley Act at the onset of the Great Depression substantially deepened the crisis. We have also discussed in detail the significant harm caused by the Hoover administration’s dogmatic rigidity in clinging to principles that, while well-suited to normal times, were highly unsuitable in responding to the Great Depression. The chances of success the next time the United States faces a similar challenge will be greater if a more coherent academic consensus on proper remedies is established.

In a world of low natural interest rates, macroeconomic policy must remain vigilant and creative—the stakes for economic growth, social cohesion, and political stability could hardly be higher.

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